

AI Effects on the Labor Market:

Is AI Disrupting the Future of Work?

How to read this presentation

This Deep Dive explains how Artificial Intelligence (AI) is reshaping the labor market from first principles.

The sections are designed to be read in order and assume no prior background knowledge.

Our goal is not to predict the future with certainty, but to give you a clear map of the mechanisms at play: how automation, augmentation, and new-task creation interact to shape jobs, wages, and productivity.

For nearly every slide in this deck, there are academic papers and policy reports that explore these topics in far greater depth.

Our goal isn't to overwhelm you with data, but to make complex dynamics legible. Omissions are intentional.

/ Learn With Me

Table of Contents

01 The Mechanics That Drive AI’s Impact on Labor	04
02 Exposure	17
03 Adoption	31
04 Labor Market Response 1: Augmentation	53
05 Labor Market Response 2: Displacement	73
06 Labor Market Response 3: Reinstatement	86
07 The AI-Work Feedback Cycle	98
08 The Case For Abundance	108
09 Conclusion	120

The Mechanics That Drive AI's Impact on Labor

Experts Agree AI Will Transform Jobs, but Disagree on How.



"I think most companies are going to be doing more or less the same things. A few occupations will be impacted, but we're still going to have journalists, we're still going to have financial analysts, we're still going to have HR employees."

- Daron Acemoglu, Nobel Laureate in Economics



"There's tremendous opportunity here. I do not think that AI means the death of all human expertise... it means amplifying in many realms and supplementing it for people who have judgment but who are not at the frontier."

- David Autor, Professor in the MIT Department of Economics, Co-Director of Labor Studies NBER



"AI will be a massive productivity boost for existing job roles and it will create many new job roles... there will be some job loss, but I think it may not be as bad as people are worried about right now."

- Andrew Ng, Adjunct Professor at Stanford University, Founder & Lead for the Google Brain Project



"You won't lose your job to AI, but you will lose your job to someone who uses AI."

- Jensen Huang, President and CEO of Nvidia



"AI could wipe out half of all entry-level white-collar jobs."

- Dario Amodei, Co-founder and CEO of Anthropic

What Is a Job?

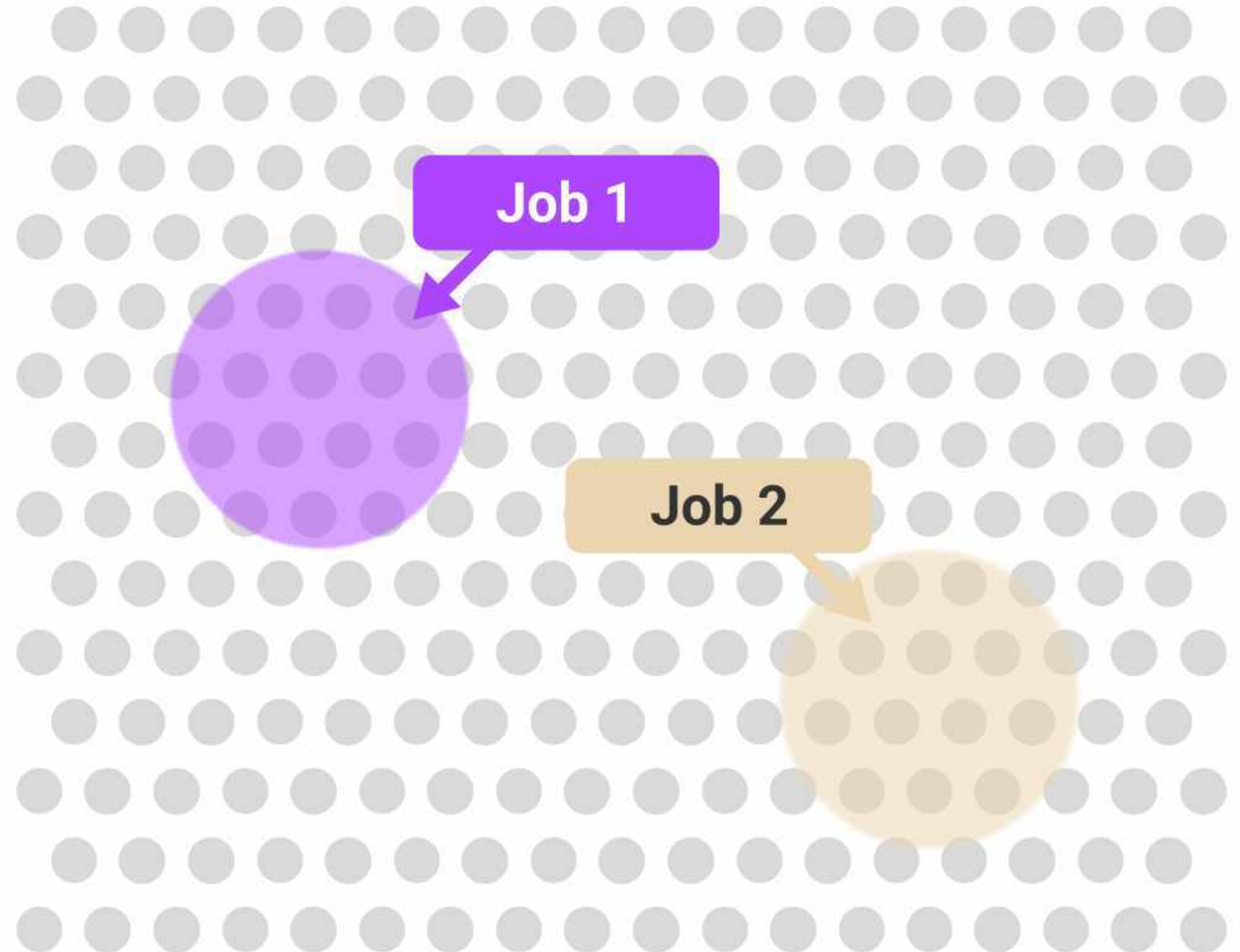
At its core, a job is a bundle of tasks.

A doctor diagnoses symptoms, orders tests, interprets results, and coordinates with specialists.

A software engineer writes code, fixes bugs, reviews pull requests, and explains trade-offs to stakeholders.

A lawyer drafts contracts, researches case law, argues motions, negotiates settlements, and advises clients.

Universe of Tasks



A Three-Layer Framework for Thinking About AI and Jobs

When a big technology arrives, like AI, it almost never wipes out whole jobs overnight.

It affects *tasks* first.

Some tasks get automated, some get supercharged, and some create new kinds of work.

To make sense of this, it helps to look at AI's impact through three layers:

(1) Exposure

(2) Adoption

(3) Labor market response

Layer 1: Exposure

This layer answers: **Given current and foreseeable capabilities, which tasks in a job are susceptible to automation, which are augmentable, and which are resistant?**

To answer, we need two components:

- 1) **The task composition of each occupation.** This allows analysis of the actual bundle of tasks in a computer programmer's or a nurse's job.
- 2) **The pattern of AI development,** for example today's models excel at text and pattern recognition but are weak at dexterous manipulation or social nuance.

Researchers combine these two components into an “exposure index,” which estimates how much of a job current AI could realistically assist or perform.

Crucially, **exposure isn't a forecast of layoff.** It just tells you where the technical potential lies, not whether firms or organizations will deploy it. That brings us to Layer 2.

Layer 2: Adoption

This layer answers: **How quickly and in what form will organizations deploy AI?**

Even if a task is technically automatable, adoption can lag because of cultural and institutional factors like legal constraints, union rules, customer backlash, or lack of complementary investments.

The degree of adoption shapes Layer 3.

Layer 3: Labor Market Response

This layer answers: **Given the pattern of adoption, what is the net effect on employment, skill demand, wages and the creation of new jobs?**

Economists talk about three response types:

- 1) Augmentation:** The labor market responds by keeping humans in their current jobs but making them more productive by “augmenting” them with AI tools.
- 2) Displacement:** The labor market responds by displacing human workers and replacing them with machines.
- 3) Reinstatement:** The labor market responds by creating new tasks and expanding human roles that complement technology. This produces new jobs such as AI engineering and other times existing occupations broaden as productivity gains lower costs and raise demand.

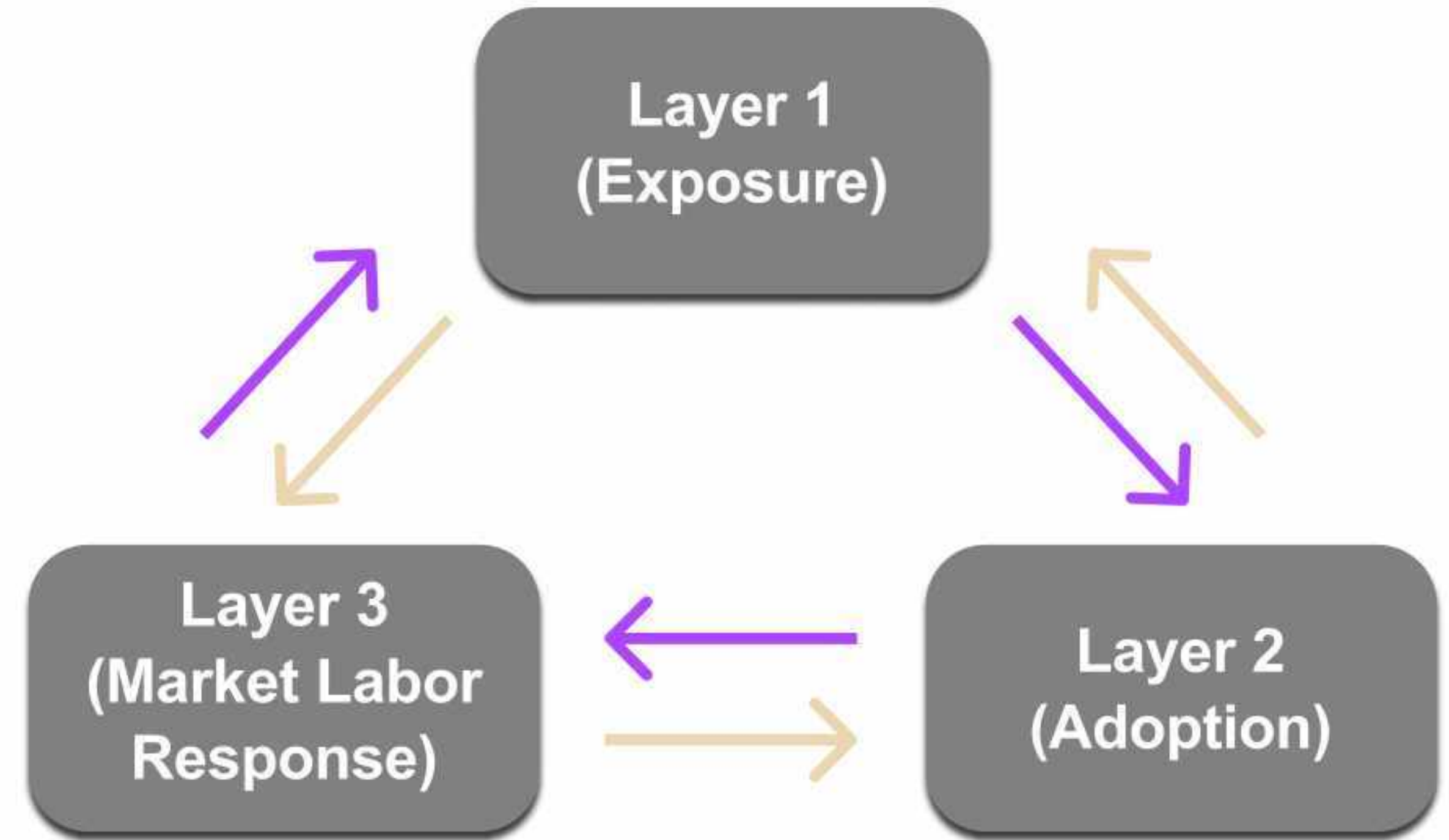
Here, labor market response reflect the collective decisions of firms, workers, and policymakers.

Feedback Loops

The three layers are not sequential. They constantly feed back into each other in multi-directional ways, as we'll see later.

This means that the same technology can yield different outcomes depending on how these factors interact. This also explains why the discussion on AI's impact on the labor market is so contentious.

If the impact of AI depends on feedback loops between technology, firms, and workers, then the first step is to trace where those loops begin.



The Scale of Potential Impact

The U.S. economy is \$31 trillion. Almost 60% of that is labor costs. So even if AI only affects a small percentage of jobs, we're talking about trillions of dollars shifting around the economy.

Globally, analysts estimate that as many as 300 million full-time jobs, roughly 27% of positions in OECD countries, will be transformed through AI automation.

However, history shows that technological revolutions rarely lead to job loss overnight.

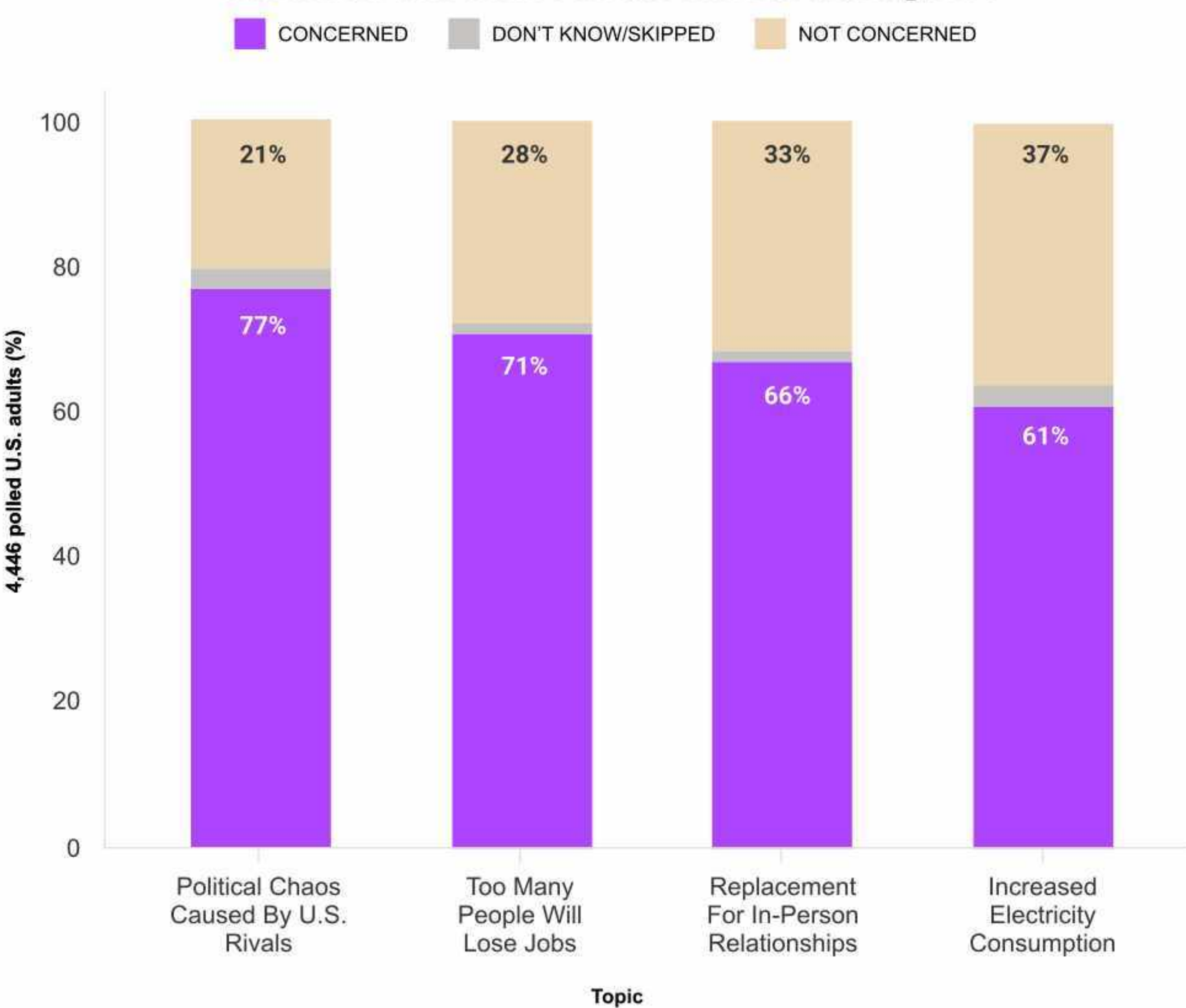
The real challenge lies in the pace and distribution of change, who adapts fastest, who benefits most, and how policy and institutions respond.

Early Concerns

71% of respondents in an August 2025 poll said they were concerned AI will be "putting too many people out of work permanently."

This captures a prevalent concern, but the actual impact is more nuanced and includes upside effects.

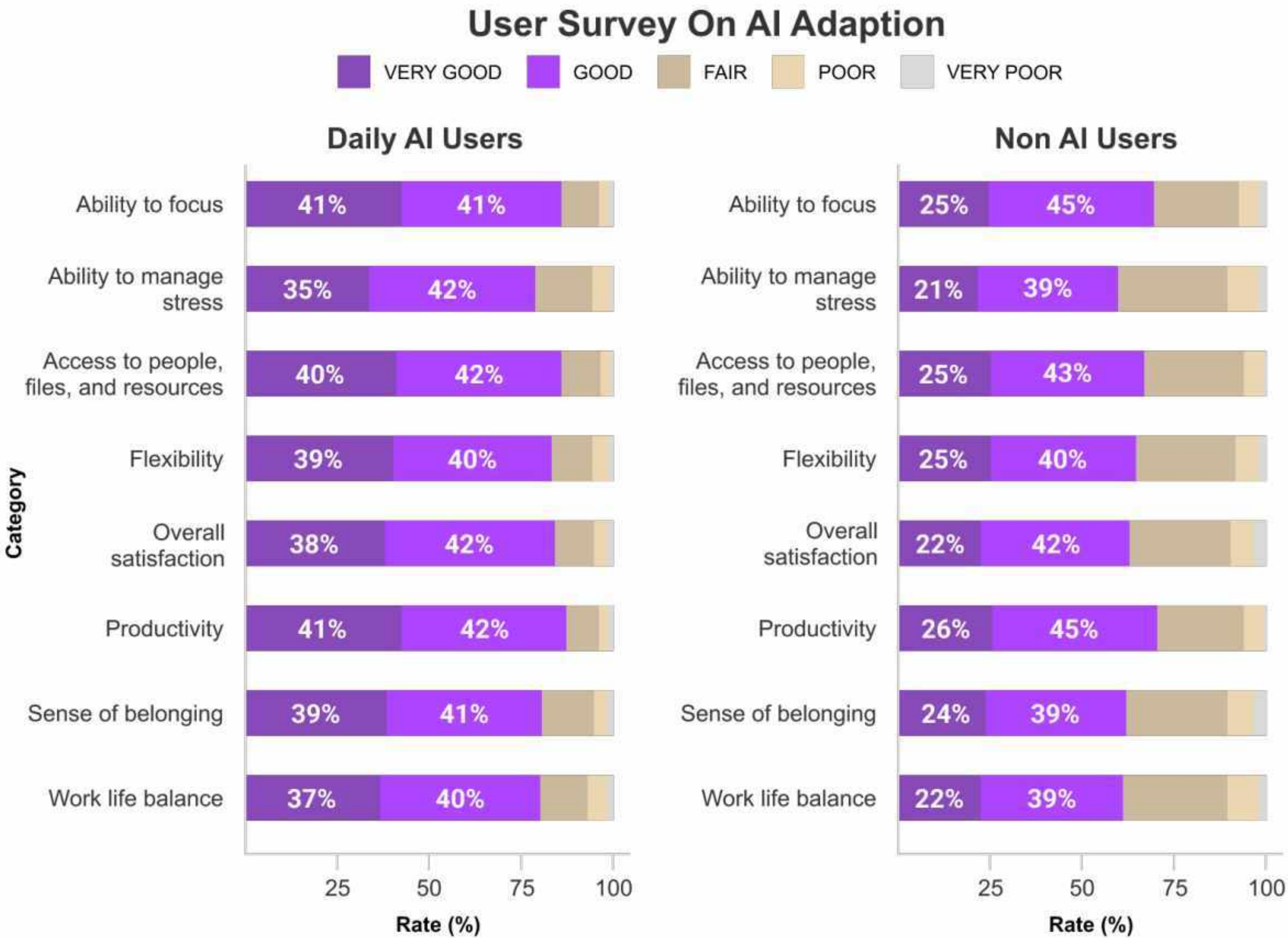
American Concerns About Artificial Intelligence



Improvement with Usage

The latest Workforce Index from Slack shows that daily AI users consistently outperform non users across key work experience metrics, including productivity, focus, stress management, and job satisfaction.

Across nearly every dimension, those who actively use AI report meaningfully better outcomes than peers who do not.



This Deep Dive

Our Approach:

This deck focuses on the past, present and the next 6-24 months rather than artificial general intelligence scenarios or robotics.

A future deep dive is coming for humanoid robotics, so keep an eye out for that.

We'll analyze AI's impact on work by breaking jobs down into tasks and by looking at three interconnected layers that determine real-world outcomes. We'll look at how exposure is measured and why exposure alone doesn't tell the full story.

This Deep Dive

Questions this Deep Dive will answer:

1. How is AI affecting the labor market today?
2. What is the scale and scope of AI adoption?
3. What to watch out for and who will benefit from AI productivity gains?

Layer 1: Exposure

The Exposure Index

Think of an “exposure index” as a measure from 0 to 100 that estimates to what extent the tasks in your job can be plausibly assisted or performed by current or near-term AI.

The higher the score, the more your job might be impacted.

How Do We Measure Exposure?

There are many different approaches to construct exposure indices, each giving a different ranking of which jobs are most open to change.

One approach is “Task to Ability Mapping.” It identifies the tasks that make up a job, estimates which of those tasks current AI can handle, and weights them by their importance in the occupation. The resulting measure is “AI Occupational Exposure” (AIOE).

This method produces tables that rank jobs and industries from most exposed, for example genetic counsellors, to least exposed, for example dancers.

Occupations with the Highest and Lowest AIOE Measures

Rank	Highest scoring	Lowest scoring
1	Genetic counselors	Dancers
2	Financial examiners	Fitness trainers and aerobics instructors
3	Actuaries	Helpers - painters, paperhangers, plasterers, and stucco masons
4	Purchasing agents, except wholesale, retail, and farm products	Reinforcing iron and rebar workers
5	Budget analysts	Pressers, textile, garment, and related materials
6	Judges, magistrate judges, and magistrates	Helpers - Brickmasons, Blockmasons, stonemasons, and tile and marble setters
7	Procurement clerks	Dining room and cafeteria attendants and bartender, helpers, clerks
8	Accountants and auditors	Fence erectors
9	Mathematicians	Helpers - roofers
10	Judicial law clerks	Slaughterers and meat packers
11	Education administrators, postsecondary	Landscaping and Groundskeeping workers
12	Clinical, counseling, and school psychologists	Athletes and sports competitors
13	Financial managers	Fallers
14	Compensation, benefits, and job analysis specialists	Structural iron and steel workers
15	Credit authorizers, checkers, and clerks	Cement masons and concrete finishers
16	History teachers, postsecondary	Terrazzo workers and finishers
17	Geographers	Rock splitters, quarry
18	Epidemiologists	Plasterers and stucco masons
19	Management analysts	Brickmasons and Blockmasons
20	Arbitrators, mediators, and conciliators	Roofers

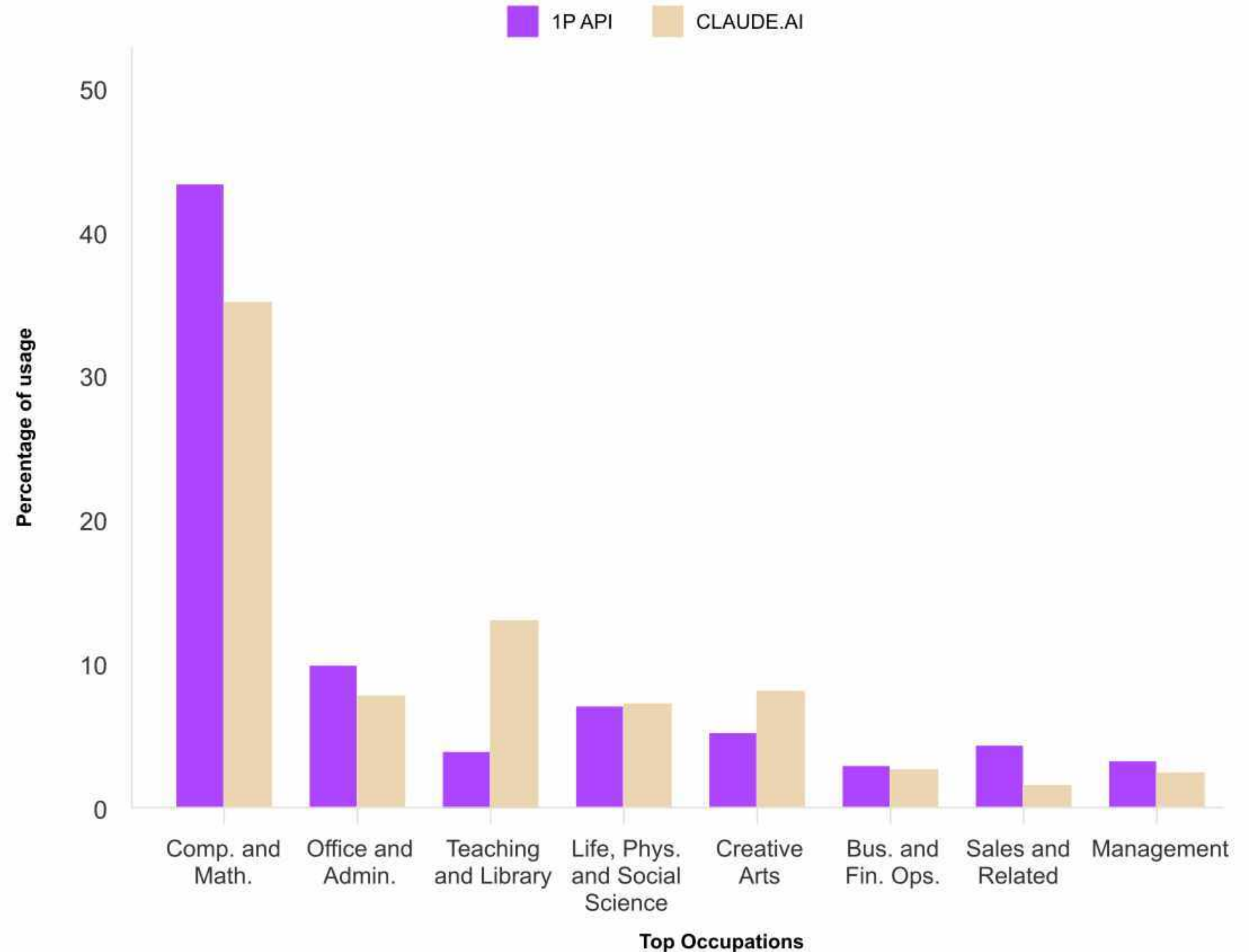
How Do We Measure Exposure?

Another approach is to analyze what workers actually use AI for today by studying millions of AI conversations and mapping them to specific job tasks, resulting in indices like Anthropic's Economic Index.

Example: Anthropic analyzes 1+ million Claude conversations to see which occupations are actively using AI for which tasks right now. Currently, 44% of AI usage is for computer and mathematical analysis.

OpenAI also launched their own performance measure with GDPval on September 2025.

Usage Shares Across Top Occupational Categories



Which Jobs are Most Exposed?

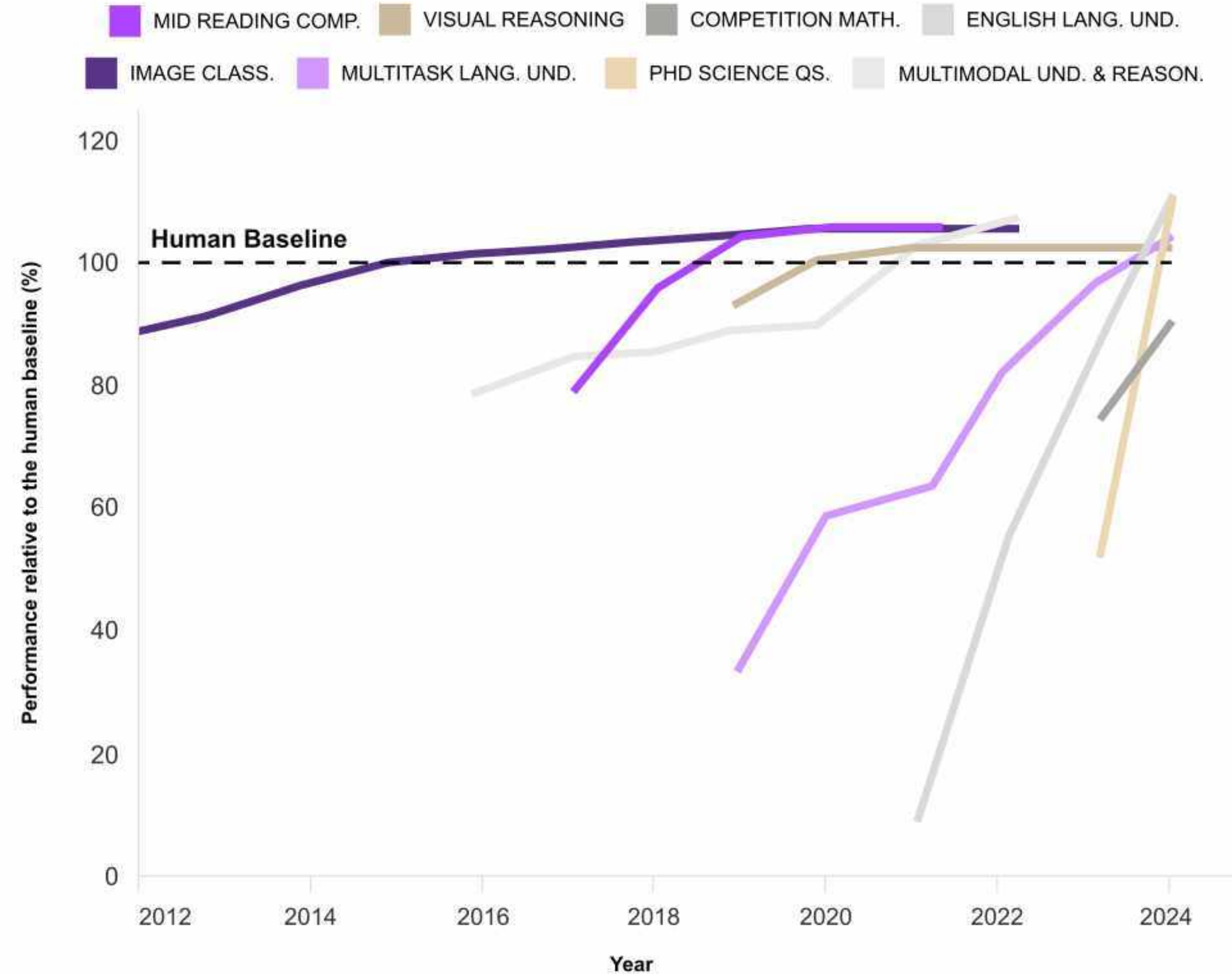
Although the technical details of the various exposure indices differ, recent studies consistently point to several patterns:

1) Text-heavy, information-processing, and communication tasks are the most exposed.

Legal research/drafting, writing/editing, customer support, market research/analytics, education content prep, finance/insurance analysis, and programming documentation/QA consistently score high on the exposure index.

Because LLMs are trained on vast text corpora and then refined through reinforcement learning from human feedback, they excel at tasks with abundant written examples and clear preference signals.

AI Technical Performance vs Human Performance
(2012-2024)



Routine Roles

2) Clerical and back-office roles are one of the most automatable within the exposed set.

Routine document processing, booking, transcription, and standardized communications are repeatedly flagged as highly affected and plausibly automatable.



“When looking for jobs that will change a lot due to AI on shorter time scales, I'd look in other places - jobs that look like repetition of one rote task, each task being relatively independent, closed (not requiring too much context), short (in time), forgiving (the cost of mistake is low), and of course automatable given current (and digital) capability.”

Andrej Karpathy
Former Director of AI @ Tesla and founding team @ OpenAI
2025

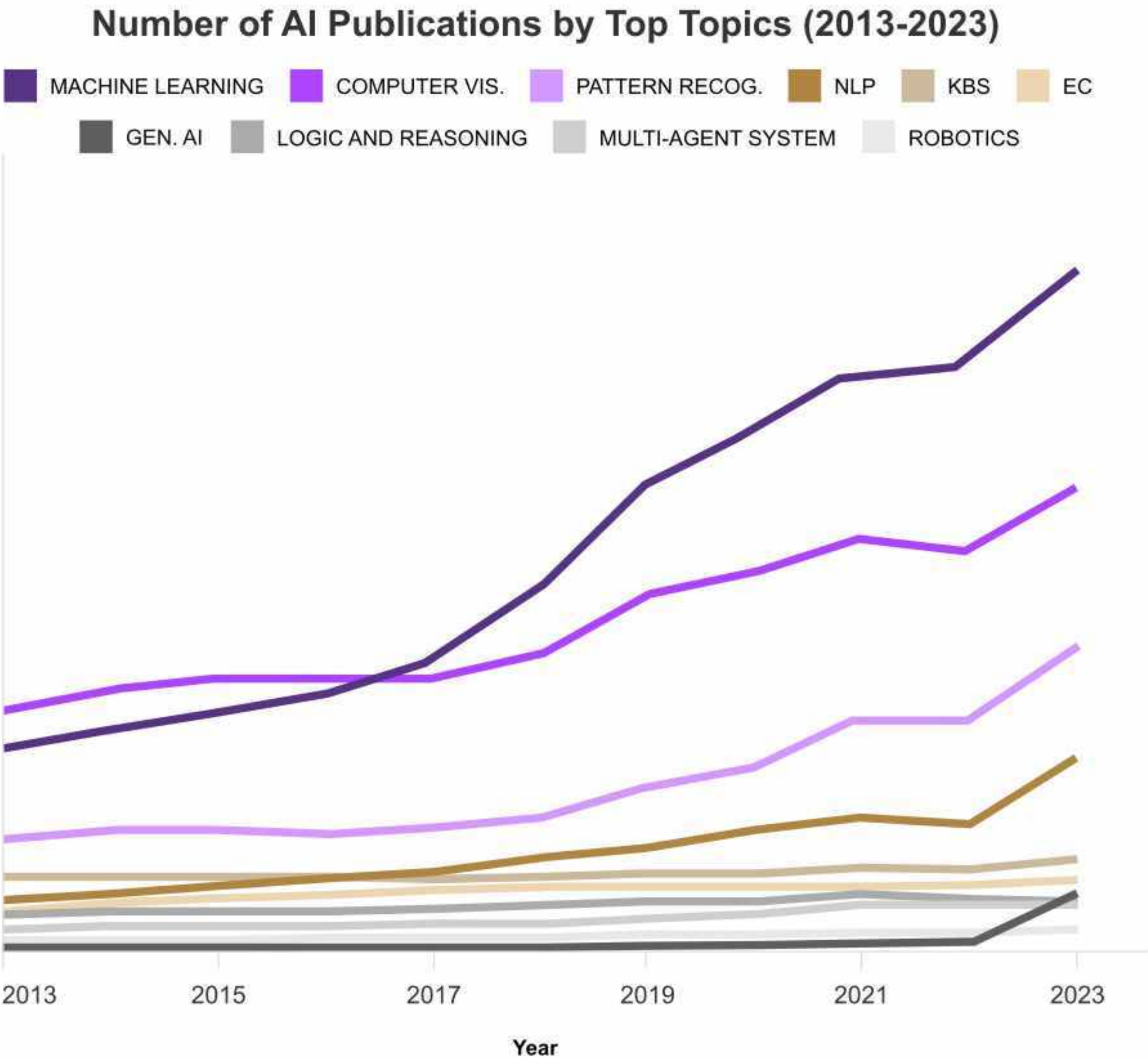
Least Exposed

3) Embodied, on-site, and interpersonal-dexterity tasks are currently least exposed.

Electricians, roofers, line workers, nursing assistants, and healthcare aides. Roles requiring physical manipulation, presence, or rich social interaction score low in current exposure indices.

This highlights the fact that most AI is currently disembodied and that the boom in AI is happening much faster than the evolution of practical robots (though that might change soon).

This is also reflected by the difference in growth and levels of publications on ML vs robotics in recent years.



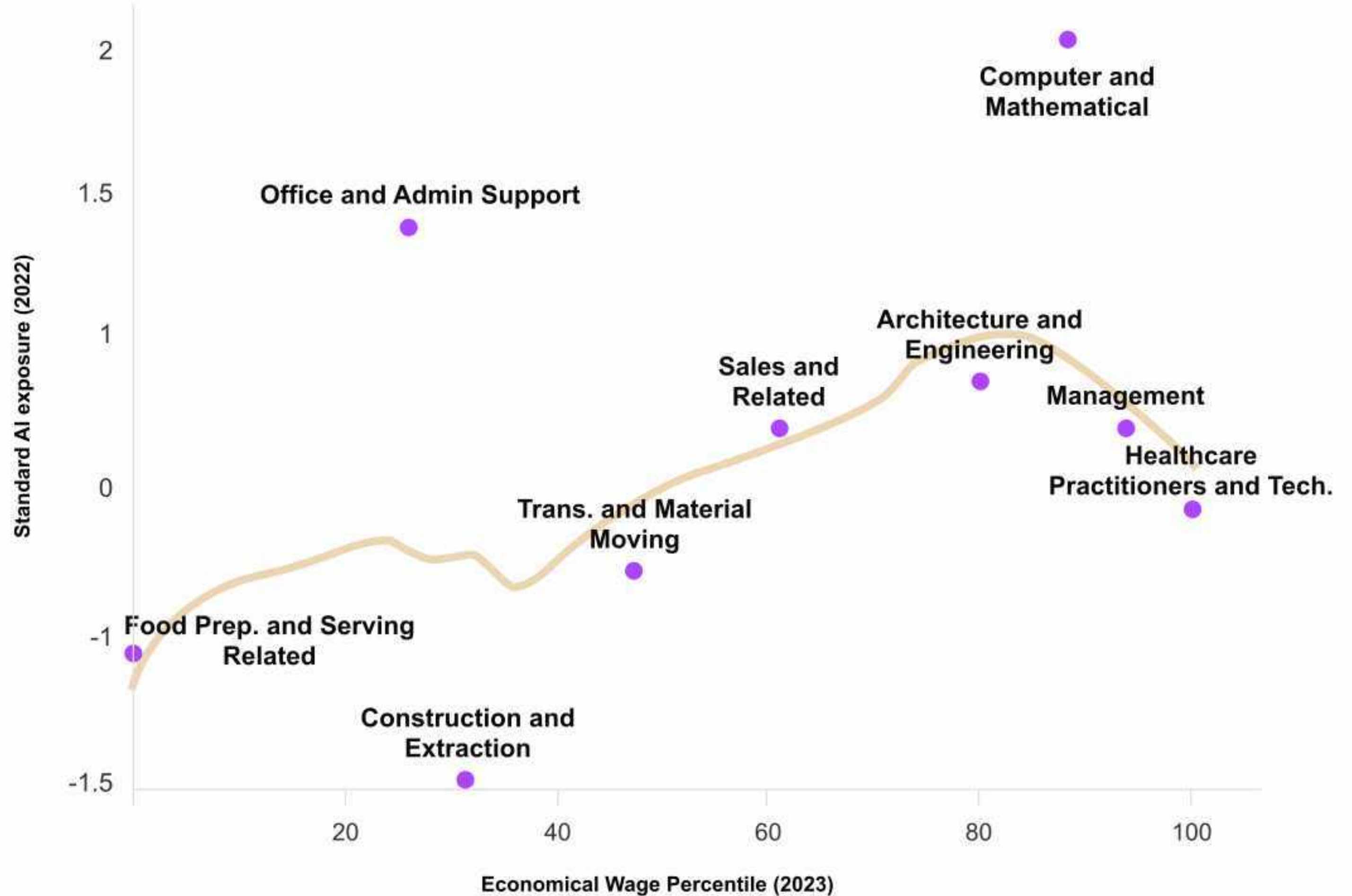
Exposure Correlation

4) Exposure is positively related to wages and education

Higher paid and higher educated occupations are more exposed to LLM era AI because their work is language and symbol dense.

In contrast to earlier tech waves that automated routine or physical tasks, Gen AI reaches cognitive non-routine work once assumed to be safe from automation.

Generative AI Exposure by Occupational Group

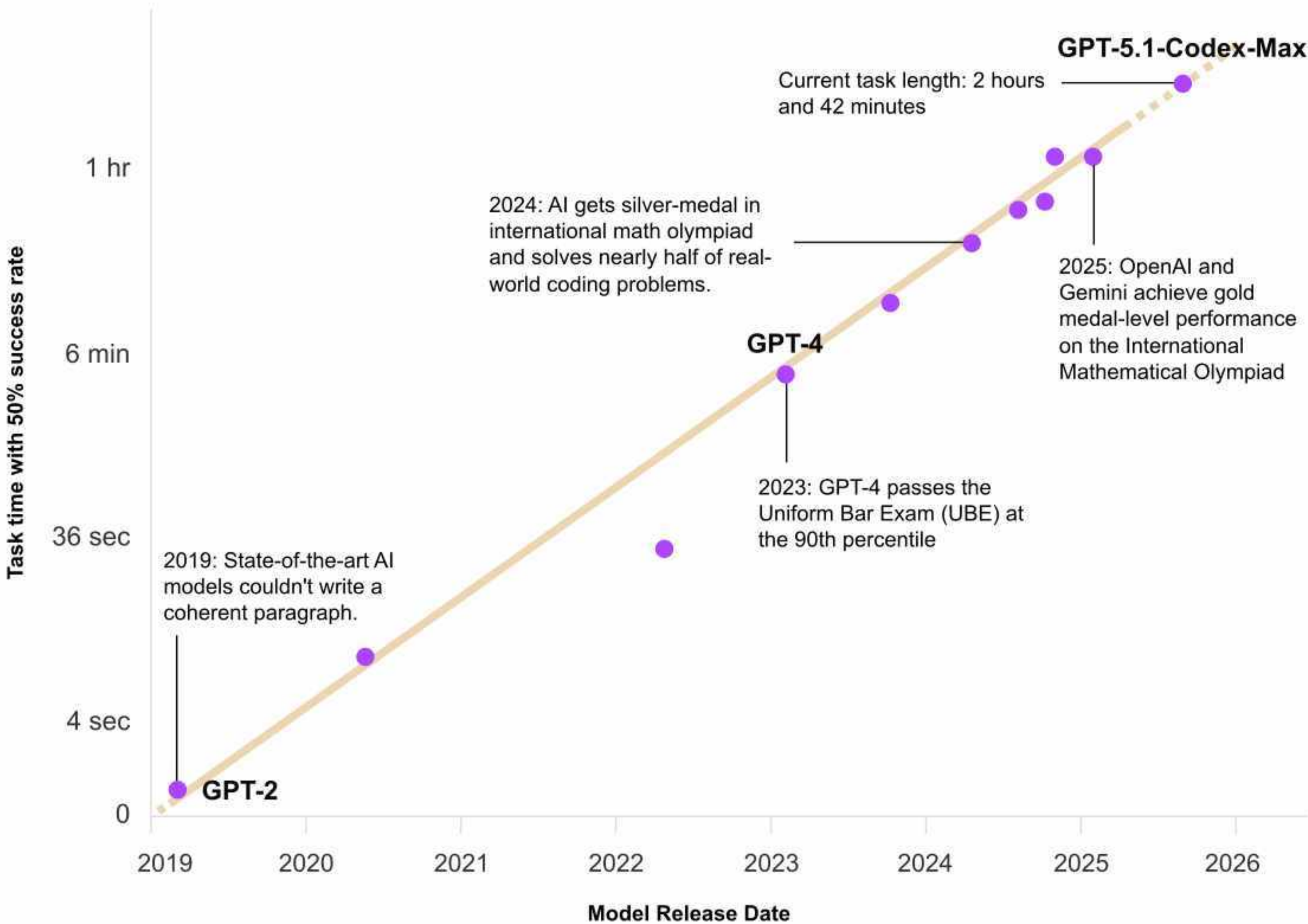


A Moving Measure

Although existing exposure indices paint a similar picture in terms of who's exposed, we should treat current tables and graphs as constantly evolving as the tech continues to advance.

Many analysts are trying to model the advancement of AI through measures of task time completion with a success rate over 50%.

AI Agents' Capacity for Autonomous Task Completion (Log Scale)



The New Moore's Law

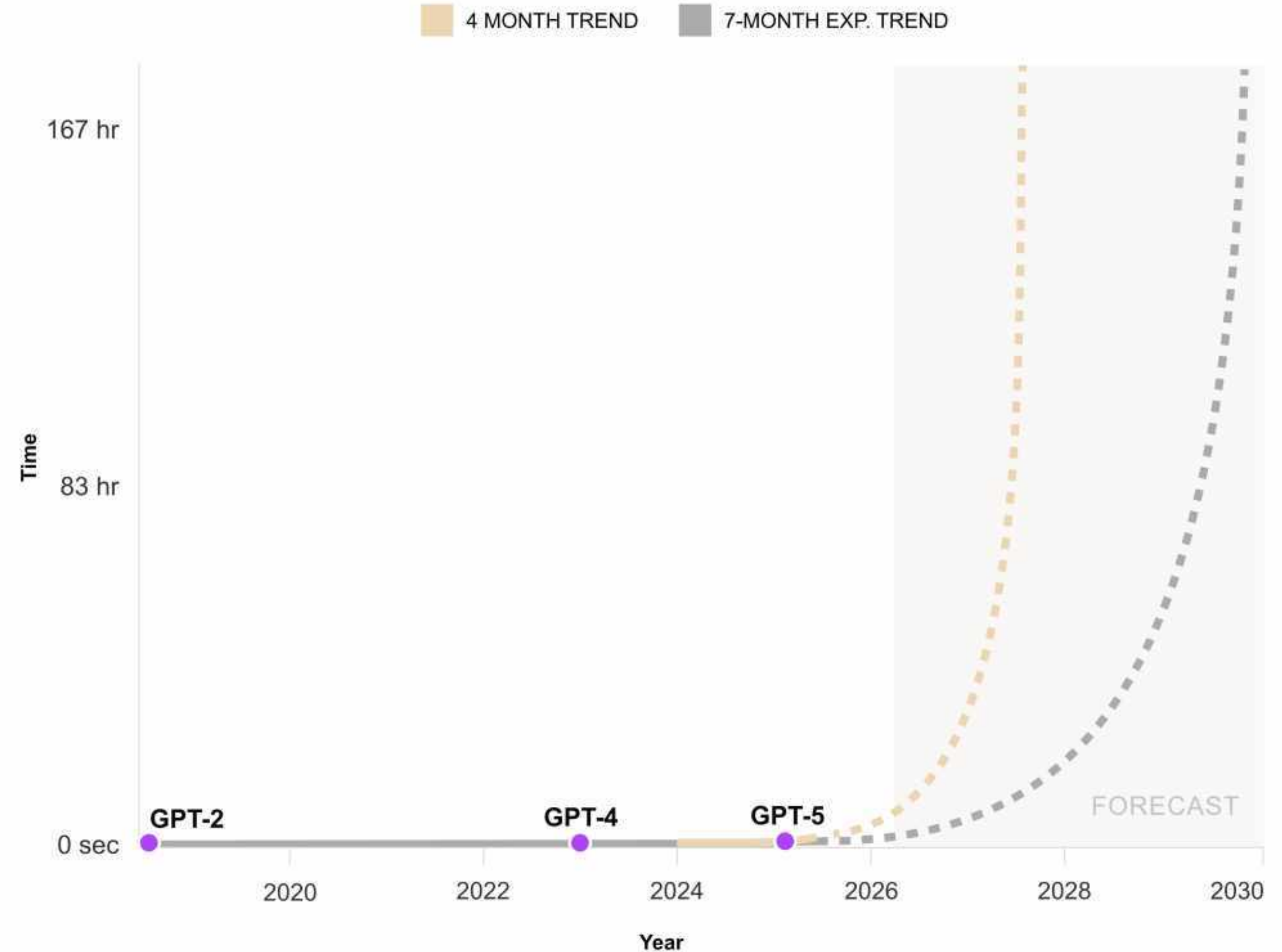
This speed of progress in the past 5 years has led researchers to formulate the “New Moore’s Law”, which says the length of tasks AI can perform doubles every ~7 months.

Assuming an exponential trend in capability growth, this trend predicts that...

- By 2026, AI can complete a 2-hour task
- By 2027, AI can complete 1 work day (8 hours)
- By 2028, AI can complete 1 work week (40 hours)
- By 2029, AI can complete 1 work month (167 hours)

Recently, there has been some evidence that the trend is ramping up to doubling every ~4 months.

Projected AI Advancement in Coding Task Autonomy



Experts Have Been Wrong In The Past

While these exponential projections are tidy in theory, experts have been wrong in both directions, underestimating and overestimating the speed of progress.

An example of underestimation: In 2021 Michael Wooldridge, a leading Oxford AI researcher, published a book listing tasks he judged nowhere near solved: understanding a story, human level automated translation, interpreting a photograph, and writing interesting stories.

These are things that your favorite LLM can easily do today.



—but others are not. To understand why AI is hard—why progress has proved so elusive over the years—it helps to look at problems that are easy to encode in this way and problems that are not, and to see why this is the case. **Figure 1** displays some tasks that we might want computers to do and shows how difficult it has proved to get computers to achieve them.

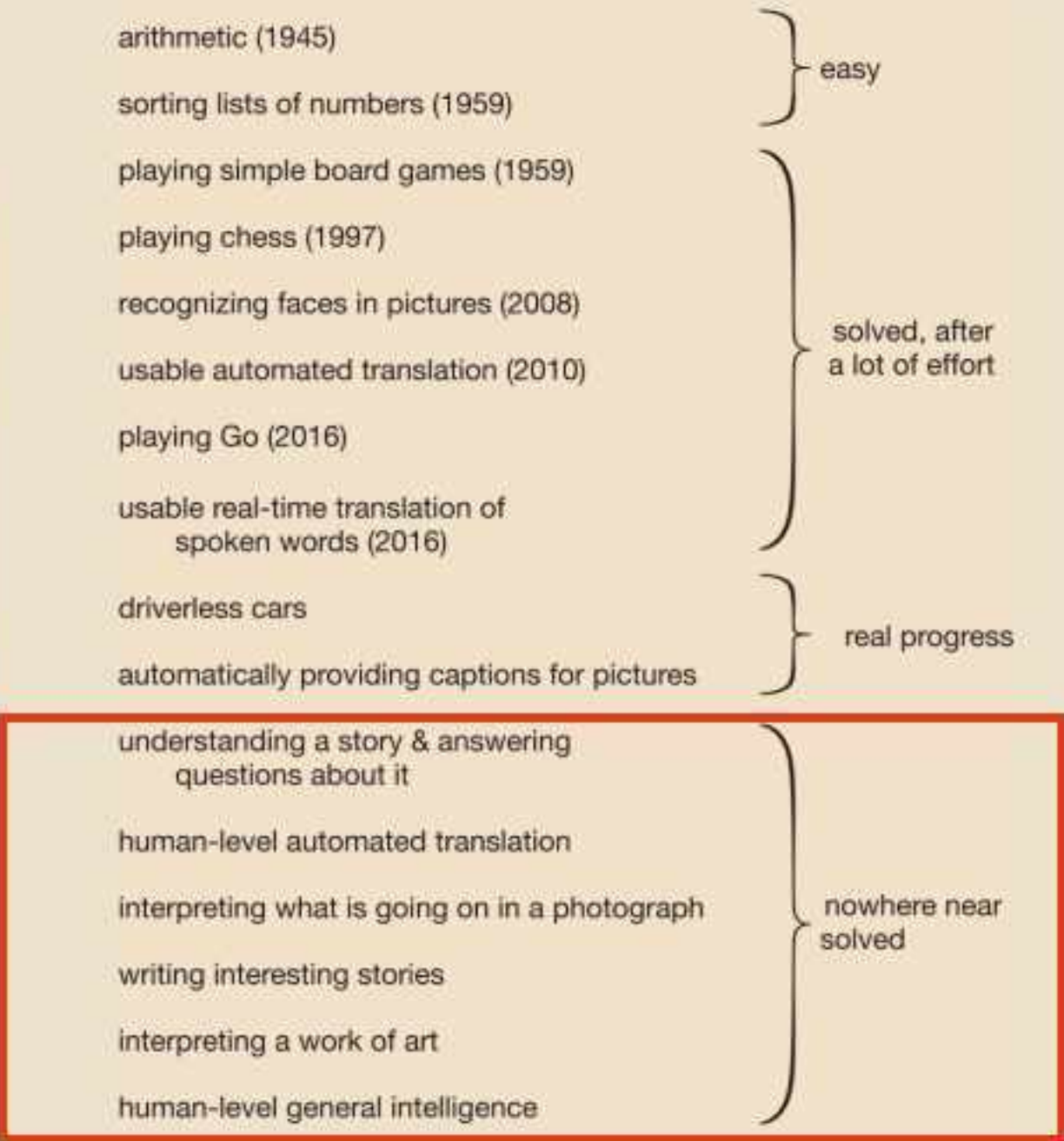


Figure 1: Some tasks that we want computers to be able to do ranked in order of difficulty. Years in parentheses indicate approximately when the problem was solved. At present, we have no idea about how to get computers to do the tasks at the bottom of the list.



Experts Have Been Wrong In The Past

“I think if you work as a radiologist you're like the coyote that's already over the edge of the cliff, but hasn't yet looked down, so it doesn't realize there's no ground underneath him. People should stop training Radiologists now. It's just completely obvious that within 5 years deep learning is going to do better than Radiologists...”

He was right about the models outperforming radiologists on benchmarks, but wrong about the demand.

The field of Radiology continues to grow and stay strong, a phenomenon we'll explore later.

Geoff Hinton “Godfather of AI”
Machine Learning and Market for Intelligence Conference
2016

Exposure Recap

Exposure is an estimated measure of how much of a job's daily work could be performed by AI.

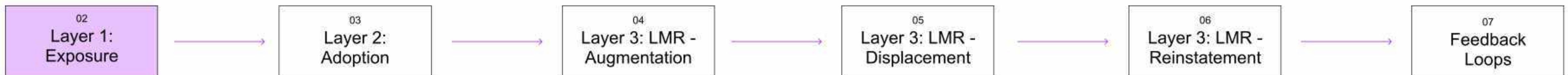
A job that is highly exposed can also be the one most likely to increase in growth as the role adapts to the rise in productivity.

Across studies, a pattern emerges: high-paying, highly educated jobs that rely heavily on language and symbols, from law and finance to marketing and software, score higher on AI exposure.

In contrast, low-wage or physically anchored jobs remain least exposed.

We've seen that AI's capabilities have raced ahead of expectations, but capabilities alone don't reshape economies.

We need to consider how AI is adopted through firms and workflows, which we'll examine in the next section.



Layer 2: Adoption

AI Adoption and Diffusion

Exposure tells you what AI could do. Adoption and diffusion tell you what AI is actually doing in the world.

Adoption refers to an individual or firm's process of deciding whether to use a new innovation.

Diffusion is the broader societal process of how that innovation spreads over time through a population.

For the purposes of these slides, we'll simplify both under the single category of adoption.

Exposure doesn't always equal adoption. A job can be highly "exposed" in theory but untouched in practice.

For example, a firm may ban their workers from using public LLMs for data confidentiality reasons or hasn't reorganized their workflows to use the new tool efficiently.

“Even if a new AI-based product is instantly released online for anyone to use for free, it takes time for people to change their workflows and habits to take advantage of the benefits of the new product and to learn to avoid the risks.”



Arvind Narayanan & Sayash Kapoor
Computer Scientists at Princeton University
AI as normal technology (2025)

AI as a General-Purpose Technology

AI sits in the family of general-purpose technologies (GPT*). These are innovations so broadly applicable that they transform not just one sector but the entire economy.

The steam engine, electricity, and the personal computer are classic examples of GPTs.

GPTs are different from narrow innovations. A new fertilizer might revolutionize agriculture but won't touch finance or healthcare.

*The abbreviation "GPT" in this section is unrelated to, and not to be confused with, the "Generative Pre-Trained Transformer" models such as ChatGPT.

Historical Adoption Rates for GPTs

The key thing to know about GPTs in the context of adoption is that they rarely produce instant productivity gains.

For example, electricity took decades to raise factory output because many factories couldn't get wired up immediately.

Building the power grid, transmission lines, motors, and wiring was expensive.

So even when electricity was available, factories often replaced steam engines with electric motors as opposed to fully electrifying operations.

The Productivity J-Curve

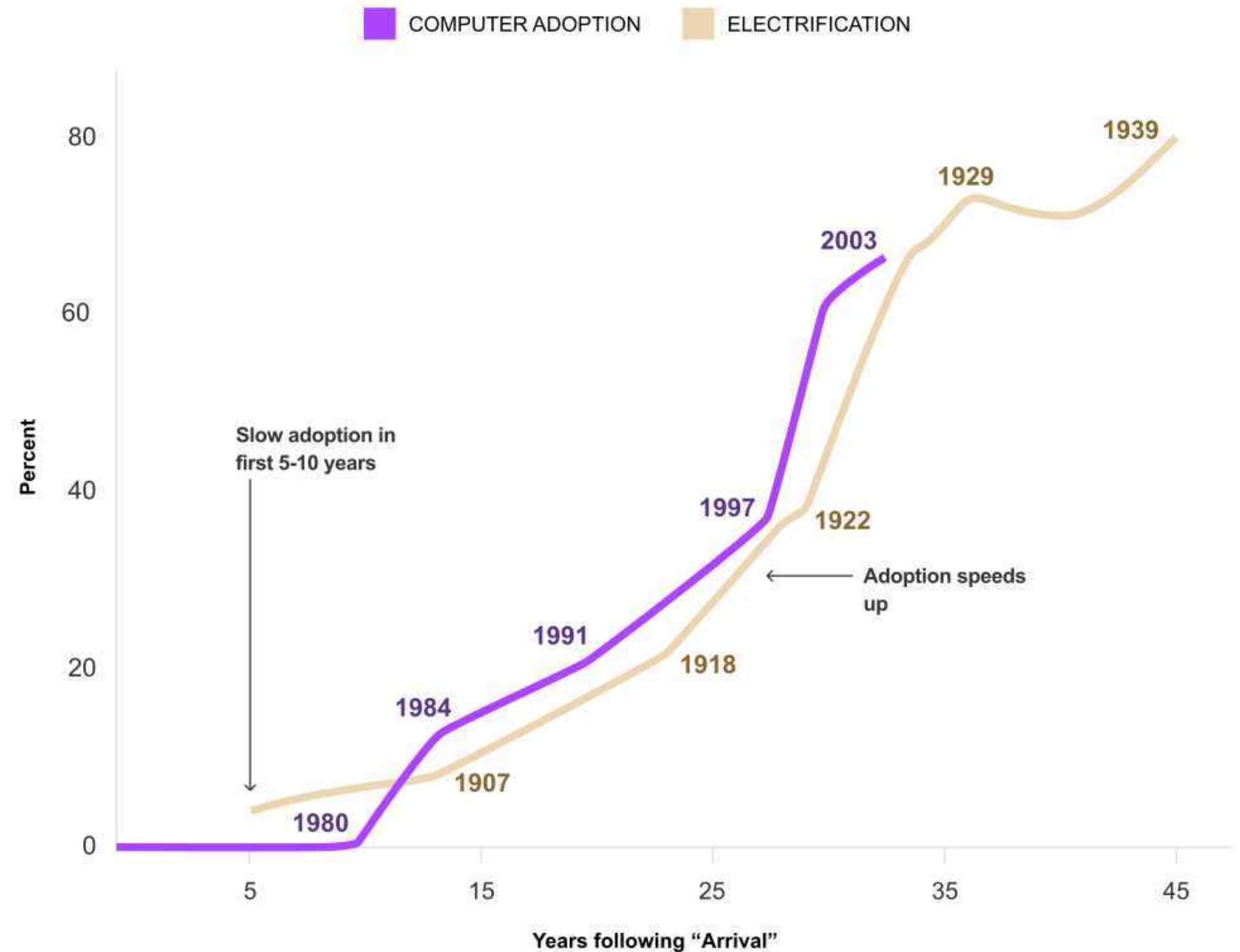
Another example is the personal computer (PCs).

Although PCs became available to firms in the early 1980s and could theoretically make some workers more productive, availability wasn't enough to lead to immediate productivity gains.

Firms needed training and complementary tools to realize the theorized productivity gains.

The broad productivity gains from personal computers became visible in U.S. statistics only around the mid-1990s, as complementary software, networking, and new work practices matured.

The Pace of Technological Adoption Since GPT Arrival





The Productivity Paradox

Economists noted the productivity paradox through the late 1980s and early 1990s.

“You can see the computer age everywhere but in the productivity statistics.”

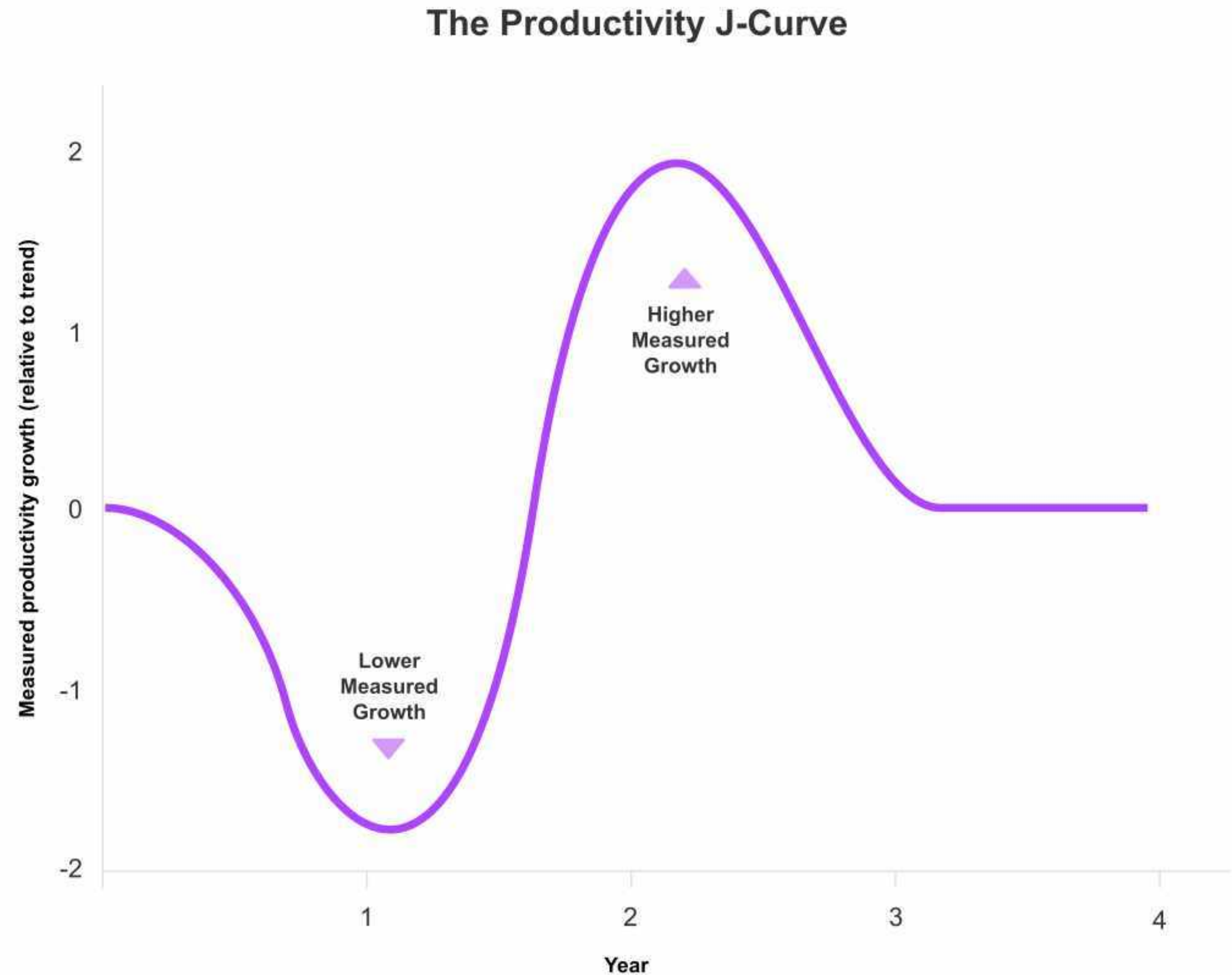
Robert Solow
Nobel Laureate in Economics
1989

The Productivity J-Curve

In early years after implementation, productivity often grows slowly or even dips due to learning costs, setup, and process re-engineering.

Historical examples of GPT adoption results in “The Productivity J-Curve”.

When we plot the relationship between productivity growth rate on the Y-axis, against time since the technology is introduced on the X-axis, we see a J-shape due to productivity appearing to dip during the investment phase, overshooting as intangible gains are realized, and then normalizing at a higher level of output.



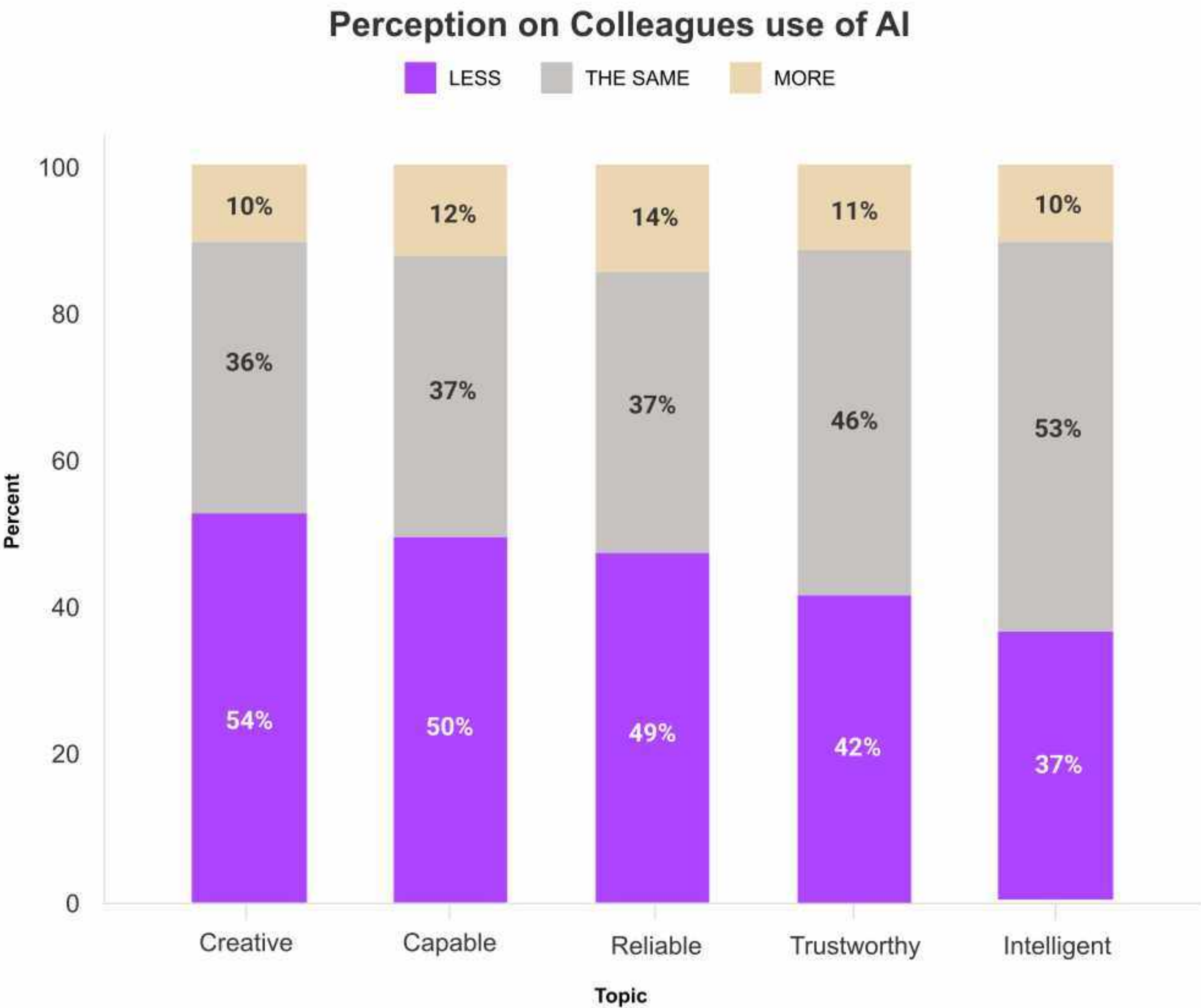
The Productivity J-Curve

A 2025 Harvard Business Review study found 40% of knowledge workers reported cleaning up such “workslop,” spending nearly 2 hours per week fixing low-quality outputs.

This suggests that early adopters might initially see higher activity, but not necessarily higher output value.

When firms invest in infrastructure, train employees, and redesign workflows, costs tend to fall as benefits accumulate.

Eventually productivity accelerates, rising sharply as adoption becomes widespread.



Pause and Performance

Another factor is the potential constant switching cost created by rapid AI launches, which can force repeated workflow resets.

But if models pause or plateau at a level like ChatGPT 5.2, Gemini 3, or Opus 4.5, productivity gains can still emerge from:

- Firms and workers getting better at using the AI tools they already have.
- Integrating AI into workflows.
- Automating tasks that are ripe for automation.
- Shifting worker effort from low-value tasks to higher-value ones.
- Scaling up adoption from early adopters to the broader economy.

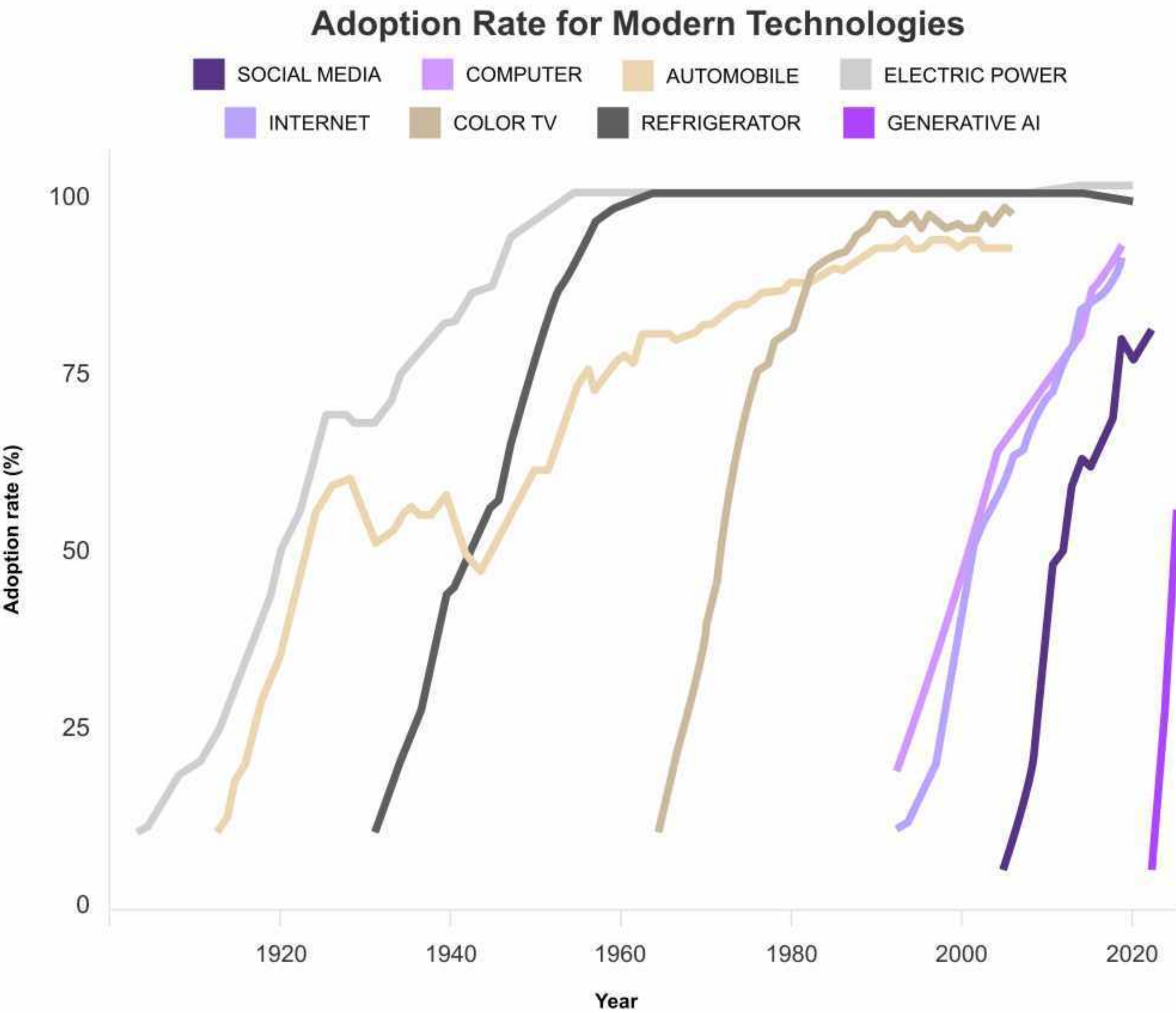
Adoption Rates For Modern Technologies

Adoption rates for modern technologies, such as social media, are faster than for older technologies like the refrigerator.

It took 50 years for US households to go from 10% adoption of electricity to 99%.

In contrast, it took just 15 years for social media to go from 5% adoption to 79%.

Within three years of ChatGPT's 2022 launch, the adoption rate of generative AI has surpassed 50%.

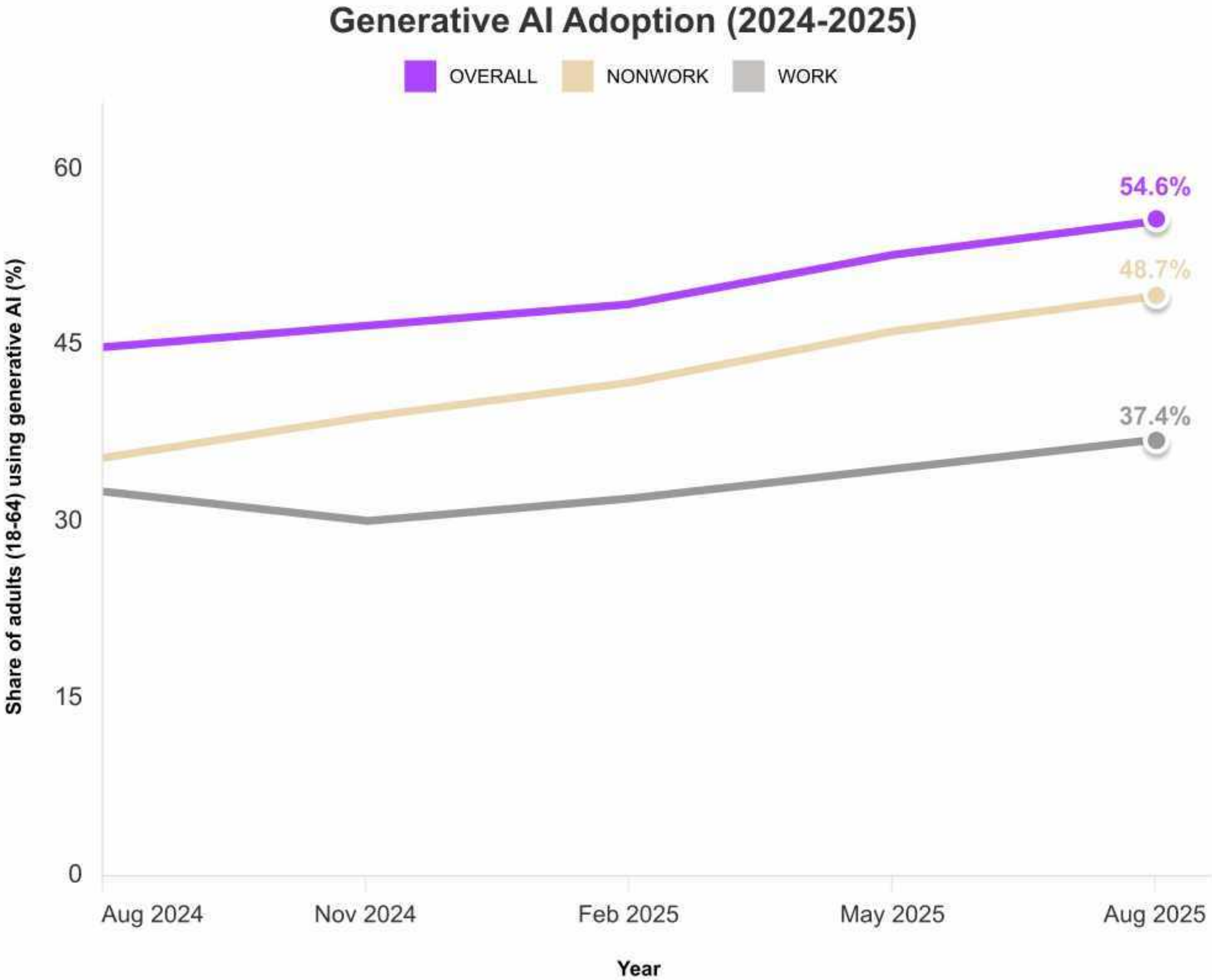


What We Know About AI Adoption So Far

Worker-Level (U.S. population age 18-64 from The Real-Time Population Survey)

The survey found that in August 2024, 44.6% of adults used generative AI overall.

Within a year, the adoption of this new technology increased by 10 percentage points to 54.6%.

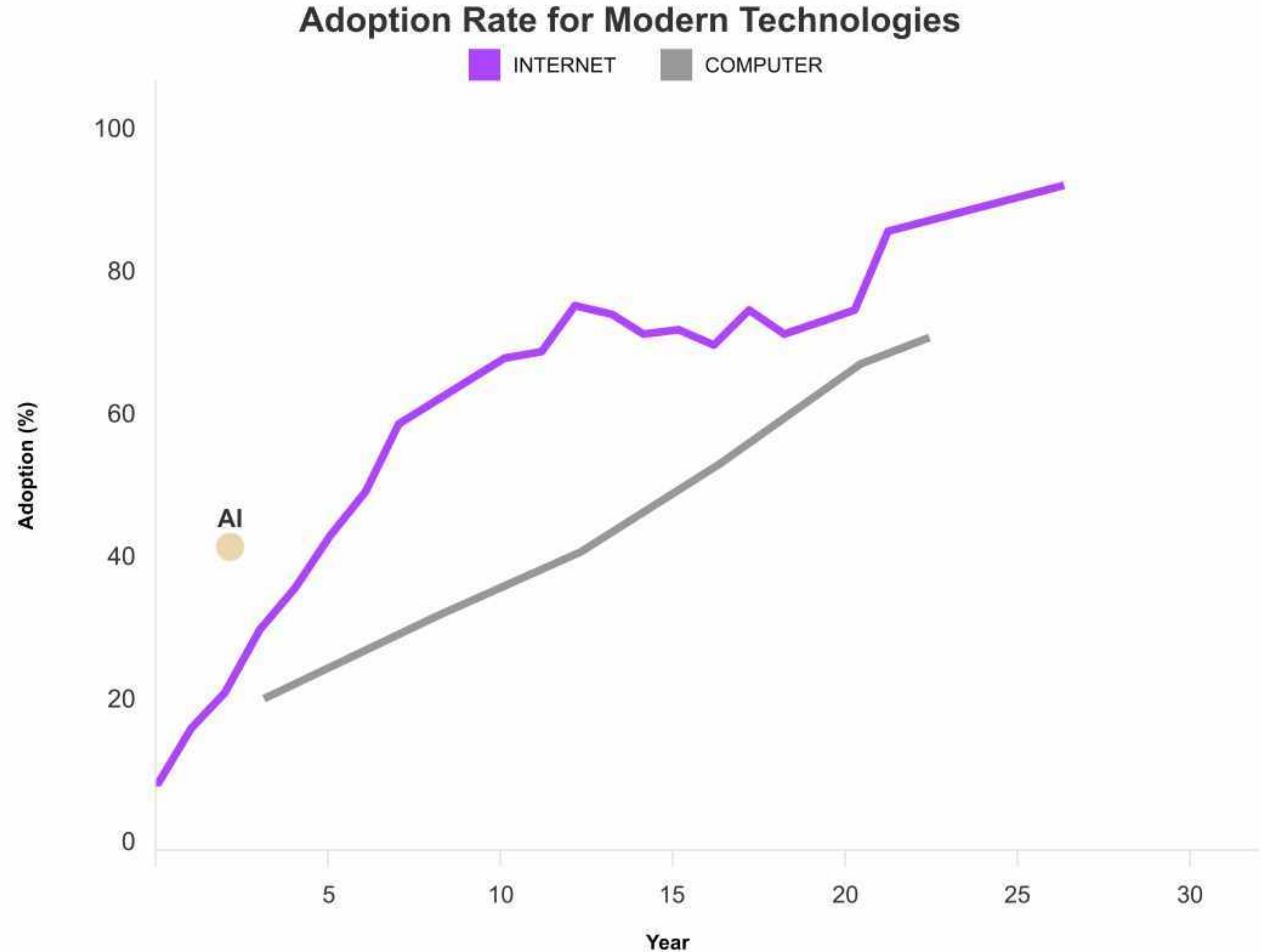


Adoption Acceleration

This trajectory is faster than both PC and internet adoption.

Assuming 1981, as the introduction year for PCs (IBM PC), and 1995 as the introduction year for the internet (when the NSF decommissioned NSFNet and allowed the internet to carry commercial traffic), and 2022 as the introduction year for GenAI (release of ChatGPT), data shows:

- **Computer:** ~13 years to reach 40% consumer adoption
- **Internet:** ~6 years to reach 40% consumer adoption
- **Generative AI:** ~3 years to reach 40% consumer adoption

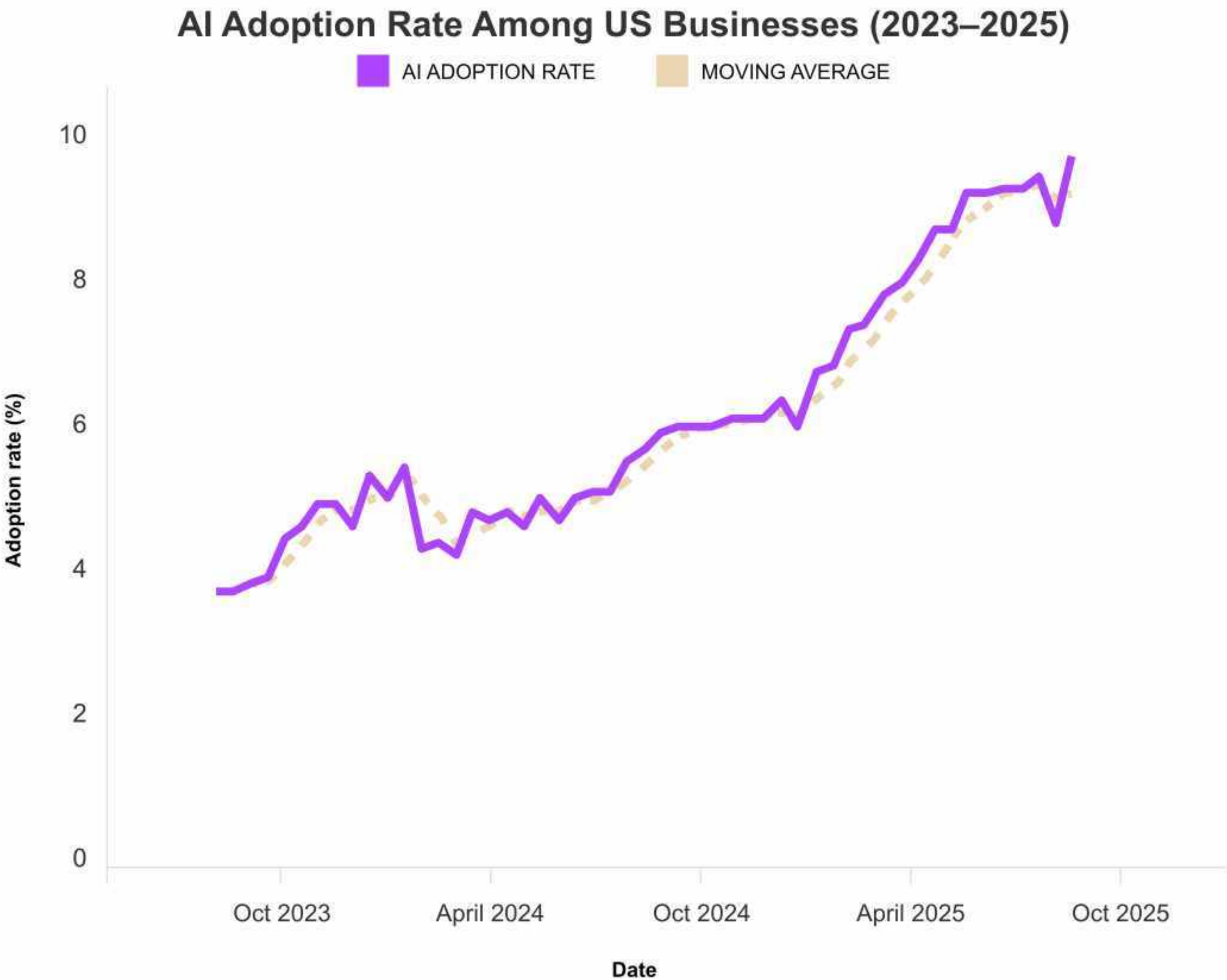


Business Adoption

Firm Adoption Rates (representative of all employer businesses in the U.S. economy, excluding farms)

Share of firms responding “yes” to the question “In the last two weeks, did this business use Artificial Intelligence (AI) in producing goods or services?” **has risen from 3.7% in fall 2023 to 9.7% in early August 2025.**

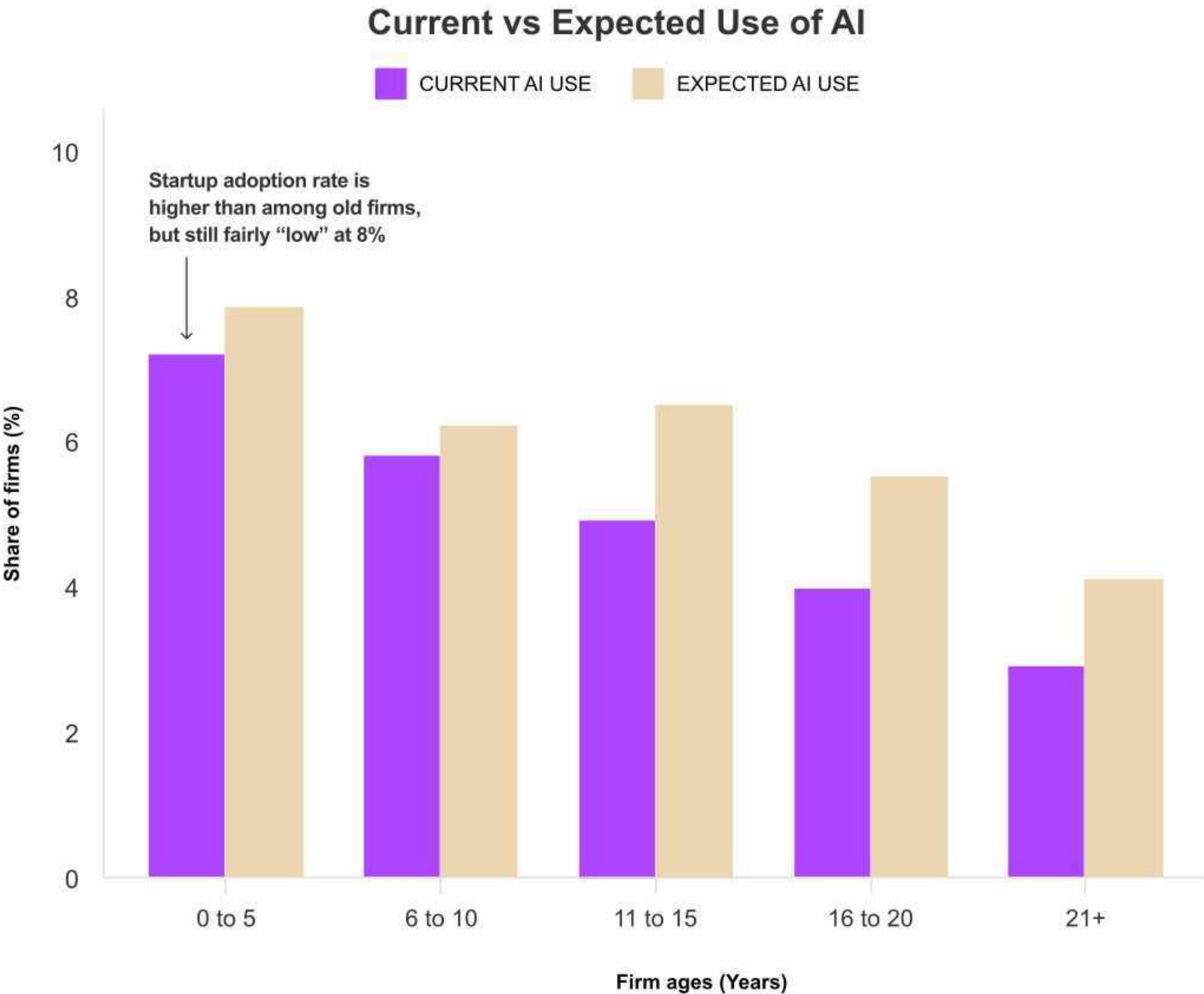
Despite this rapid rate of growth, the vast majority of firms in the US do not report using AI in their production processes.



Disparity in Business Types

Adoption is uneven by firm age.

Start-ups and younger firms are more likely to experiment with generative AI, partly because they have fewer legacy systems and lower reputational risk.



Disparity in Sector Types

Adoption is uneven by sector.

Although software engineering, legal services, and creative work are all “exposed” to AI, adoption patterns look different across these industries.

Modular, testable tasks (coding, data cleanup) see faster adoption because outputs can be checked cheaply.

High-stakes, low-error-tolerance tasks (e.g. legal briefs) face heavier verification costs and slower adoption.

Let’s take a deeper look...

Sector	Adoption Pattern
Software Engineering	High adoption (e.g. Cursor and Claude Code)
Legal Services	Mixed: In-house counsel are more trusting ; law firms are cautious.
(“Traditional”) Creative Arts	Wary or cautious (lawsuits from Disney, authors, songwriters)

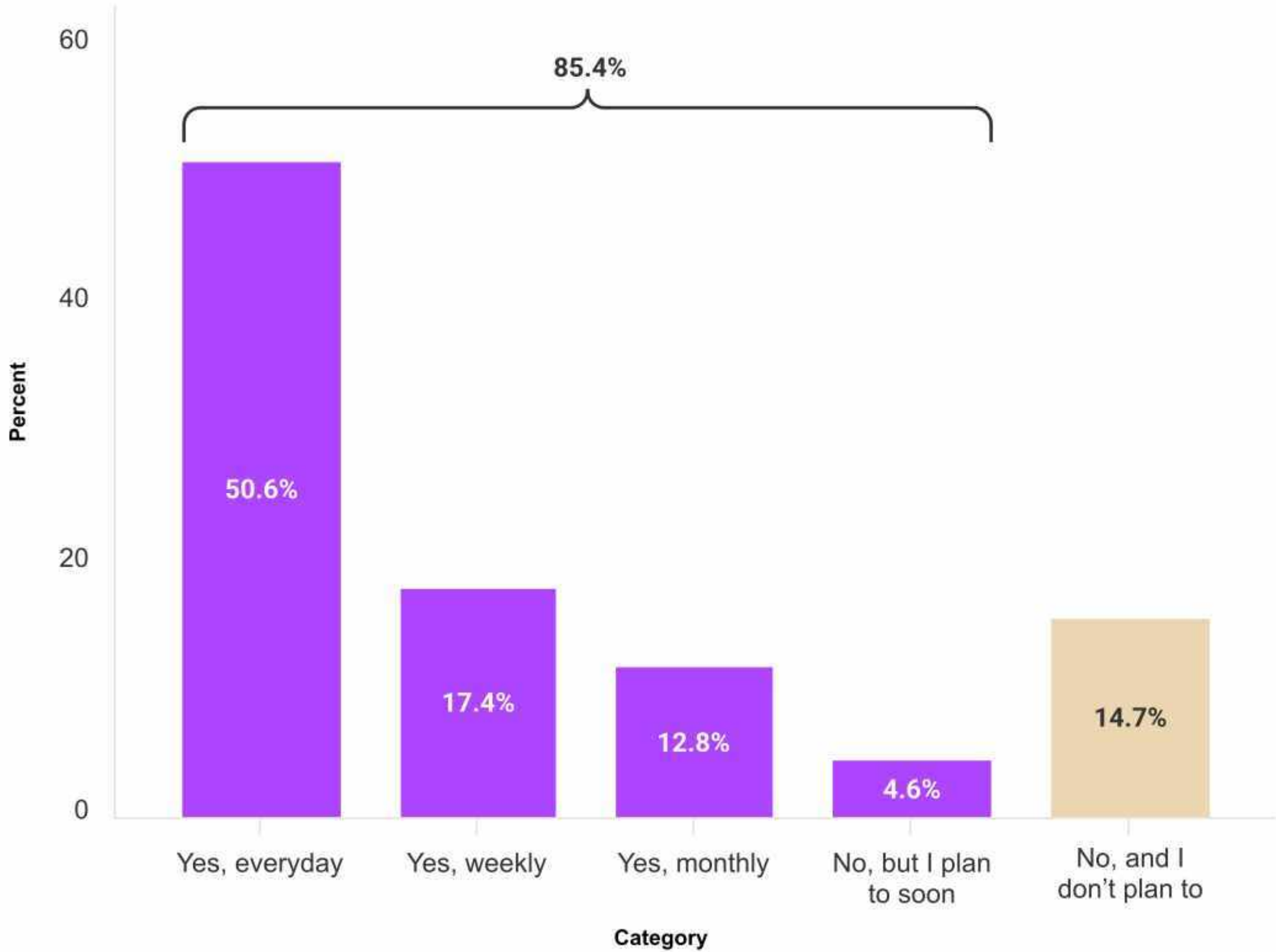
Tech and Software

Software engineering has high AI adoption.

Claude Code, Cursor, GitHub Copilot, and similar tools are now standard for autocomplete, boilerplate, and testing.

The Stack Overflow 2025 Developer Survey shows that 85.4% of developers are using or plan to use AI tools in their development process. Also, about 51% of professional developers said they use AI tools daily.

How Often Professional Developers Use AI in Their Work



Legal Services

Generative AI can already draft memos, summarize contracts and search case law, but law firms are adopting it more cautiously than in-house legal teams.

The cost of mistakes in law is high.

After high-profile incidents of “hallucinated” cases, U.S. judges have issued standing orders requiring lawyers to disclose AI use and certify citations.

For example, in Mata vs Avianca, lawyers for the plaintiff used ChatGPT to help prepare a legal filing, only for the AI to “hallucinate” and generate non-existent case citations.

In addition, under the American Bar Association’s Model Rule 1.6, lawyers must protect client confidences and take reasonable precautions against disclosure, so many law firms pilot AI only behind secure firewalls or under contracts that guarantee data protection.

Legal Services

The result is private deployments among large firms, which take time to build and are costly, making them accessible mainly to organizations with substantial resources.

For example, Allen and Overy rolled out one of the first enterprise-scale GPT-based legal AI systems, under firm control, for thousands of lawyers.

Cleary Gottlieb acquired Springbok AI to build custom internal tools. This is rare in BigLaw but signals that firms prefer owning the stack rather than relying on public bots.



World ▾ Business ▾ Markets ▾ Sustainability ▾ Legal ▾ Commentary ▾ More ▾

Law firm Cleary buys legal tech company in AI bid

By Sara Merken

March 18, 2025 1:02 AM GMT · Updated March 18, 2025



Creative Industry

Generative AI is technically powerful in creative work, but its uptake is slower than in software.

Creative works sit under some of the strongest copyright regimes in the economy.

Just in the first half of September 2025 alone, a few high-profile creative lawsuits are underway:

- Studios like Disney, Universal and Warner Bros. have sued Midjourney for enabling AI images of their characters. (In the meantime Disney struck a \$1 billion deal with OpenAI to bring their licensed characters to the Sora short form video tool).
- Anthropic agreed to a proposed \$1.5 billion settlement with authors over the use of books in training data.
- In Sweden, a new AI music license was created to make lawful training possible - an example of how regulation shapes adoption.

These lawsuits create a cloud of uncertainty over who owns AI-generated content.

Unlike code, which is modular and easily reviewed, creative outputs are subjective and style-driven.

Small similarities in character or style can spark infringement claims, and reputational damage can be immediate.

Creative Industry

In 2023, American Actors' Union went on the longest strike, demanding fair pay for repeats and rules so companies can't use AI to copy actors' voices or faces without asking and paying them.

The settlement redirected adoption toward licensed, consent-based, and compensated uses.

It shows how labor market pressure can influence AI adoption.



Adoption Recap

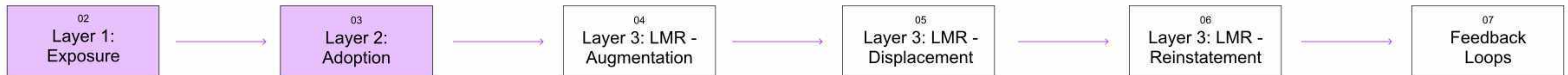
AI behaves like a general purpose technology, similar to electricity or personal computers, whose full effects unfold gradually as firms learn to reorganize around it.

Historically, such technologies take decades to reach scale. Generative AI appears faster on the surface: individual adoption has surged, with roughly 40% of U.S. workers using AI tools in some capacity by the end of 2024.

But firm-level adoption remains slow and uneven. Business surveys show fewer than 10% of U.S. firms have meaningfully integrated AI into workflows, with adoption clustering in software.

Elsewhere, legal and institutional barriers, from data privacy to client confidentiality, continue to hold deployments back.

In the next section, we will examine what happens as adoption deepens, looking at how firms, workers, and markets respond to automation pressure, augmentation opportunities, and the creation of new human tasks.



Layer 3: Labor Market Response - Augmentation

The 3 Labor Market Responses

When AI gets adopted, it doesn't simply "take" or "make" jobs. It changes the tasks inside jobs.

Some tasks get supercharged, some slowly fall off over time, and some new ones emerge.

Economists usually talk about three responses: augmentation, displacement, and reinstatement.

Let's discuss each of these responses in turn and how they create dynamic feedback loops.

What is Augmentation?

Augmentation happens when AI tools help humans do the tasks they already perform, faster and better, without transferring ownership of the task to a machine.

The user's role adapts as they verify the inputs and outputs, while the AI assists in the middle-to-middle tasks.

Output then rises due to increased speed, scale, reduced error, and reduced cognitive load.

In the next few slides, we'll first review case studies showing augmentation in action; then we'll synthesize what the aggregate data suggest about how large effects are, who benefits most, and where augmentation falls short.

Who Benefits From Augmentation?

Anyone who has spent time working with AI tools doesn't need convincing that they can change how work gets done. The real question is how they change it and for whom.

In some settings, AI acts as an equalizer or “expertise multiplier”. It lifts the floor by helping less-experienced workers perform like seasoned experts, narrowing performance gaps and speeding up learning.

In others, the effect is uneven or even counterproductive. When tasks are ambiguous or when AI output is misleading, experienced workers learn to course-correct but novices may not. In these cases, AI can widen performance gaps instead of closing them.

In other words, augmentation isn't a single story. It's a spectrum that depends on the task, the tool, and the worker's judgment.

Let's take a look at the evidence →

Case Study 1: Knowledge Workers

Who: 7,137 knowledge workers across 66 large firms, spanning multiple industries and roles that rely heavily on Microsoft Office tools.

When: Field experiment ran Sept 2023 – Oct 2024; last revised Nov 2025.

Design:

- 6-month randomized access to integrated AI
- Outcomes measured via product telemetry (actual behavior, not surveys)
- Focusing on post adoption with steady-state effects (months 4–6)

Tool: Microsoft 365 Copilot, embedded directly into Outlook, Teams, and Word.



Case Study 1: Knowledge Workers

Findings: Beneficial, but narrow

Email work fell materially:

- Copilot users spent ~2 fewer hours per week on email ($\approx 17\%$ reduction).
- Email work became more consolidated, creating longer uninterrupted work blocks.

Work spills less into off-hours:

- Time spent working outside regular hours fell by $\sim 9\%$, especially after-hours email.

No meaningful change in task mix or volume:

- No increase in meetings attended, documents produced, or email threads handled.
- AI made existing tasks faster, but did not reallocate workers toward new tasks.

Coordination work is largely unaffected:

- No statistically significant changes in meeting time or document activity on average.

This study showed that generative AI delivers real but narrow augmentation: it speeds up individual, text-heavy tasks (especially email), and reduces fragmentation, but did not yet trigger task reallocation or workflow redesign at the individual level.

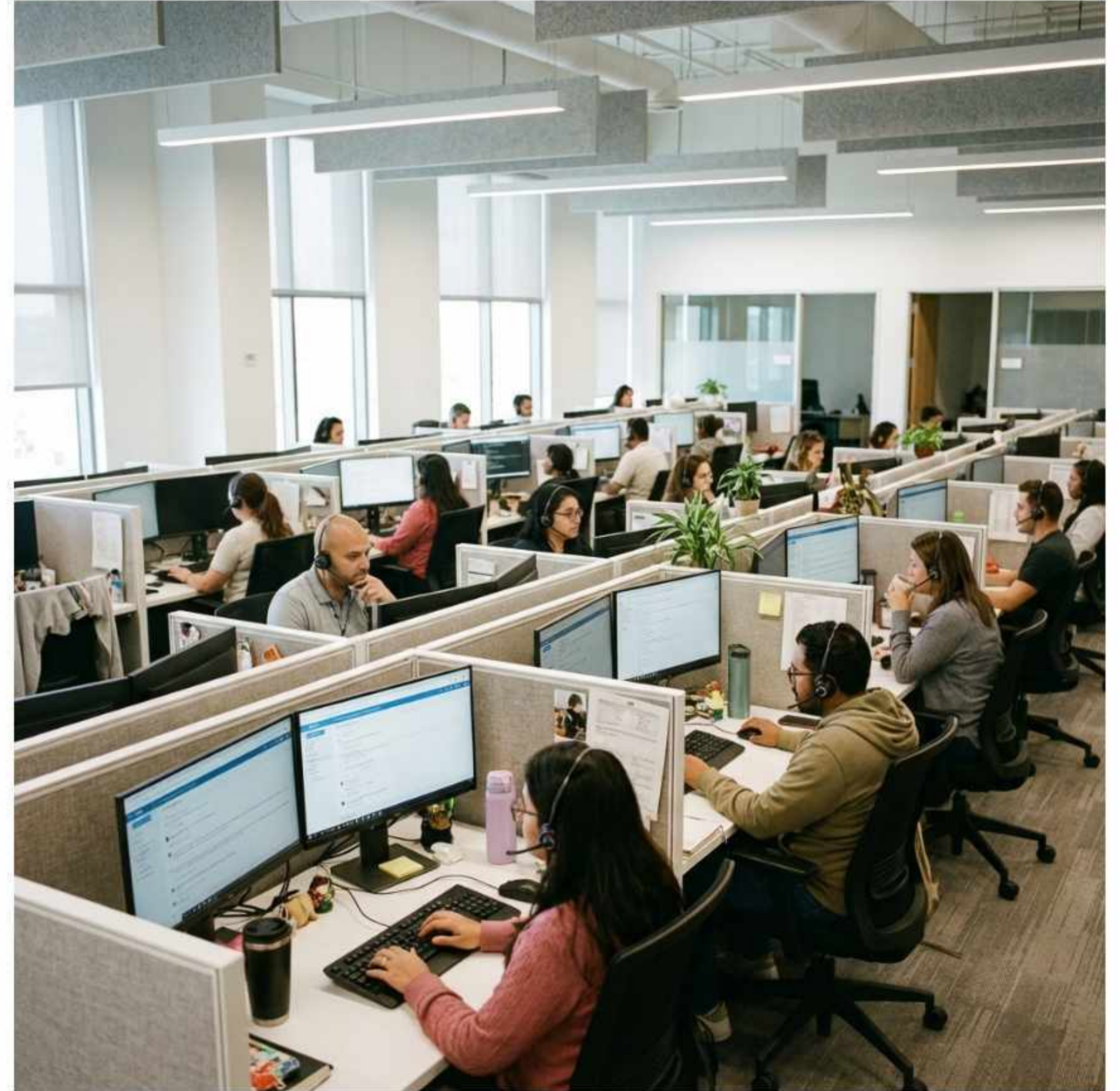
Case Study 2: Chat Support Agents

Who: 5,172 customer-support agents working for a Fortune 500 firm that sells business-process software.

When: Late 2020/Early 2021, Published Feb 2025

Design: Randomized control trial where half of the agents got access to the AI tool. Half of them didn't.

Tool: GPT-3–class assistant watching live chats; suggested replies/steps/tone; agents could ignore/edit (human-in-the-loop).



Case Study 2: Chat Support Agents

Findings: Beneficial overall

The AI tool resulted in a 15% increase in the number of chats that an agent successfully resolves per hour.

Less skilled and less experienced workers improve significantly across all productivity measures, including a 30% increase in the number of issues resolved per hour.

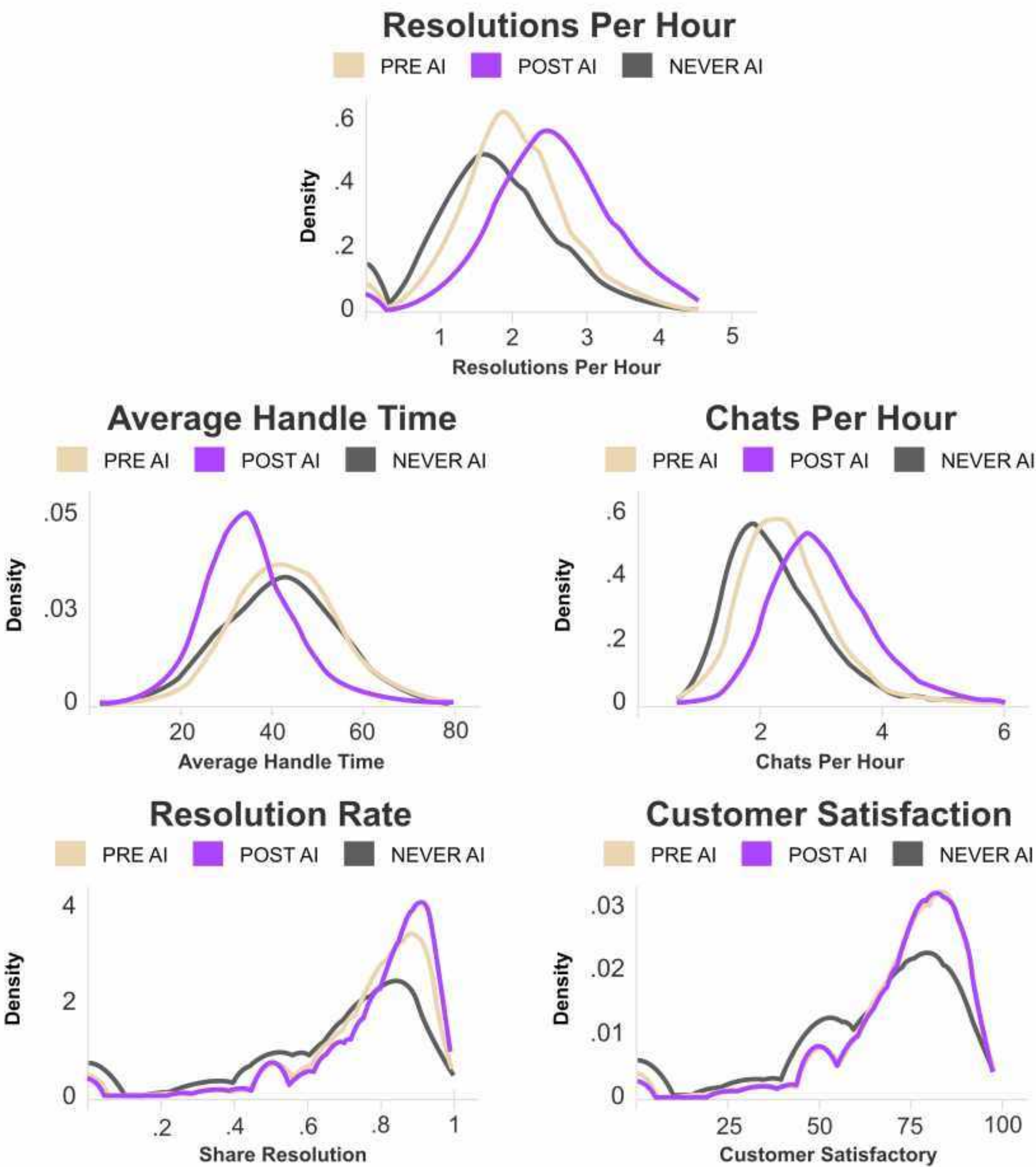
The AI tool also helped new agents gain experience quicker. AI-assisted workers at two months behaved like No-AI workers who've been at the firm for 6 months.

In contrast, AI has little effect on the productivity of higher-skilled or more experienced workers.

Case Study 2: Chat Support Agents

- = Group 1 (Treatment) - Before receiving AI
- = Group 2 (Treatment) - After receiving AI
- = Group 3 (Control) - No AI ever

Takeaway: Augmentation allows more efficiency more of the time, especially at the bottom of the skill distribution.



Case Study 3: Consultants

Who: 758 consultants from Boston Consulting Group (BCG), representing ~7% of their global individual contributor workforce.

When: Spring 2023

Design: Randomized control trial with 3 groups of consultants that were assigned 18 realistic consulting tasks

One group worked manually, one with GPT-4, and one with GPT-4 plus prompt engineering training.

Tasks: Idea generation, creative product innovation, analytical market segmentation, press releases, and one “trick” task.



Case Study 3: Consultants

Findings: Good for juniors on “simple” tasks

Consultants using GPT-4 completed more work, faster, and at a higher quality on tasks like idea generation, market segmentation, and press release writing.

The gains were larger for “junior” consultants.

Below-average consultants improved performance by 43%, while above-average consultants improved by 17%.

AI helped everyone, but it especially lifted lower performers.

Case Study 3: Consultants

Findings: Mixed on “sophisticated” tasks

The researchers also included a “trick” task that the AI should find easy, but which was carefully designed to confound it (strategy advice using misleading financial data and interviews)

On this task, GPT-4 gave poor recommendations.

But even when AI output was wrong, most consultants still leaned on it rather than override it, leading to worse outcomes than those who didn't use AI.

Takeaway: Gains are large for tasks that align with what the model is good at. For those, productivity, speed, and quality all improve (12.2% more tasks, ~25.1% faster, and produced ~40% higher quality outputs vs no-AI control). People who are weaker to begin with gain much more. But outside that zone, AI can mislead or lower output for those without expertise or systems to verify the results.

The Jagged Frontier

The BCG consultant study shows that AI augmentation delivers modest-to-strong productivity gains in many bounded, knowledge-work tasks, often in the range of 10-30% improvements in speed or output, sometimes more in quality.

But these gains are concentrated. Augmentation happens when tasks are well-specified (e.g. customer support), when AI's competencies already match or overlap task demands, and when users have less skill or experience to begin with.

In contrast, where tasks are ambiguous, novel, or outside what current models handle well, AI can mislead, reduce accuracy, or produce worse outcomes.

This is captured by the concept of a “jagged frontier” →

The Jagged Frontier

Inside the Frontier

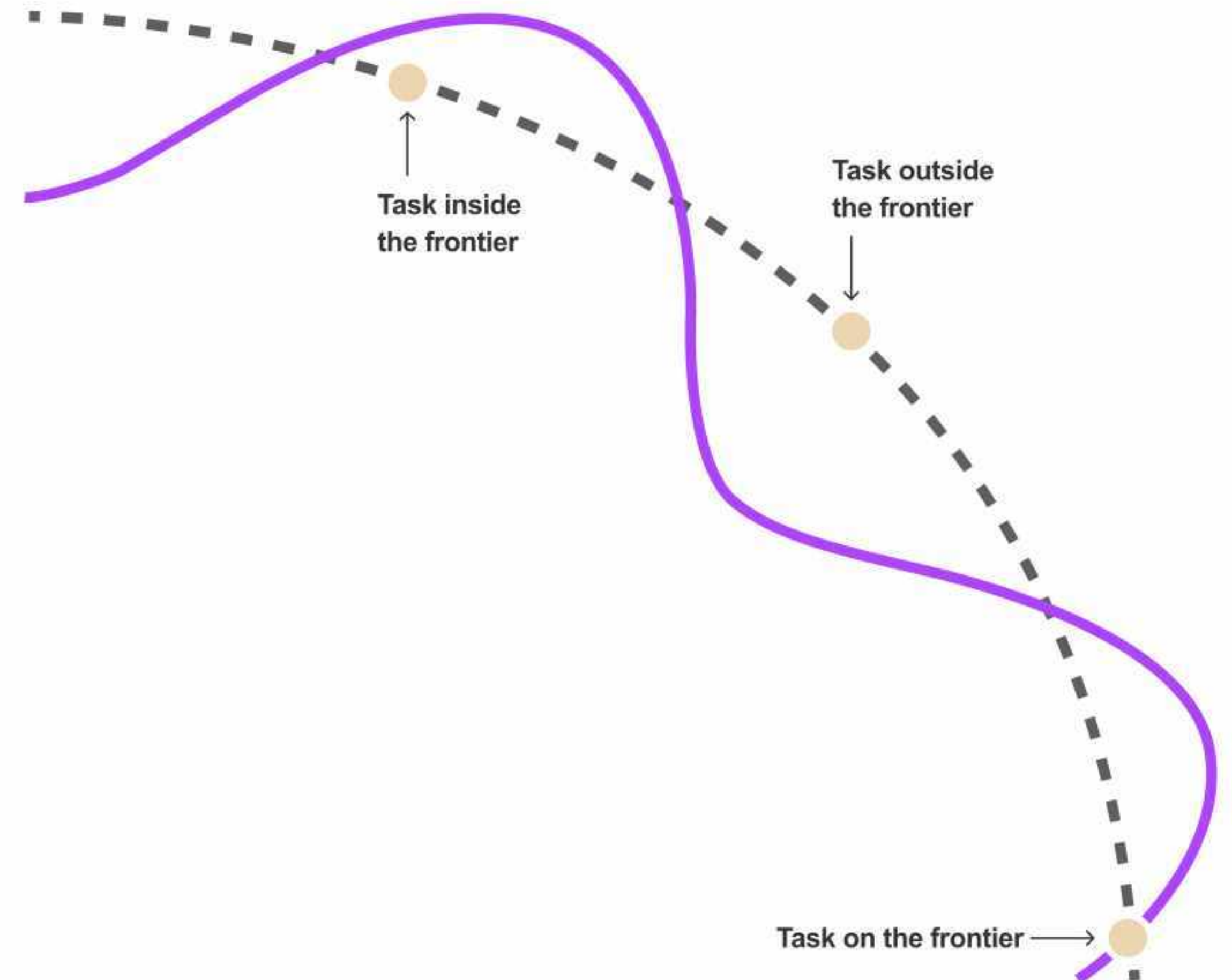
- Tasks tend to be ceilinged or standardized where there is a “right” answer, and lots of prior examples.
- AI captures and broadcasts tacit knowledge.
- Creates the biggest gains for lower-skill workers, with little room for the top to improve. (Call-center; much of the BCG consultants “simple” tasks are inside the frontier.)

Outside the Frontier

- Tasks tend to be open-ended and outside training data (novel ideation, research strategy, complex management decisions).
- Performance depends on the users ability to judge AI output.
- Experts can sift good from bad, so top workers may benefit the most, but only if the task is within the model’s competence.

The Jagged Frontier Diagram

■ AI CAPABILITY ■ EQUAL DIFFICULTY TASK



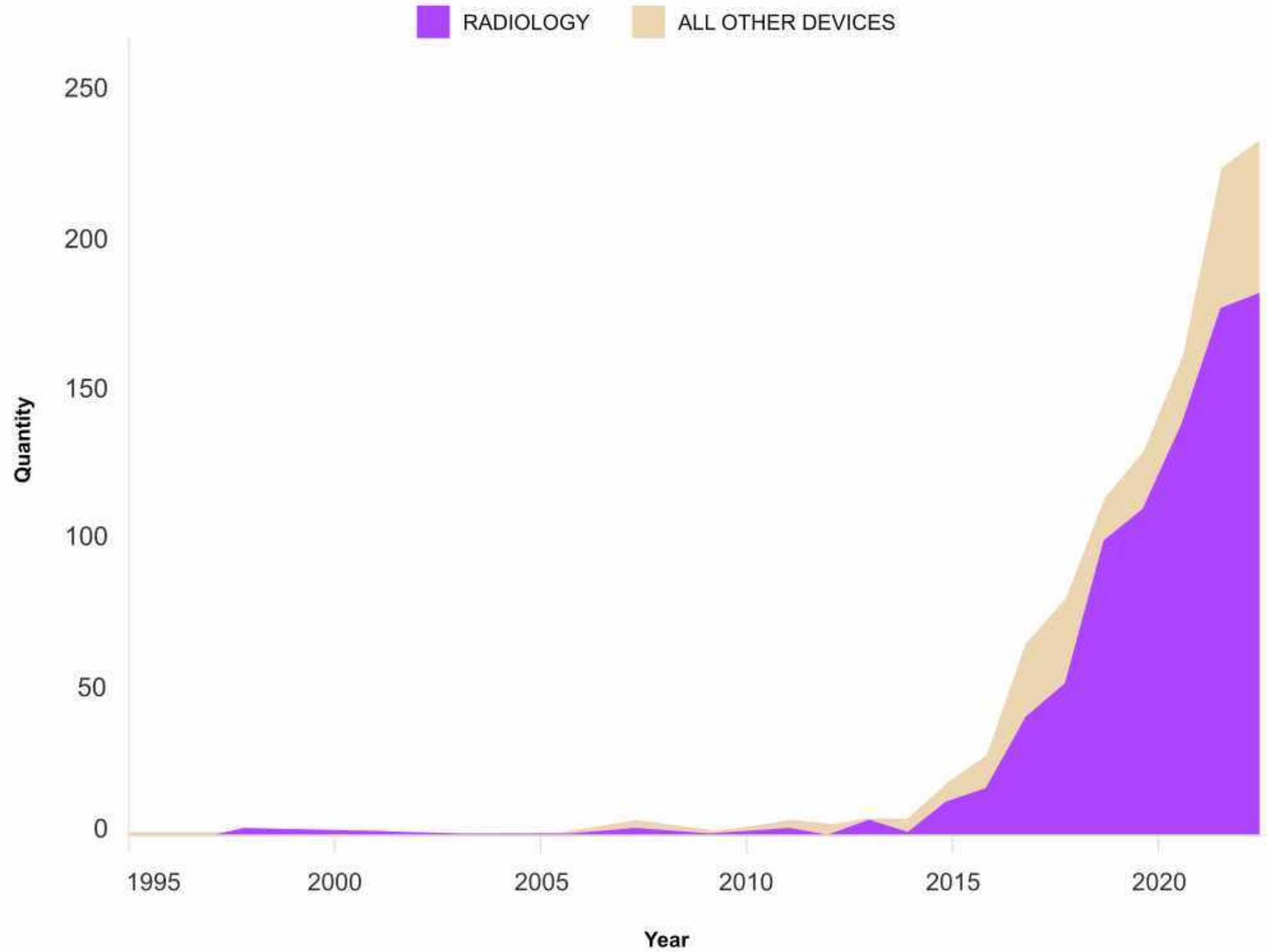
Case Study 4: Radiologists

As mentioned earlier, in 2016 Geoffrey Hinton argued that “we should stop training radiologists now” because AI would soon surpass them.

In benchmark tests, AI models do outperform human radiologists on narrow tasks, and some are even FDA cleared to operate without human review under strict conditions.

Yet radiology jobs sit at record highs, with an average annual salary of \$520,000.

Annual Number of FDA-Cleared AI-Enabled Medical Devices (1995-Present)



Case Study 4: Radiologists

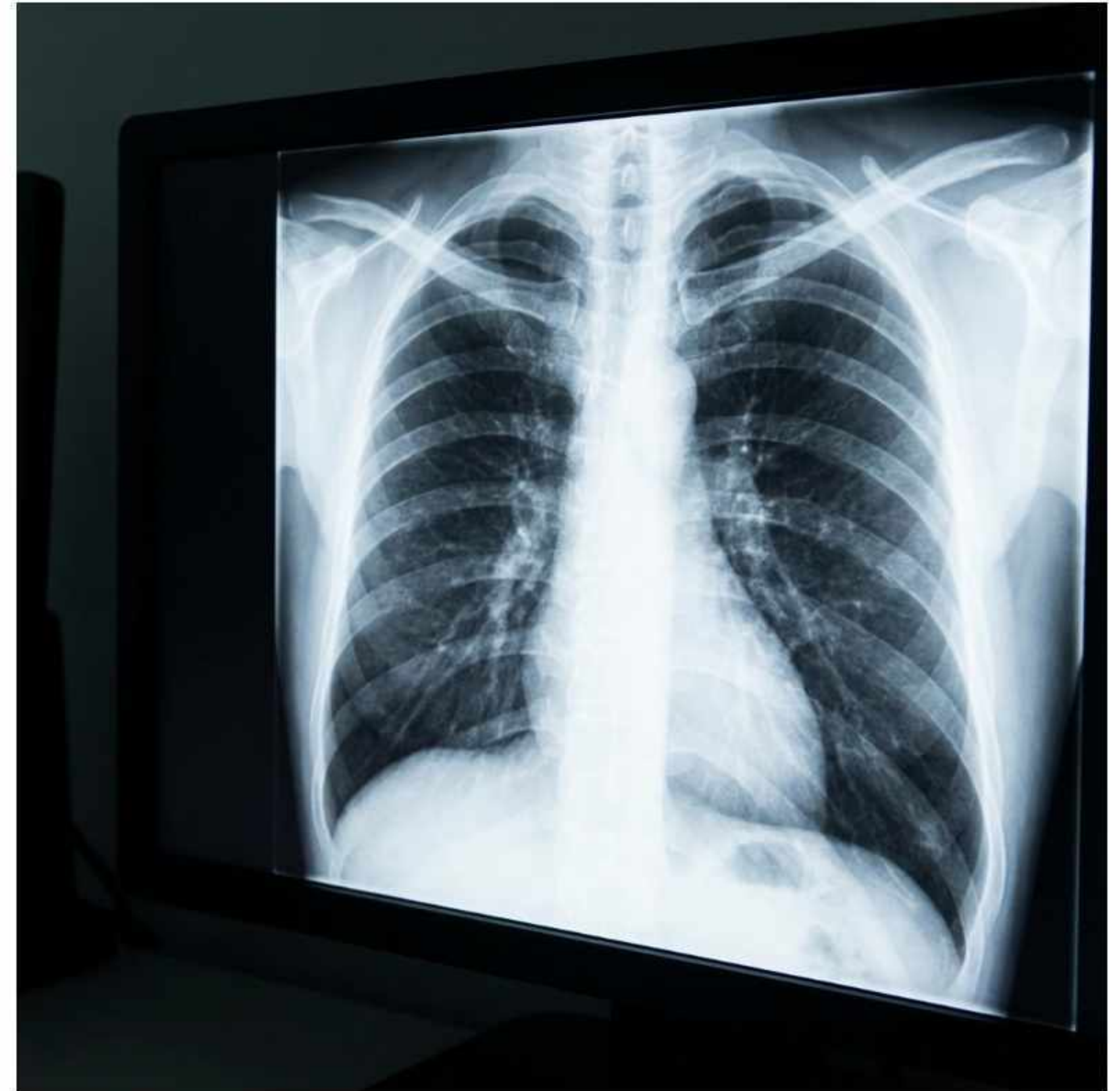
Who: 140 radiologists reading 324 patient cases across 15 diagnostic tasks.

Design: Each radiologist read with and without AI assistance.

Findings: Mixed

Effects varied by person: Some radiologists improved with AI, others got worse. Conventional predictors (years of experience, subspecialty, prior AI use) did not explain who benefited.

When the AI's own predictions were less accurate, it pulled human performance down (on aggregate and ~half of the individual pathologies).



Case Study 4: Radiologists

Takeaway: Radiology illustrates both sides of the augmentation spectrum.

Inside the frontier, AI tools act as productivity amplifiers. Most clinical imaging tasks (e.g. detecting lung nodules, flagging fractures, or triaging scans) are structured, repetitive, and supported by large, labeled datasets. These are inside-frontier tasks: well-defined problems with abundant training examples.

Outside the frontier, models struggle outside training data. These include rare pathologies, mixed-modality interpretation, or image sets with artifacts and incomplete data.

Case Study 4: Radiologists

Combined, this explains why radiology jobs haven't been wiped out contrary to Hinton's claims.

(1) AI systems excel at the narrow, standardized tasks they were trained on. But they degrade sharply when faced with new imaging devices, patient demographics, or rare pathologies that weren't represented in training data.

(2) Many AI tools are deployed as assistive or triage, meaning a human must review or override the output. Fully autonomous readings require much stricter regulatory and liability standards.

(3) A radiologist's work extends beyond image interpretation. Diagnostic reading occupies only a small portion of their time. They also design clinical workflows, train medical staff, consult with physicians, and review imaging orders. Even if an AI system achieves perfect accuracy in reading images, radiologists remain essential for these broader responsibilities.

(4) Even when AI boosts productivity accurately within the frontier, it doesn't reduce the total number of radiologists needed.

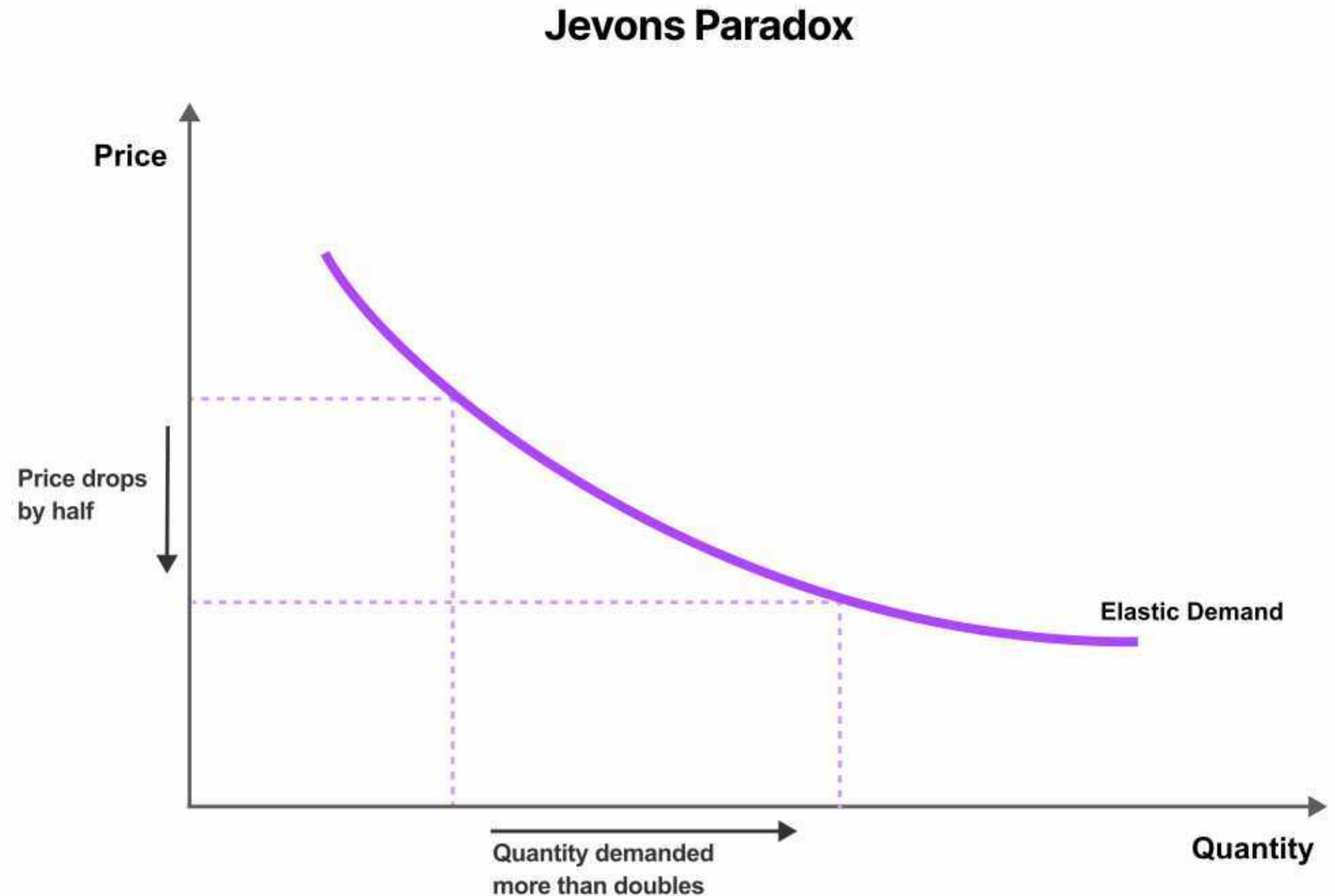
That's partly because of **Jevons Paradox** →

Jevons Paradox

In 1865, economist William Stanley Jevons observed that as coal-burning engines became more efficient, total coal consumption actually rose.

Jevons paradox now refers to the pattern in which making something cheaper or faster increases overall demand rather than reducing it.

In the case of radiology, as AI lowers the time and cost per scan, hospitals order more scans. Imaging demand has risen dramatically since the digitization of radiology and AI is accelerating that trend. The result is more work for radiologists, not less.



Augmentation Recap

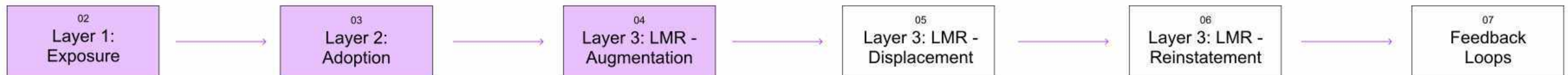
Firms are pairing AI with humans on structured, verifiable work. Productivity per worker rises on these tasks and quality holds or improves when outputs are checked.

Inside the frontier, goals are clear, examples are plentiful, and feedback loops are short. These are great conditions for human-in-the-loop AI tools.

Mid-career professionals who take ownership of decision quality, and juniors who show judgment advance fastest. Teams that treat AI outputs as verdicts (not drafts) on open-ended tasks might see quality slip.

There are also signs of the Jevons Paradox at play as AI increases productivity and lowers prices, which increases the demand for services like radiology instead of replacing it.

Although some tasks that are augmented soon get automated which leads us to our next chapter, displacement.



Layer 3: Labor Market Response - Displacement

From Augmentation to Automation to Displacement

We saw that AI tools can raise individual productivity - e.g. allow customer-support agents to resolve more tickets per hour and radiologists to read more scans per hour.

The radiology case shows that superior benchmark performance doesn't necessarily lead to automatic displacement.

But other jobs may not be as robust. For example, if one designer with Midjourney (AI image generation) can do the work of five concept artists and the company no longer needs the remaining workforce, that's augmentation for the one but displacement for the other four.

Historical Example: 1990s Telephone Operators

In the early 1900s one of the most common jobs for young American women was telephone operator. Nearly every call had to be routed manually through a human at a switchboard.

Then came automation. Between 1920 and 1940, AT&T, the largest employer of phone operators in the US, rolled out mechanical switching across the country.

This was one of the largest automation projects in history. In city after city, machines replaced operators, and the once-ubiquitous role largely disappeared as operator jobs dropped rapidly by 50 to 80%.



Historical Example: 1990s Telephone Operators

The labor market adjusted quickly.

Young women entering the workforce found new opportunities as secretaries, clerks, and service workers, often with comparable or better pay.

But the women who had built careers as operators were left behind. Their hard-won experience in a now-extinct occupation did not translate, and many suffered long-term earnings losses.

What's Happening Today?

Stories of software engineers getting laid off have gone viral.

Example: Shawn K shared his experience on Substack of getting laid off at FrameVR.io as AI took over the company

“We had been reorienting the company towards AI, adding on AI features all throughout the software and trying to capitalize on AI to our customers, and then shortly after that kind of restructuring and strategizing...I got laid off,” he told the Independent.

The Great Displacement Is Already Well Underway

It's Not a Hypothetical, I've Already Lost My Job to AI For The Last Year

 SHAWN K
MAY 04, 2025

 1,280

 345

 208

Share

...

Coverage of Corporate Rebalancing

Amazon:

“It’s hard to know exactly where this nets out over time, but in the next few years, we expect that this will reduce our total corporate workforce as we get efficiency gains from using AI extensively across the company.” - Andy Jassy, CEO of Amazon

Atlassian:

“If we make call center staff more productive, people aren’t going to call more, and we’ll probably need less call center staff.” - Scott Farquhar, citing increased customer self-service and AI-powered tools as the reason.

Duolingo:

In January 2024, Duolingo announced it would be off-boarding 10% of its contractor workforce (technically not “layoffs”), as the company pivoted to AI translation.

Salesforce:

CEO Marc Benioff stated in a podcast that the company was able to reduce its customer support headcount from 9,000 to 5,000, thanks to the implementation of agentic AI agents.

The Challenge

In many cases companies do not state that a job was eliminated because AI can perform it entirely, and when one does announce AI-related restructuring, the media pushes for coverage of layoffs, while companies quietly hiring don't get the same attention.

This creates systematic bias in anecdotal evidence.

The stories suggest AI plays a major role, but its effect is tangled with economic pressure, competition, and other macro trends.

A stronger case requires evidence that controls for confounders such as firm-level growth or decline and broader macro pressures.

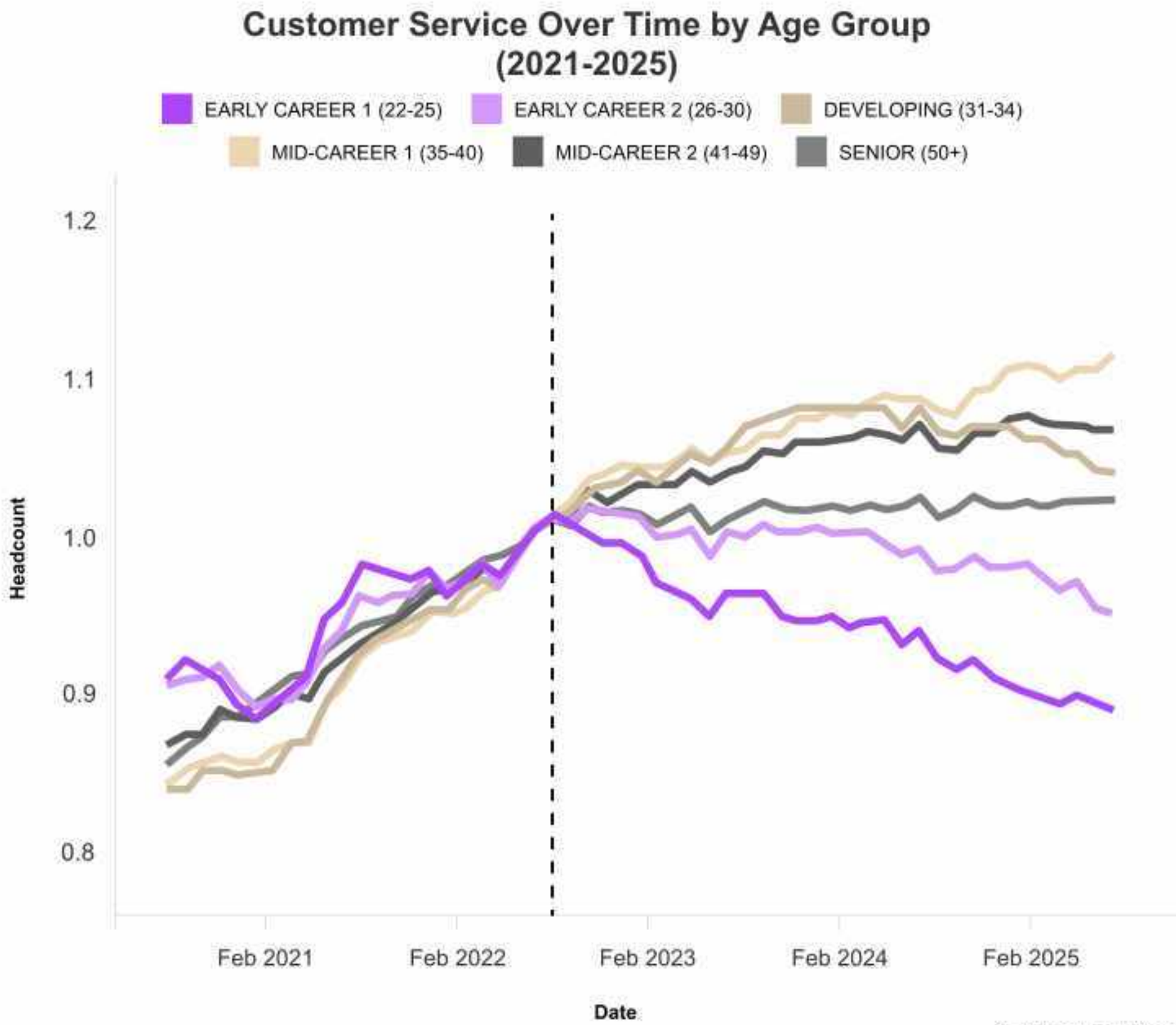
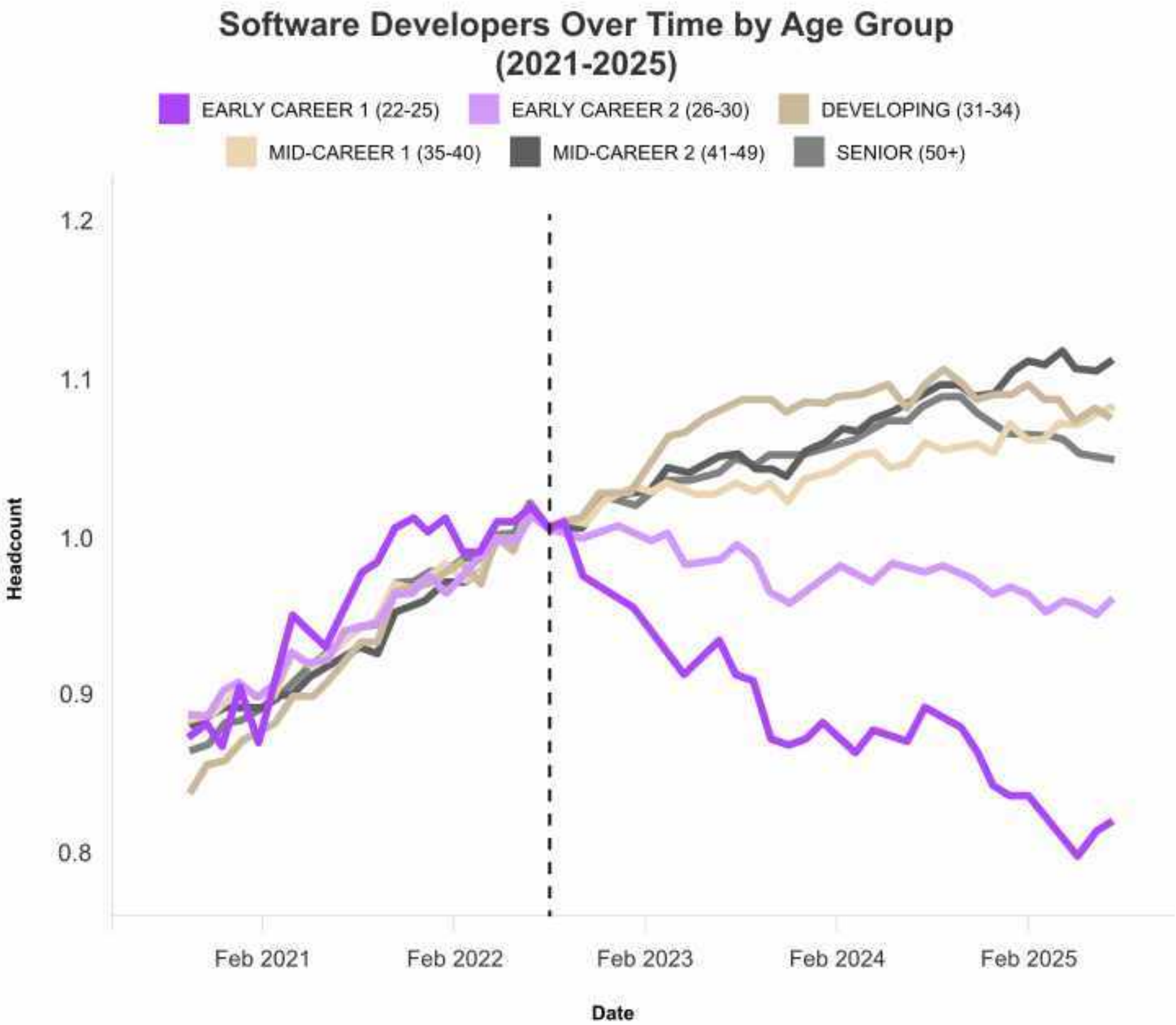
Recent evidence from Stanford's Digital Economy Lab does this.

Let's see what they find...

The Evolution of Entry-Level Work

Employment for 22-25 year-old developers fell ~20% from peak in 2022. We see a similar pattern for customer service reps, another job highly exposed to AI.

For both roles, the decline is sharpest for the 22-25 age group, with older, more experienced workers less affected.

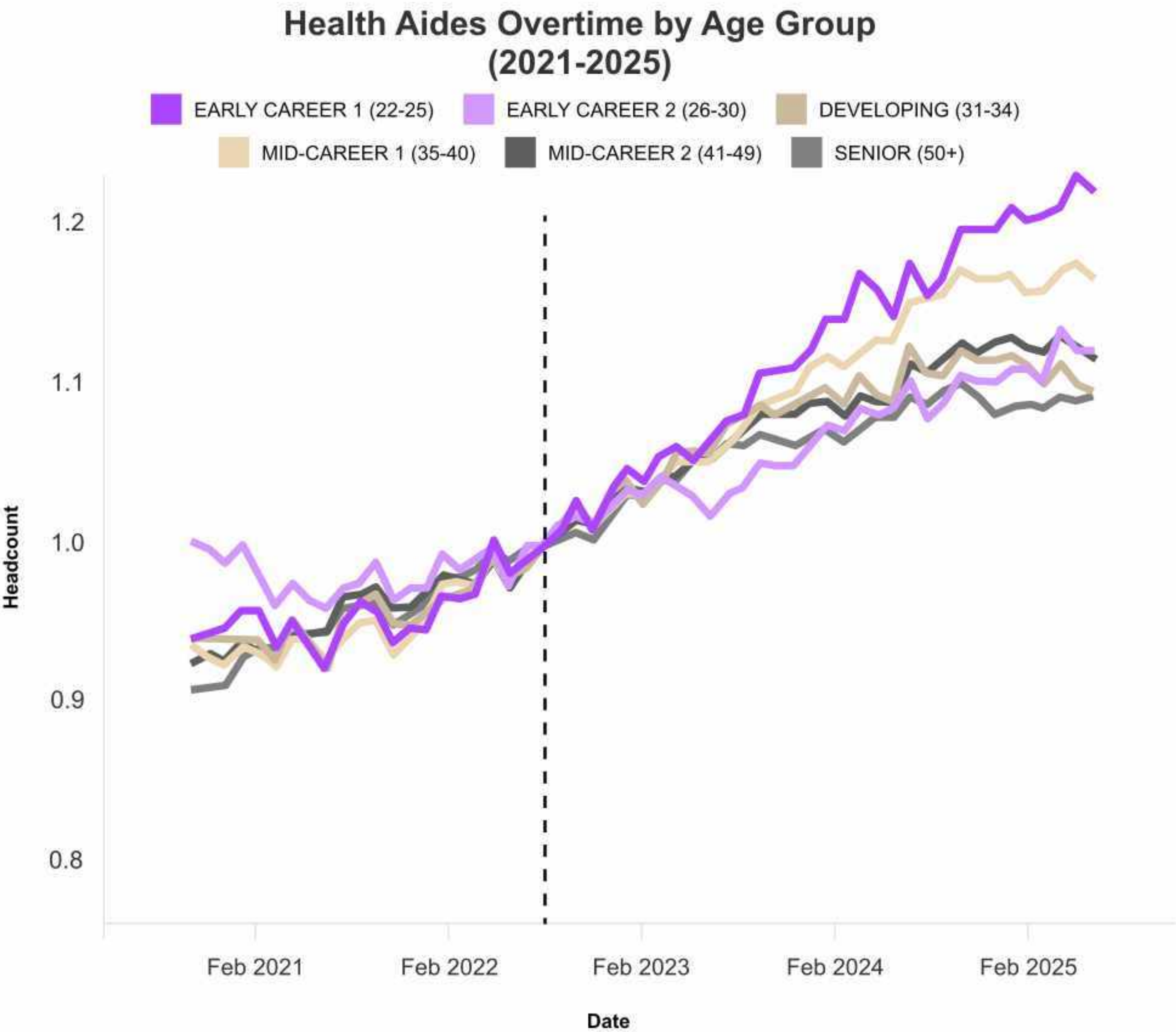


Source: [Brynjolfsson et al. \(2025\)](#)

In-Person Resilience

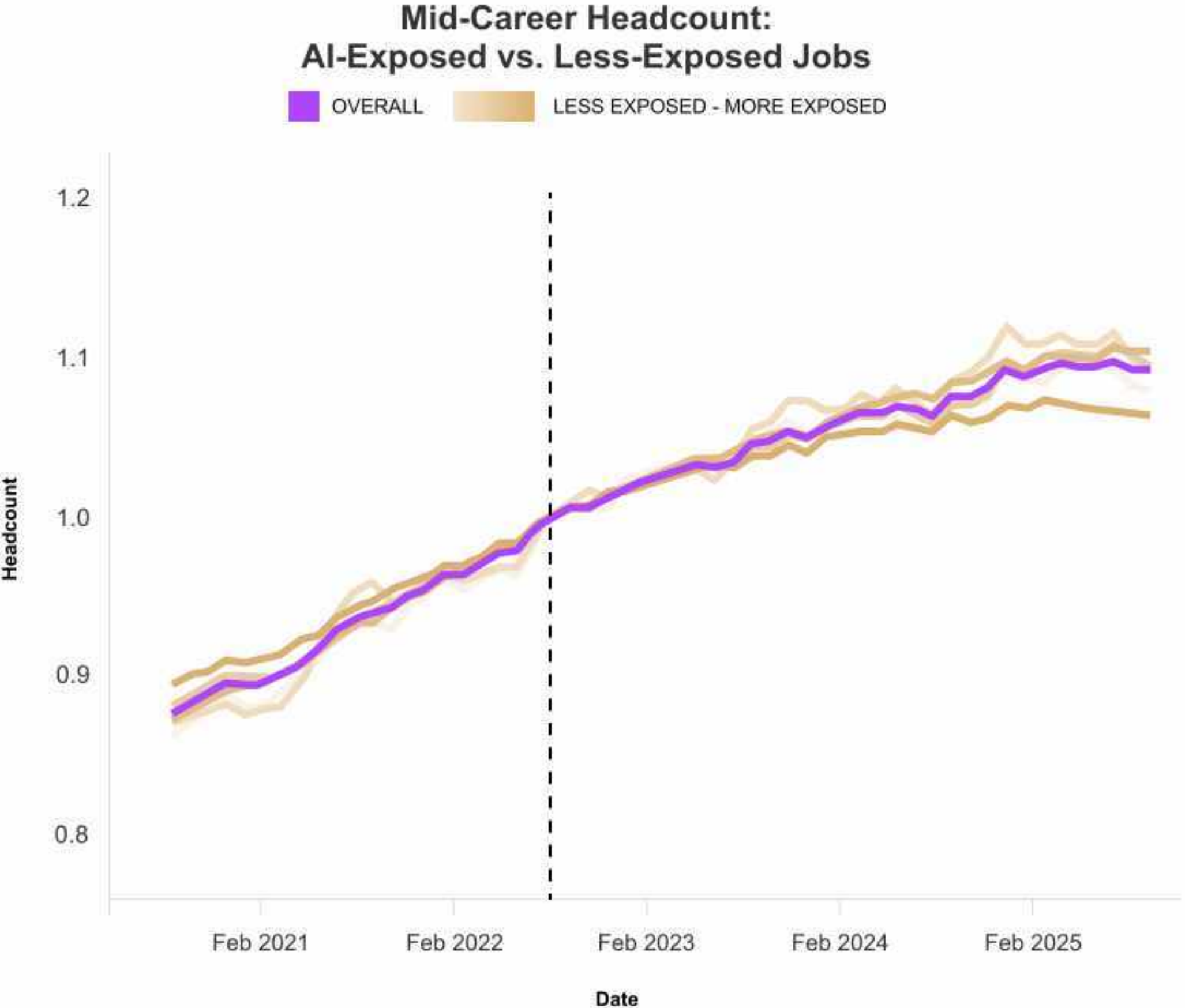
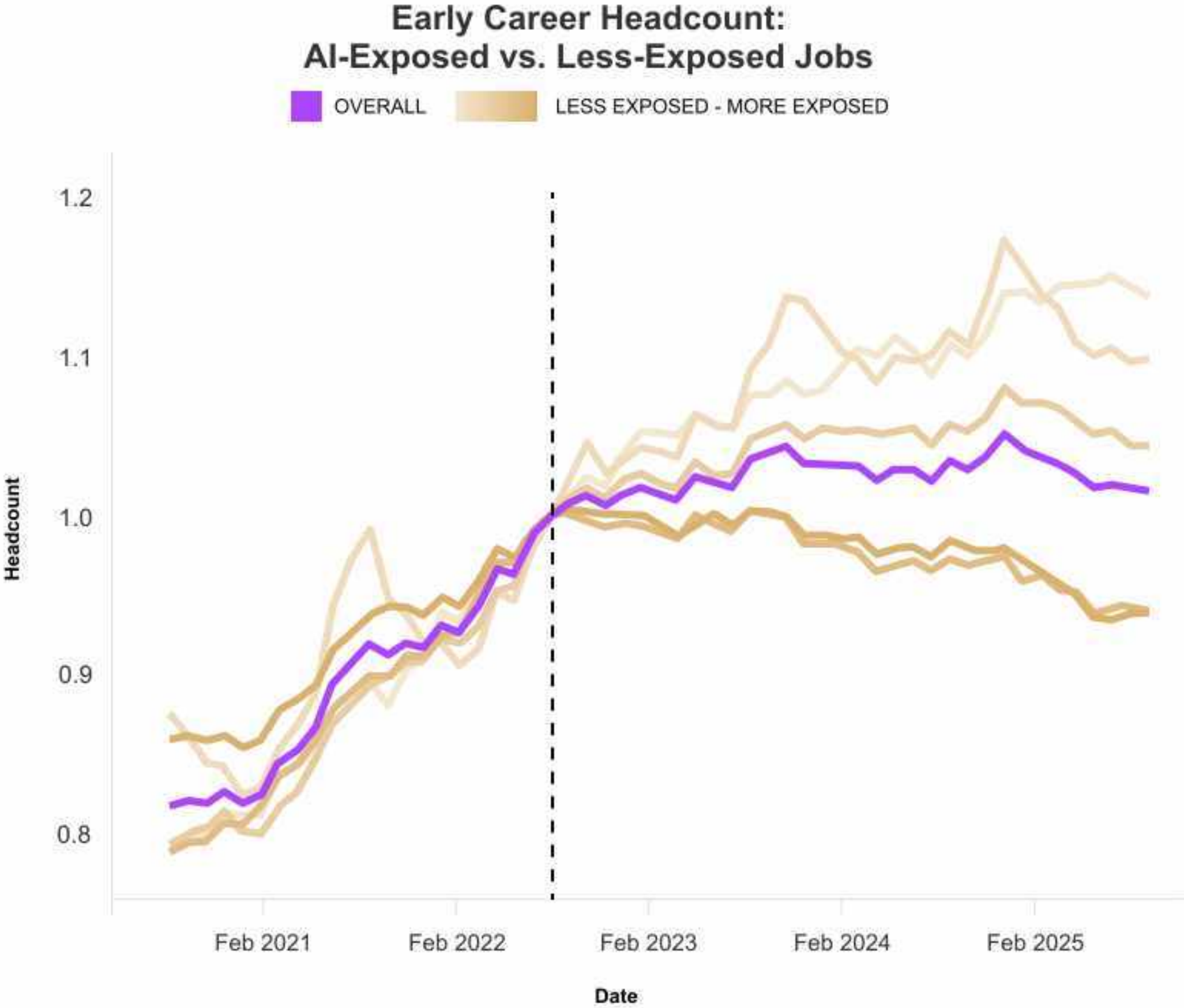
In contrast, jobs less exposed to AI, like health aides, show the opposite trend.

These jobs, which require in-person physical tasks, have seen the fastest employment growth among youngest workers.



Entry-Level Jobs

Overall the data shows that 22-25 year-olds' employment is falling in the most AI-exposed jobs and rising in the least exposed jobs. For more mature workers, there are small differences in employment trends based on AI exposure.



The Challenge

1. Could it just be tech overhiring?

Tech firms did expand aggressively during the pandemic, then cut back. But the results hold even after excluding the tech sector and computer-related jobs.

2. Return-to-office effects?

Unlikely. The pattern persists even in roles that can't be done remotely.

3. Declining education quality?

The same trend appears in occupations where most workers don't have college degrees.

The author's conclusion:



“Overall, we do not want to claim that AI explains all of the patterns we show in the data but our results are consistent with the notion that AI is having a tangible impact on the labor market for entry-level workers.”

Erik Brynjolfsson
Economist at Stanford University
2025

Displacement Recap

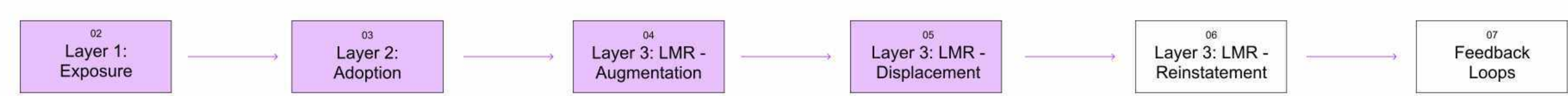
Entry-level roles remain the most exposed, with hiring stabilizing below pre-AI peaks as managers backfill tasks and oversight needs rise. While some fields like the healthcare space, continue to see growth across all age brackets, with the strongest rise on the youngest cohort.

LLMs overlap with junior task bundles, creating some bottom-of-the-pyramid pressure and incentivizing firms to freeze junior hiring first.

Firms realize short-term cost savings, while mid-level employees gain leverage as their judgment and integration skills become comparatively more valuable.

New grads are most likely to bear the brunt if they are in the most automatable functions and don't adapt to the new tools being implemented.

These tools are also transforming skills, training and learning, opening up new sectors, which leads us to reinstatement.



Layer 3: Labor Market Response - Reinstatement

Reinstatement

Reinstatement refers to the process by which technology doesn't just displace human labor, but creates new tasks that eventually make up (either partially or fully) for the jobs lost through automation.

If reinstatement is strong, then over time, the labor market may absorb displaced workers into these new roles. If it is weak, then displacement may dominate, leading to longer-term employment declines, skill mismatches, etc.

In this section, we dive into evidence of two kinds of AI-driven job creation:

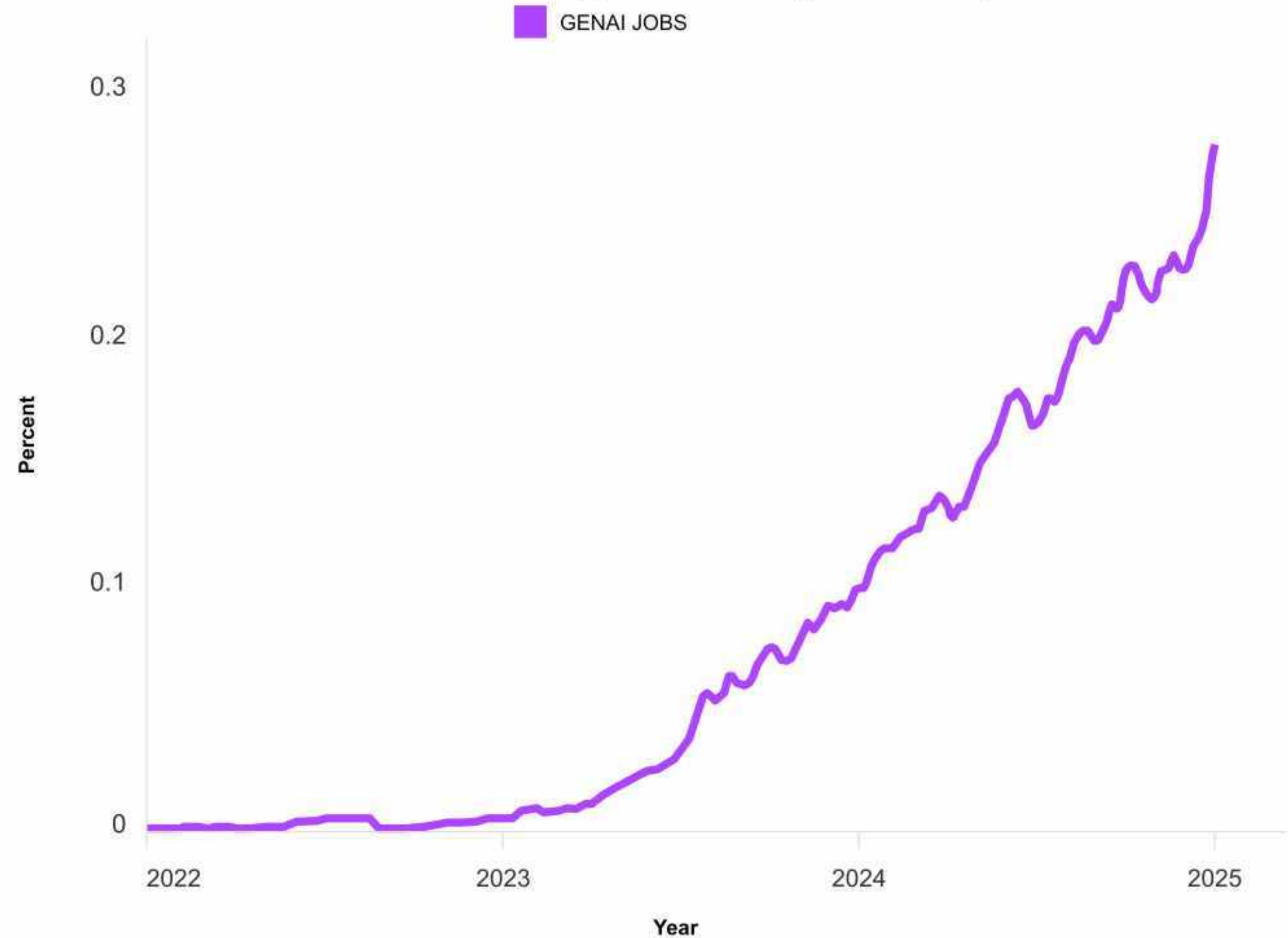
- (1) New occupations born from AI
- (2) Resurgent roles driven by AI's infrastructure growth

(1) New Occupations Born From AI

Advances in AI, especially modern machine learning and generative AI, have given rise to brand-new job titles and specializations.

These occupations did not meaningfully exist before the AI boom, and their emergence is direct evidence of AI creating jobs.

GenAI Job Posting Over Time (2022-2025)



Examples of New Roles

AI-Related Engineering

“Artificial Intelligence / Generative AI Engineer” is among the fastest-growing roles on LinkedIn’s Jobs on the Rise 2025 lists, particularly in the U.S., U.K. and Netherlands, as demand surges for people who can build, fine-tune, and integrate AI systems.

Prompt engineering emerged as a highly valued specialization during the generative AI boom, with some listings in 2023 offering very high salaries (e.g. US\$335,000) for roles whose focus was primarily optimizing prompts and AI outputs.

While dedicated prompt engineering titles are now less dominant as tools have gotten good at prompting, the skill remains important and is increasingly embedded within other roles like ML engineers, product managers, and content designers.

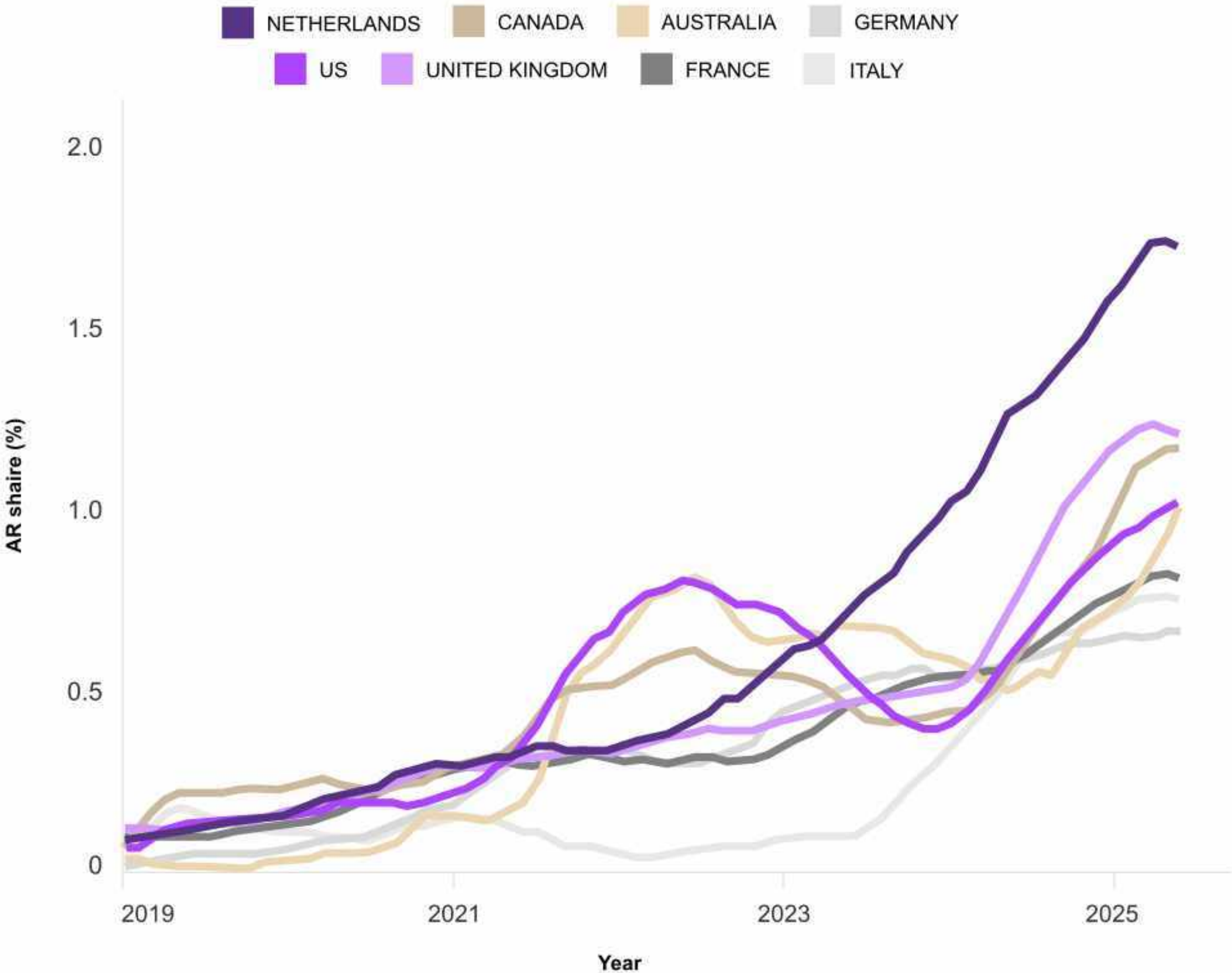
Examples of New Roles

AI Safety / Trust & Risk Roles

Positions with titles such as “Responsible AI Lead,” AI Governance” , “AI Policy Advisor” or “AI Ethics Program Manager” were virtually unheard of a few years ago but now number in the hundreds on U.S. job boards.

Companies in finance, healthcare, technology and government are hiring such roles to develop ethical guidelines, audit algorithms for bias and ensure compliance with emerging AI regulations.

AI-related Job Postings Across Selected Markets (2019-2025)



Examples of New Roles

There has been a rise in jobs for AI red-teamers and adversarial researchers who stress-test models for flaws, security vulnerabilities and dangerous behavior.

The executive title Chief AI Officer (CAIO) or equivalent “Head of AI” has also become much more common.

LinkedIn reports that the number of companies with such roles has roughly tripled over the past five years. **These roles often combine strategy, governance, investment priority, and oversight of AI deployment.**



Cisco Careers

[Search jobs](#)[Careers home](#)[Emerging Talent](#)[Events](#)[Stay In Touch](#)

[< Back to search results](#)

Director of Engineering (Adversarial AI – Red Team, Threat Intelligence)

Apply

Share



Job notifications

Sign up to receive notifications of similar jobs

Notify Me

(2) Resurgent Roles

AI's proliferation is revitalizing demand for existing occupations, especially those that build and maintain the physical and digital infrastructure enabling AI.

So, instead of eliminating all blue-collar work, AI is creating more demand for skilled trade roles.

These are not brand-new occupations but existing ones whose demand is surging because of AI deployment.

In many cases, AI is indirectly creating these jobs by driving massive investments in infrastructure and industry expansion.



Examples of Resurgent Roles

Construction Workers for Data-Center Build-Outs

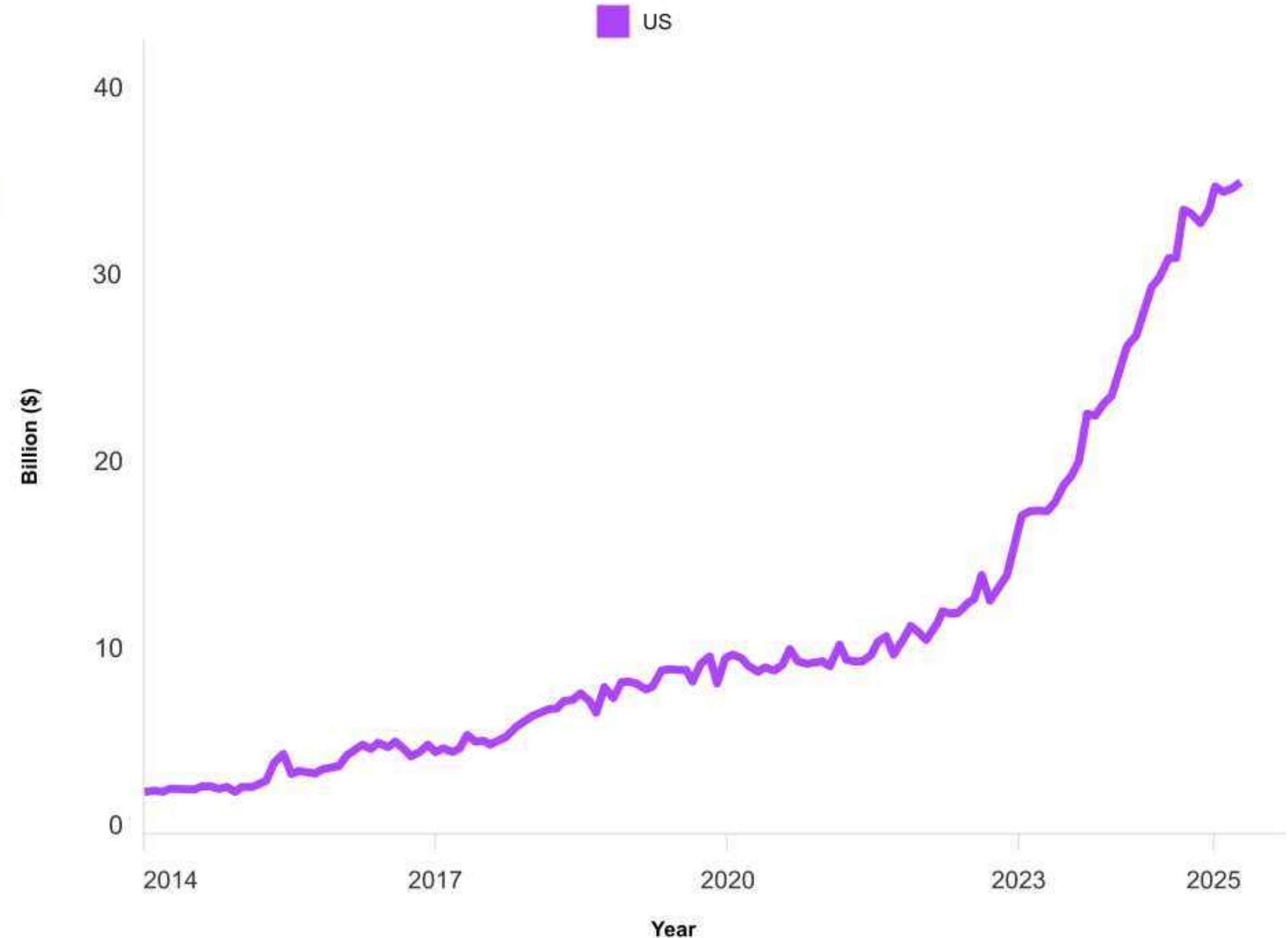
U.S. data center construction spending hit a record \$40 billion annual rate in June 2025, up from a 50% increase in 2024, driven by AI's computing needs.

For example, Microsoft's new \$3.3 billion AI data-center project in Wisconsin will create about 2,300 union construction jobs during the build-out phase.

Meta is building an AI hub in Louisiana spanning four million square feet that will require over 5,000 construction workers at peak.

Each of these mega-projects create a "city" of trade workers.

U.S. Data Center Construction Spending
(2014-2025)



Examples of Resurgent Roles

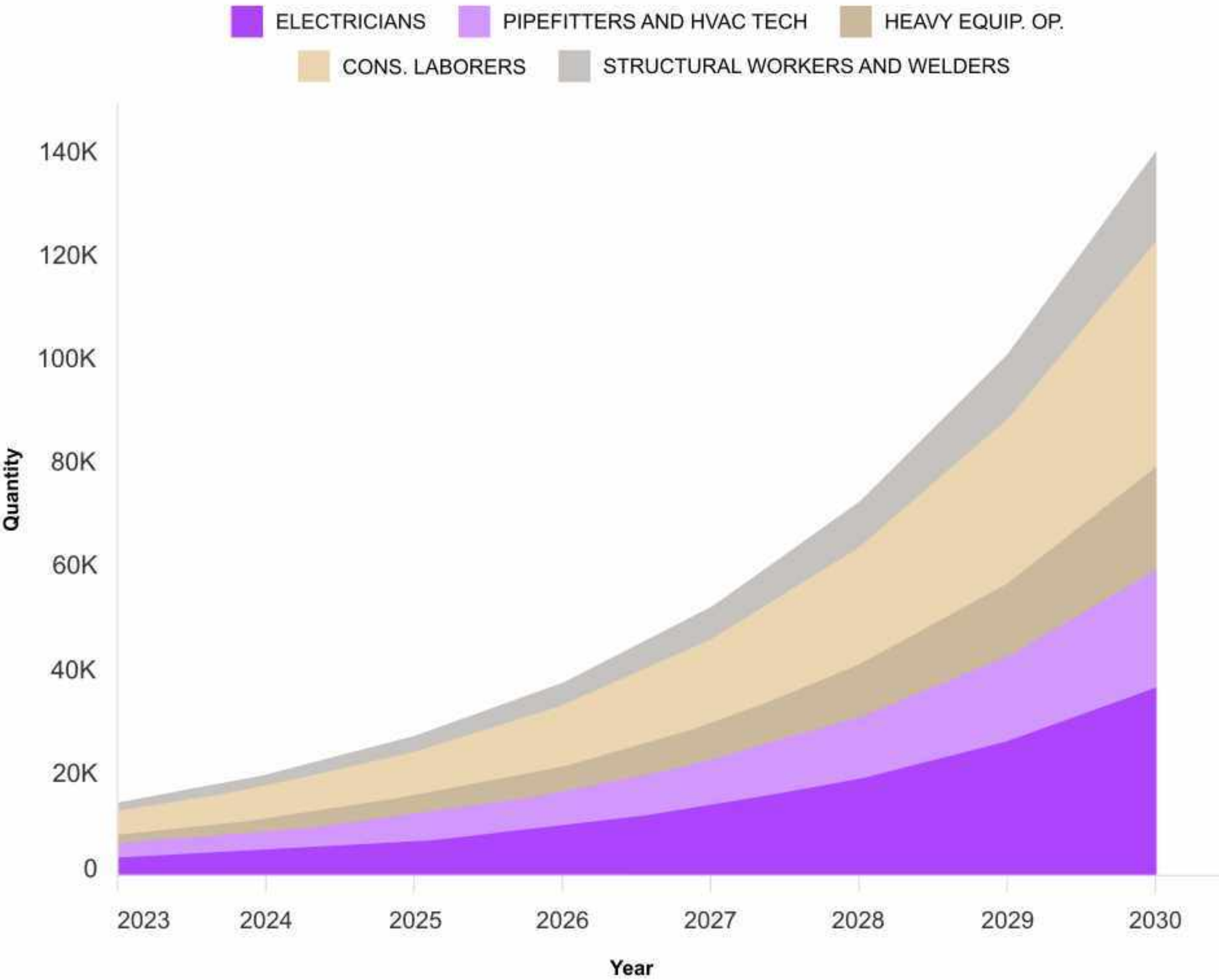
Electricians and Power-System Technicians

AI data centers and GPU clusters demand vast amounts of electricity and specialized wiring. So, wherever AI infrastructure rises, armies of tradespeople are needed on the ground.

Data centers now compete with factories and manufacturing plants for electricians and plumbers.

Workers building data centers are seeing pay increases of 25-30%, with some breaking into the multiple six figure salary range.

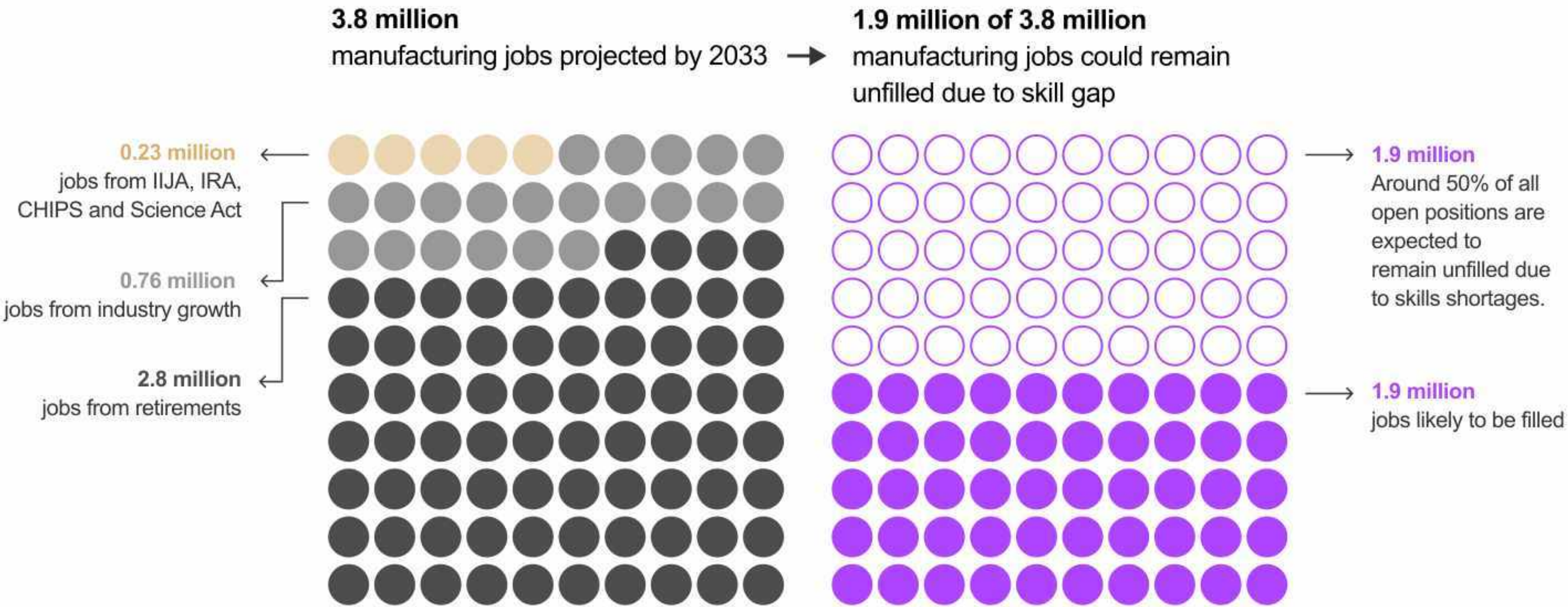
Resurgent Roles Needed in Upcoming Years (2023-2030)



Examples of Resurgent Roles

Today, there are roughly 400,000 unfilled skilled-trade roles in the U.S., as data centers demand more electricians, HVAC technicians, and skilled labor. **By 2033, that figure is projected to rise to about 1.9 million.**

Projected Manufacturing Job Vacancies in the U.S. by 2033



Examples of Resurgent Roles

Industrial Machinery and Robotics Technicians

Every automated warehouse, fulfillment center, and AI-enabled factory requires people to install, calibrate, and repair the robots and autonomous systems.

Amazon, for instance, now employs more robotic maintenance technicians than ever to keep its AI-guided logistics running.



Reinstatement Recap

AI growth creates new demand for “soft” and “hard” infrastructure, including evaluation/oversight teams, governance and model-risk functions, and large-scale data-center buildouts requiring electricians, technicians, and skilled trades.

Reinstated roles emerge where human judgment, compliance, and domain expertise remain essential, such as researchers, engineers, safety specialists, and red-team professionals.

Construction and technical trades benefit from AI-driven capital expenditure, while workers in adjacent fields may need to reskill or shift into areas where demand is accelerating.

With exposure, adoption, and labor-market responses now mapped, we can see how each layer interacts to create new feedback loops.



The AI-Work Feedback Cycle

Feedback Loops

As we've seen, AI's impact on work isn't a straight line from "what models can do" to "who keeps their job." It's a living system with feedback loops.

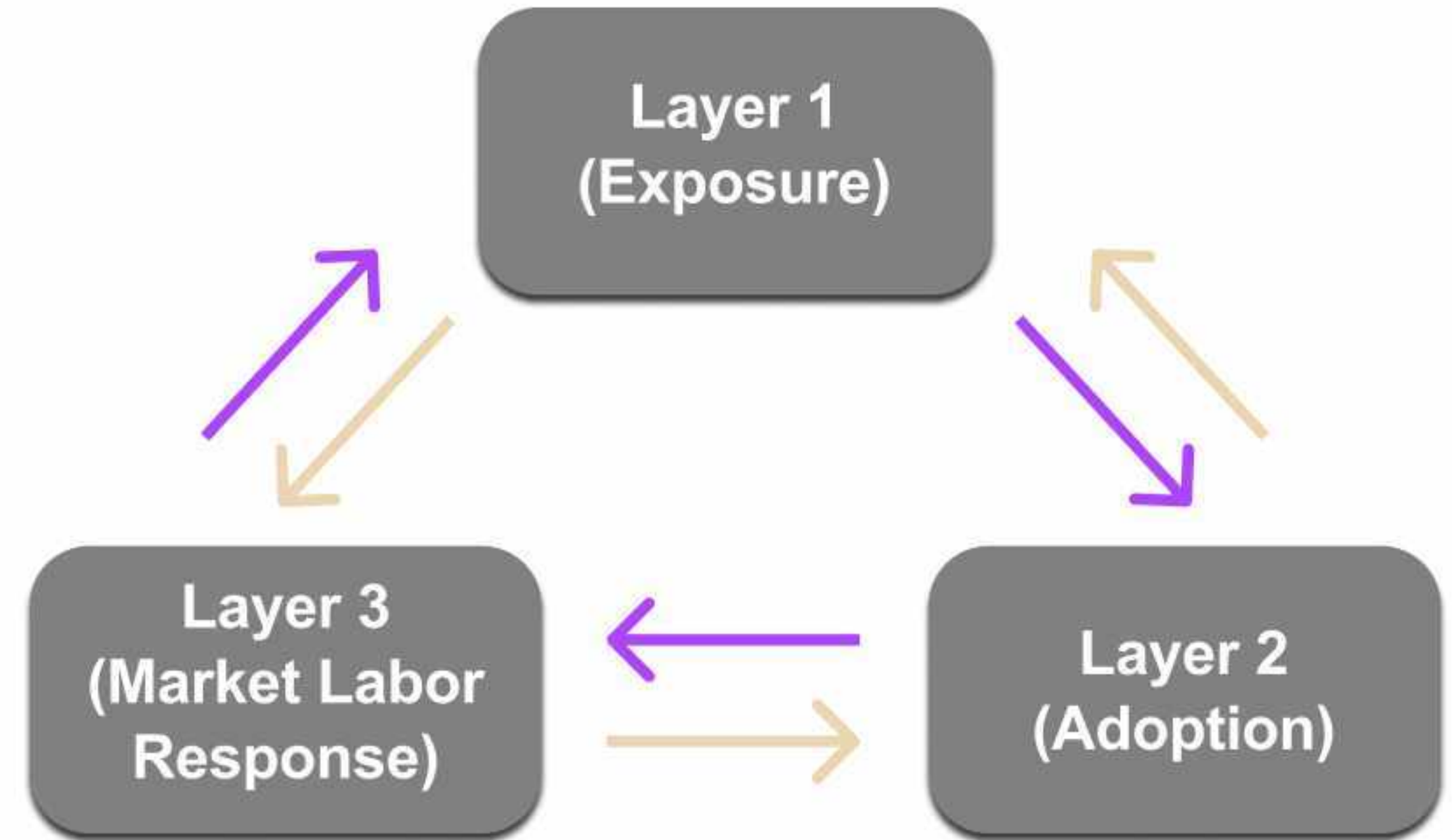
New capabilities determine exposure, which can invite or deter adoption.

Adoption choices reshape hiring, skills, and wages.

Those labor market outcomes push back on what gets built next.

The loop also runs in the other direction.

Let's explore →



(1) Exposure Leading to (2) Adoption

When a model proves useful, firms try it, especially on modular, testable, and low-liability tasks.

This pattern is visible across industries.

Software: As LLMs improved at coding-related tasks, developers quickly adopted them and usage surged.

Law: By contrast, frequent hallucination errors such as fake citations and misquoted cases have made legal services far more cautious, prompting new ethics guidance and court sanctions.

(2) Adoption Leading to (3) Labor Market Response

How firms use AI, whether to automate or augment, shapes who gets hired, what they earn, and which career paths endure.

When AI augments rather than automates, junior employees often gain the most, provided the task remains “inside the frontier.”

When firms pursue replacement, roles built around routine or substitution tasks face fewer jobs, lower pay, and limited advancement.

(3) Labor Market Response Leading to (1) Exposure

What workers and clients demand, and what regulators allow, steers R&D.

Examples:

After Irish DPC concerns, Meta paused plans to train on EU users' data in 2024, then in 2025 resumed with a narrowed approach.

OpenAI was temporarily blocked in Italy in 2023 and later fined €15m by Italy's Garante in Dec 2024 over GDPR issues.

The National Policy Framework for AI launched in 2025, directs Congress to ensure that there is a national standard, not 50 discordant State ones.

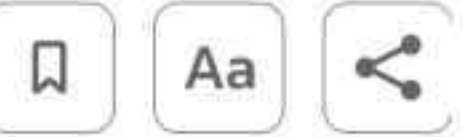


World ▾ Business ▾ Markets ▾ Sustainability ▾ Legal ▾ Commentary ▾ More ▾

Italy fines OpenAI over ChatGPT privacy rules breach

By Elvira Pollina and Alvisè Armellini

December 20, 2024 8:51 PM GMT · Updated December 20, 2024

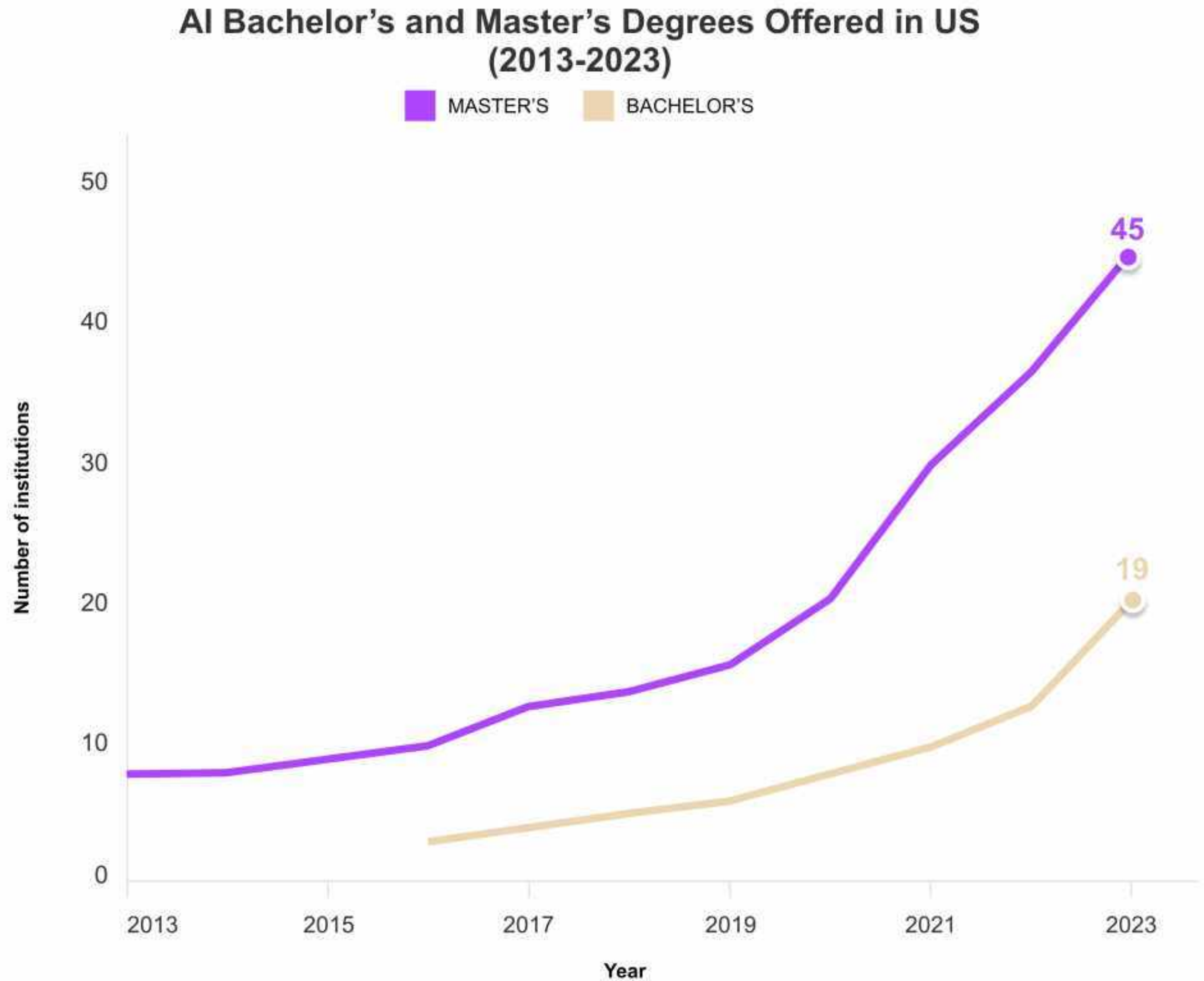


(1) Exposure Leading to (3) Labor Market Response

We can also see the cycle running in reverse.

Anticipated effects of AI on jobs nudge students, workers, and schools to make different decisions.

The number of institutions in the U.S. that offer an AI-specific bachelor's degree nearly doubled between 2022 and 2023, while the number of institutions offering an AI-specific master's degree has sharply increased as well.



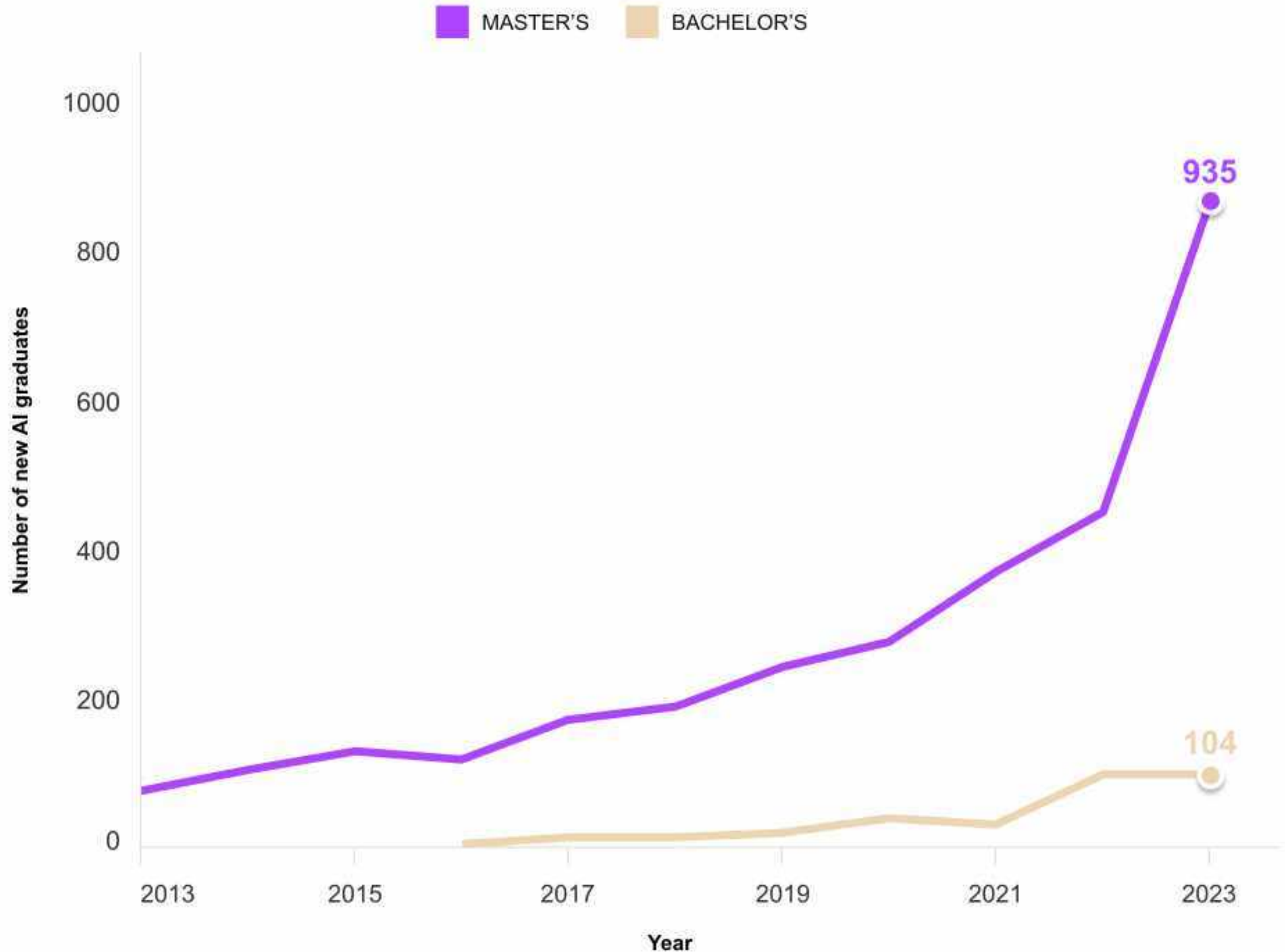
(1) Exposure Leading to (3) Labor Market Response

There was also a sharp increase in students graduating with master's degrees in AI between 2022 and 2023.

Carnegie Mellon University, which graduated more AI majors than any other institution, doubled its number of graduates.

Outside of formal education, Coursera reports a 1,060% year-over-year surge in GenAI course enrollments from 2023 to 2024, and that by 2024 a learner was signing up for a GenAI course every minute, indicating strong self-upskilling pressure.

AI Bachelor's and Master's Graduates in US (2013-2023)



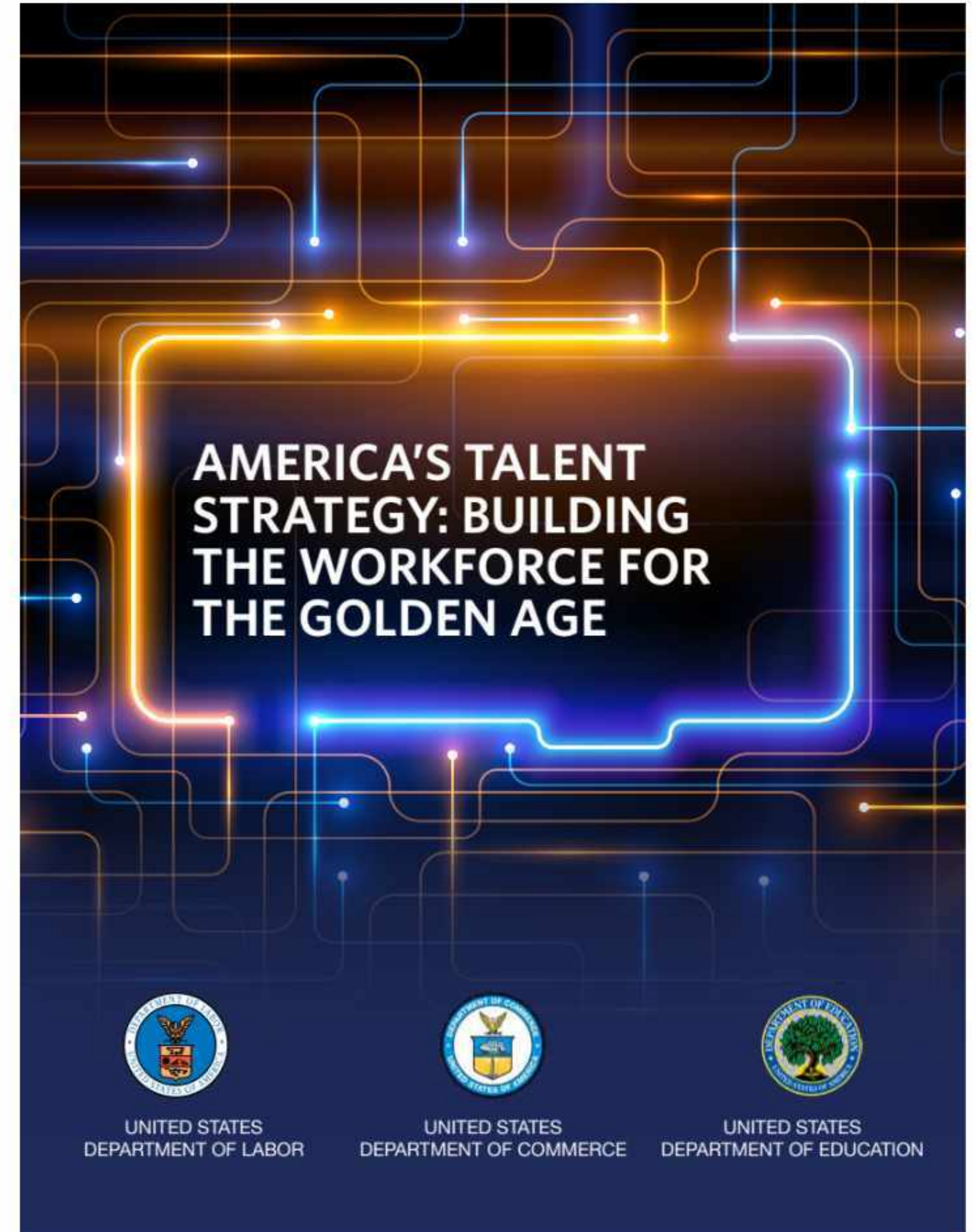
(3) Labor Market Response Leading to (2) Adoption

When layoffs trigger backlash, firms might adjust how fast and in what form they adopt this new technology.

We saw some examples earlier, whereby actors and screenwriters' protests led the industry to adopt new standards towards the use of AI in creative content.

Governmental response also influences market adoption. Recently with Trump's EO "Advancing Artificial Intelligence Education For American Youth", the Department of Labor launched an initiative to promote AI literacy across the American workforce.

By exercising conditional federal funding, procurement, and grant design, the government is able to bring businesses and workers together to aim for an AI golden age that benefits all.

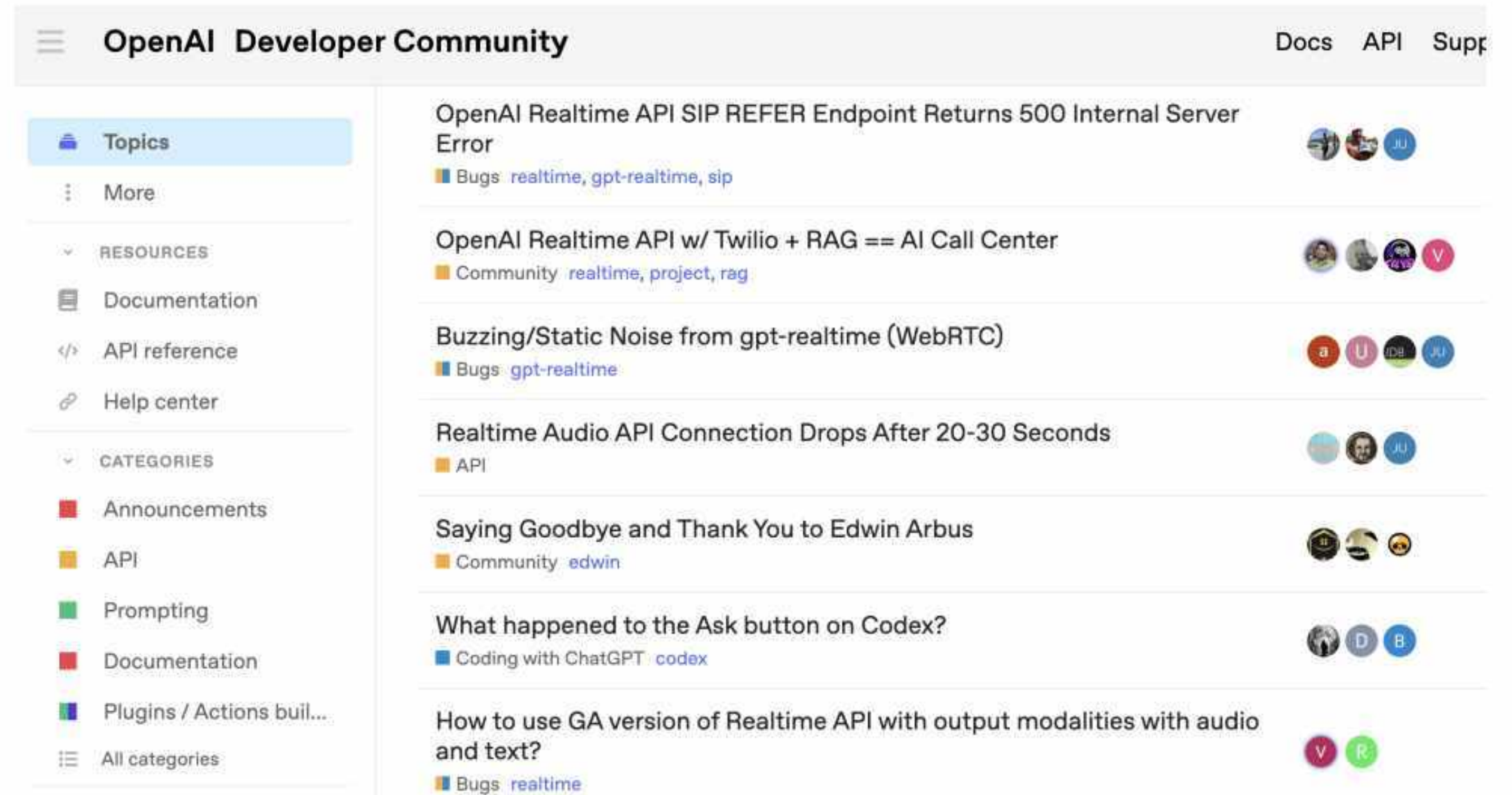


(2) Adoption Leading to (1) Exposure

What's deployed gets tested and improved.

For example, massive developer adoption means errors get reported back more quickly to model developers. Open AI's developer forum allows developers to meet and chat with other developers while building with OpenAI's APIs and developer platform.

Many Fortune-500 firms are using internal chatbots or virtual assistants. Those deployments highlight issues like “when to escalate to a human,” tone appropriateness, and how well internal knowledge is retrieved. Providers then refine model behavior to reduce escalation, improve context recall, refine tone, etc.



Feedback Loop Recap

The tasks and roles most exposed to AI determine where firms experiment first, which tools they adopt, and how aggressively automation pilots are run.

As firms incorporate AI into workflows, workers respond through augmentation (using the tools), displacement (task substitution), or reinstatement (new demand for oversight, safety, and infrastructure).

As workers upskill, resist, negotiate limits, or shift into new roles, they influence which tasks stay exposed, which become protected, and which newly open up.

Task exposure, adoption choices, and labor-market reactions shape one another in all directions.

This means that the same technology can yield different outcomes depending on how these links play out.

Understanding these loops shows what opportunities to watch out for.



The Case for Abundance

Historical Precedent for Abundance

When 90% the labor force shifted away from agriculture in the 20th century, employment didn't collapse, instead it transformed into entirely new industries, often in unexpected and unanticipated ways.

Additionally we continue to see Jevon's Paradox play out in multiple fields when innovation introduces a potential replacement for a bundle of tasks.

The Legal Discovery Paradox:

“While electronic discovery software has become a billion-dollar business since the late 1990s, jobs for paralegals and legal-support workers actually grew faster than the labor force as a whole, adding over 50,000 jobs since 2000, according to data from the U.S. Census Bureau.” - James Bessen, Executive Director of the Technology & Policy Research Initiative at Boston University

Historically, every major technological transition has produced short-term displacement in some area and long-term abundance overall.



“We estimate that about six out of 10 jobs people are doing at present didn’t exist in 1940... A lot of the things that we do today, no one was doing at that point. Most contemporary jobs require expertise that didn’t exist back then, and was not relevant at that time.”

David Autor
Professor of Economics at MIT
MIT News, 2024

Expertise

Technology, like computers, has complimented people who have judgment, analytical skills, and interpersonal skills. AI can also help to develop expertise via research and tutoring at an accelerating rate, **becoming a force multiplier for new types of expertise.**

New research shows that automation that eliminated expert tasks (decreasing expertise requirements), can reduce wages, while increasing the total number of less expert workers.

Contrarily, automation that raises requirements will raise wages, while reducing the set of qualified workers, as more expertise is required for the role. This has an overall positive effect on the market as it opens up more opportunities for one group of people, while also increasing wages for others.

David Autor, an MIT researcher, believes that people should take the implications of automation seriously, but does not see running out of jobs as one of them. From his analysis, the United States, as in much of the industrialized world, is in a demographic crunch and is more likely to see a shortage of workers, not a limitation on work.

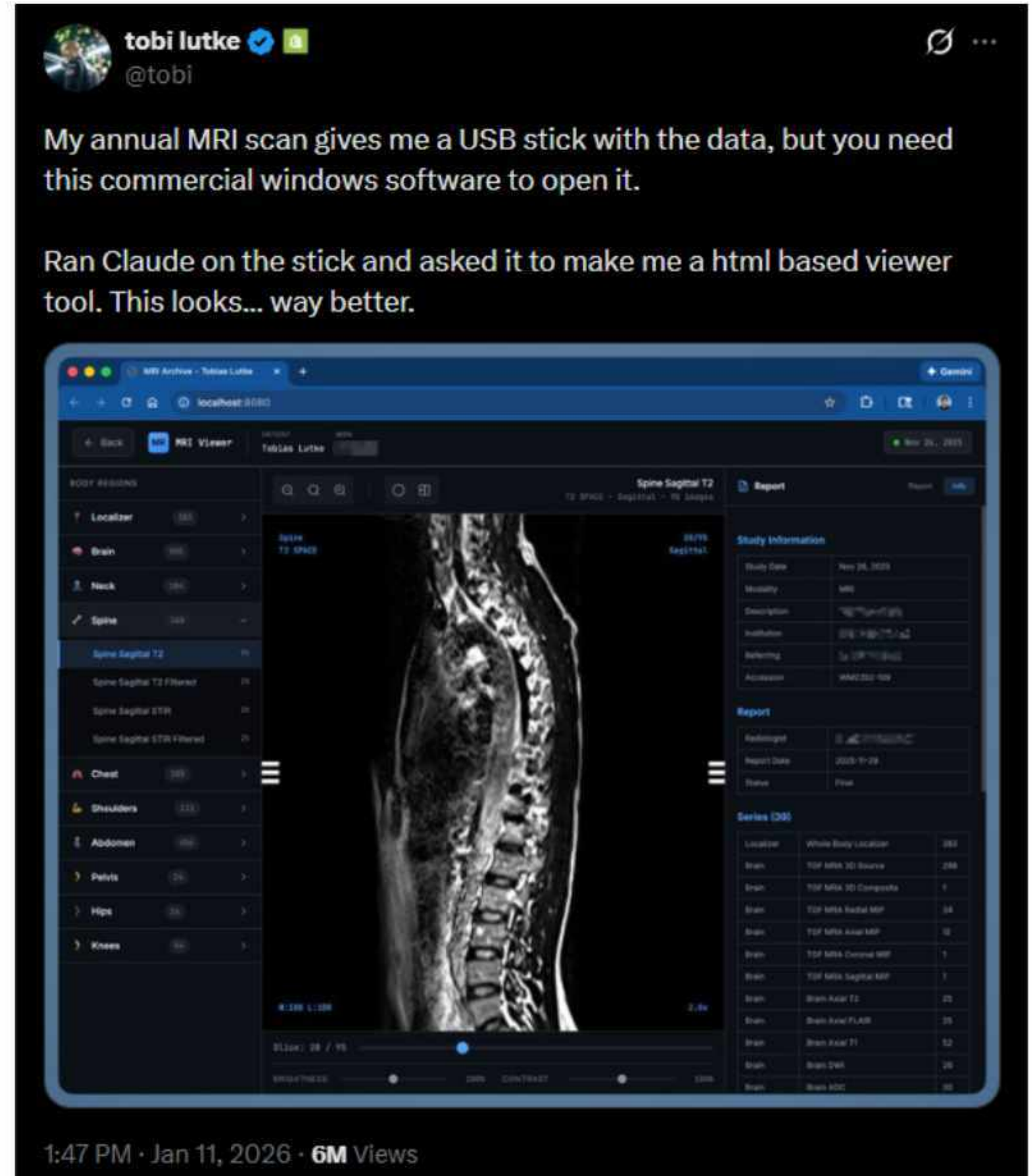
Building Towards Abundance

As the world adopts AI and gets familiar with the tools, people start to build an intuition for what it can do and how it can supplement their daily work and personal life.

Software used to be built by companies with lots of engineers, and now it can be built by individuals. Accessibility is improving everyday with no-code apps that allow beginners to start their journey.

Tobi Lutke, the CEO of Shopify, recently shared his experience using AI coding agents to develop a custom application to view his MRI scan data at home.

This type of decentralized exploration is leading to personal and professional productivity increases across the world as access to these models get cheaper and more distributed.



Thoughtful Expansion

AI can expand opportunity as it raises productivity in ways that lower costs, expand demand, and create new work.

The productivity gains will require clear workflow integration, organizational change, and skills. It currently isn't enough to just rely on model capability. This means abundance and growth is a choice made through execution and well thought through diffusion.

“Ultimately, an effective approach to leverage generative AI appears to be through human-AI collaboration that considers both context and user expertise, with AI serving as a complement rather than a replacement for human capabilities. As its adoption continues to grow, the importance of critical thinking, human oversight, and responsible integration will be pivotal in ensuring that AI delivers sustained, meaningful impacts across industries.” - OECD Artificial Intelligence Papers No. 39

In General

(1) Get AI-fluent by building

Early evidence suggests that the people who learn fastest in this transition are those who actively use AI tools and build with them.

Practical experiments like building a lightweight workflow agent or regularly working alongside AI move workers from surface familiarity to genuine understanding of the jagged frontier in their field.

(2) Stay alert to augmentation vs. automation exposure

As AI systems evolve, the line between augmentation and automation keeps shifting.

Recent data shows that new opportunities are emerging in oversight, exception handling, and stakeholder communication. These tasks absorb the value automation creates rather than being displaced by it.

But these safe zones aren't fixed. As tools improve, areas of complementarity today could become areas of substitution tomorrow, so ongoing awareness is a crucial skill.

Entry-Level Workers

(1) Watch the rise of new AI-adjacent roles

Early evidence suggests entry-level work is shifting. Listings for AI operations, model evaluation, and AI governance have multiplied. Those who adopt, implement and master this surge in AI productivity will see the largest improvements to their opportunities.

The fastest growth now appears in “AI-adjacent” product and coordination roles, where human oversight complements automation.

(2) Keep an eye on alternative and entrepreneurial paths

When entry-level hiring tightens, “unconventional” career paths like entrepreneurship or portfolio careers (a mix of freelance, contracting, and personal projects) become comparatively more accessible and attractive.

The cost of self-enterprising is at historic lows due to the rise of internet-first businesses, the creator economy, no-code automation tools, and e-commerce platforms.

Mid-Career Workers

(1) Operate on the jagged frontier (see augmentation section)

Studies show that generative AI improves performance on structured, well-defined tasks but offers mixed results on open-ended or judgment-based work.

For mid-career professionals, opportunities lie at the “jagged frontier” where human oversight remains essential. Workers who know when to lean on AI and when to override it become important “translation layers” inside their organizations, guiding how automation scales.

(2) Shape how AI fits into your function

Early evidence from firms adopting AI, such as customer support teams, law firms, and consultancies, suggests that productivity gains depend as much on workflow design as on model capability.

This opens space for mid-career workers to define how AI is used day to day: setting quality thresholds, clarifying when to escalate to a human, and refining feedback loops. These are emerging coordination tasks that reward experience and contextual understanding, not just technical skill.

Senior Leaders

(1) Help companies get over the “dip” in the J-curve through AI investments (see adoption section)

Recent enterprise surveys find that a significant majority of gen-AI pilots are failing to show measurable impact.

The limitation isn't model horsepower, but rather from poor integration, weak ownership, and reluctance to face operational friction.

This creates an opportunity for senior workers to improve on and codify AI investment policies.

≡ Search **FORTUNE** [Subscribe Now](#) [My Account](#) ▼

MIT report: 95% of generative AI pilots at companies are failing

By **Sheryl Estrada**
Senior Writer And Author Of CFO Daily

[Add us on](#)  

August 18, 2025, 6:54 AM ET



Senior Leaders

(2) Set Operational AI rules

Early evidence from firms adopting AI suggests that clear governance determines whether performance gains materialize safely. Senior leaders now face the task of defining what “good, better, best” AI use means in practice.

Many are publishing internal guidelines that distinguish between what must remain human-in-the-loop, what can proceed as “AI-first” with oversight, and what’s prohibited outright (especially in regulated domains).

Others are institutionalizing explainability and escalation by giving teams lightweight playbooks, capturing when to trust, when to test, when to defer.

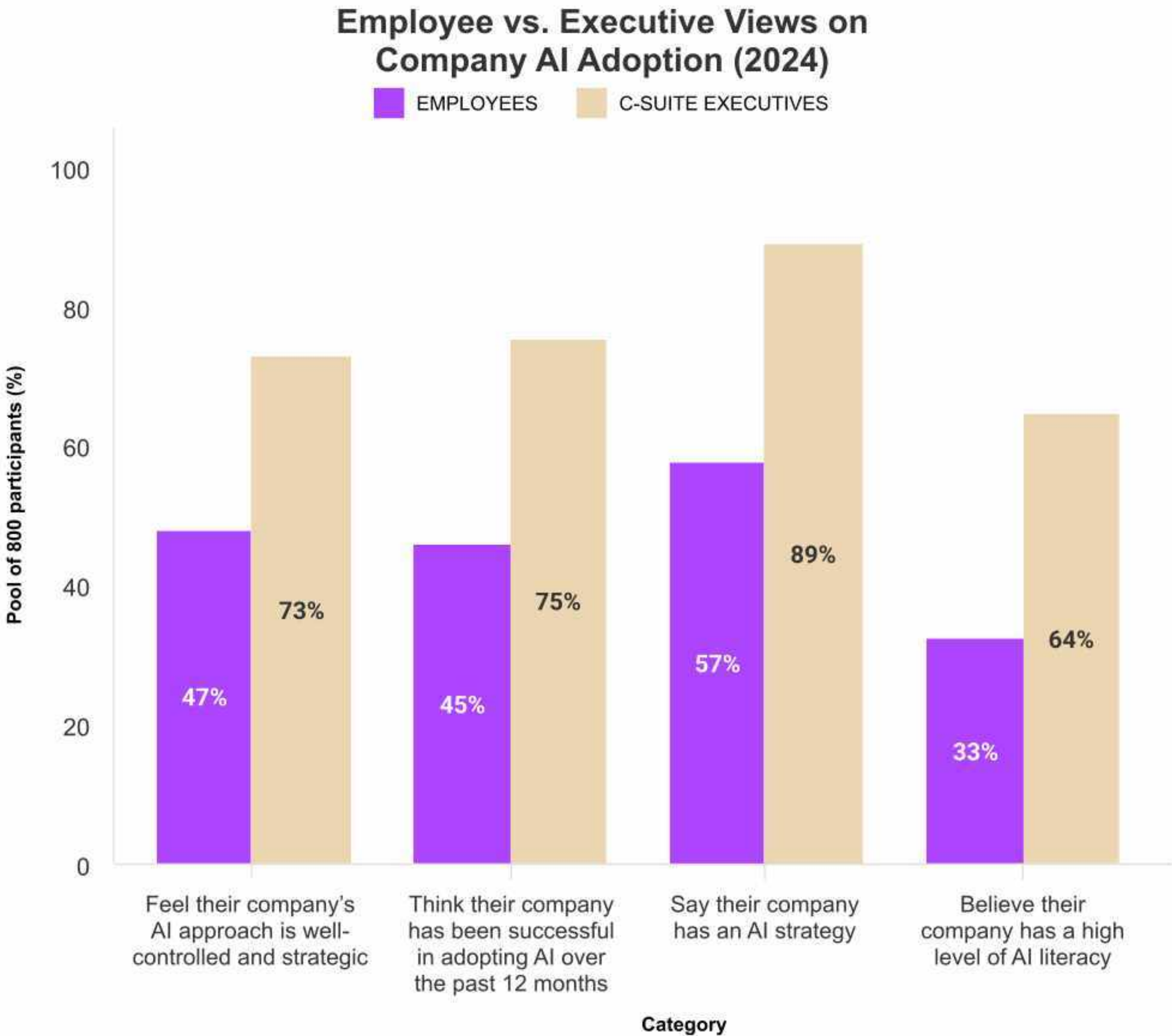
Forward-leaning firms are also tying AI outcomes to clear success metrics (accuracy, error rates, P&L impact) so that deployment decisions stay accountable and auditable.

Senior Leaders

(3) Setting the culture around AI adoption and deployment in your company

Surveys show AI can generate significant internal friction - e.g. executives push while employees resist. **This is a culture and change problem.**

This suggests there's an opportunity for senior leaders to shape their company's culture and attitude towards AI through initiatives like budgets for training, revising incentives (e.g. reward time saved and errors reduced, not tool usage), and empowering mid-career/line managers to encourage adoption.



Conclusion

Diversity of Opinions

Within the AI space today we see a vast range of opinions from entrepreneurs who are integrating AI into their systems, customers using the software in their daily lives, enterprises running tests and pilots, and research scientists who are pushing the frontier.

What sounds like contradicting takes on the state of AI come from the fact that within each layer we see multiple scenarios playing out. The conversation is nuanced when each person is experiencing a different state of AI based on their level of expertise, field, and position within a sector.

Another factor is due to how rapid the change happening is within the technology itself. Electricity as a concept is consistent, and core functions of computers got better and faster, but AI continues to advance generally across new fields and soon into the physical world with robotics.

How is AI Affecting the Labor Market Today?

Evidence of augmentation: AI boosts productivity and quality most on “inside-frontier” tasks (structured, verifiable work with clear goals and abundant examples). But on “outside-frontier” tasks (open-ended, ambiguous, or strategic work) augmentation is less reliable. It amplifies human performance where success is measurable and feedback loops are short and demands human supervision where stakes are high and truth is uncertain.

Evidence of displacement: The clearest signs of disruption appear at the entry-level in highly exposed, cognitive jobs. For now, this process is selective rather than systemic with concentration in roles with clear and repetitive deliverables.

Evidence of reinstatement: AI is creating new occupations and task bundles rather than eliminating them. Demand for AI engineers, governance specialists, and model evaluators has grown sharply. Meanwhile, the physical infrastructure supporting AI (e.g. data centers) is driving sharp growth in construction, electrical, and technician jobs.

What is the Scale and Scope of AI Adoption?

At the worker level, adoption is fast: roughly 50% of U.S. adults have used generative AI, with usage rising sharply among younger, educated workers.

At the firm-level, diffusion remains slow. As of summer 2025, fewer than 10% of businesses report regular AI use in production. Across firms, adoption is uneven. Startups and modular tasks see rapid uptake; high-stakes, regulated professions (law, healthcare, creative IP) are slower to adopt.

This mismatch explains why productivity gains appear modest and employment can remain strong. AI's reach is broad and growing, but its depth of integration is relatively shallow.

The Bureau of Economic Analysis GDP data also showcases the second and third order consequences of AI infrastructure buildout. In the first half of 2025, “information processing equipment” totaled about 1.11 percentage points to real GDP growth, while “software” added about 0.99 points. Together they explain a large portion of the 3.8 percent change in annualized growth.

What to Watch Out For and Who Will Benefit From AI Productivity Gains?

How firms reorganize around AI. Adoption speed matters less than how firms reorganize once they adopt. Known change is coming to all jobs done on software, the focus will be on how these jobs evolve and where new ones will emerge.

Education, training, and policy adaptation. Monitor whether retraining programs and higher-education systems evolve to implement AI widely, emphasizing oversight, evaluation, and contextual reasoning. On the policy side, watch for how governments incentivize productivity gains and the continuing public-private apprenticeship models that match workers to new AI-adjacent roles.

The past is the prologue and despite short term volatility, technological development has led to higher prosperity and standards of living. Those who embrace the change the fastest and dive deepest will have the best chance in creating the next era of abundance.

Disclaimer

This document is provided for educational purposes only. Nothing contained in this document is investment advice, a recommendation or an offer to sell, or a solicitation of an offer to buy, any securities or investment products. References herein to specific sectors are not to be considered a recommendation or solicitation for any such sector. Additionally, the contents herein are not to be construed as legal, business, or tax advice.

Statements in this document are made as of the date of this document unless stated otherwise, and there is no implication that the information contained herein is correct as of any other time. Certain information contained or linked to in this document has been obtained from sources believed to be reliable and current, but accuracy cannot be guaranteed.

This document contains statements that are not purely historical in nature but are “forward-looking statements” or statements of opinion or intention. Any projections included herein are also forward-looking statements. Forward-looking statements involve known and unknown risks, uncertainties (including those related to general economic conditions), assumptions and other factors, which may cause actual results, performance or achievements to be materially different from those expressed or implied by such forward-looking statements. Accordingly, all forward-looking statements should be evaluated with an understanding of their inherent uncertainty and recipients should not rely on such forward-looking statements. There is no obligation to update or revise these forward-looking statements for any reason.

This document also contains references to trademarks, service marks, trade names and copyrights of other companies, which are the property of their respective owners. Solely for convenience, trademarks and trade names referred to in this document may appear without the ® or ™ symbols, but such references are not intended to indicate, in any way, that such owner will not assert, to the fullest extent under applicable law, its rights or the right of the applicable licensor to these trademarks and trade names.